Module 5
Lecture Notes

Competencies 133 to 193. One section per competency, each headed with its number and what you should be able to do, with worked exam questions at the end of every topic.

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The urinary system

The written notes for the urinary system lecture: the kidney and its layers, the sectioned kidney, the nephron, the JGA and filtration membrane, renal blood flow, the ureters, bladder, and urethra, and the urinary disorders. Module 5 competencies 133 to 140, with nine worked questions.

Competency 133. Kidney gross anatomy and layers

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

The urinary system is made of two kidneys, two ureters, the urinary bladder, and the urethra. The kidneys filter the blood and make urine. The rest of the system carries the urine away, stores it, and releases it.

Structure	What it is
Kidneys	The paired organs that filter the blood and form urine
Ureters	The paired tubes that carry urine from the kidneys to the bladder
Urinary bladder	The muscular sac that stores urine
Urethra	The tube that carries urine from the bladder to the outside
Roles of the kidney	Filtering wastes from the blood, balancing water, ions, and pH, and helping regulate blood pressure

Where the kidneys sit

- The kidneys are bean-shaped organs on the posterior abdominal wall, one on each side of the vertebral column, roughly between the T12 and L3 vertebrae. The lowest ribs partly cover their upper ends.
- The kidneys are retroperitoneal. This means they sit behind the parietal peritoneum, outside the peritoneal cavity, against the muscles of the back.
- The right kidney sits a little lower than the left kidney because the liver takes up the space above it.
- A suprarenal (adrenal) gland sits on top of each kidney like a cap. The adrenal gland is part of the endocrine system, but the same renal fascia wraps it together with the kidney.

The hilum

Each kidney has a rounded lateral border and a concave medial border, and the medial border faces the spine. The notch in the medial border is the renal hilum. The renal artery, renal vein, nerves, lymphatic vessels, and ureter all enter or leave the kidney here. Just inside the hilum is the renal sinus, a space filled with fat that holds the renal pelvis, the calyces, and the branching vessels.

The three tissue layers, from the inside out

Three layers of tissue surround each kidney. From the kidney outward they are the renal capsule, the adipose capsule, and the renal fascia. Each layer has its own job.

Layer	Tissue	Job
Renal capsule (innermost)	A thin, tough layer of dense fibrous connective tissue on the kidney surface	Holds the kidney's shape and protects its surface
Adipose capsule (middle)	A layer of fat around the renal capsule, also called perirenal fat	Cushions the kidney and helps hold it in place
Renal fascia (outermost)	A sheet of dense connective tissue	Anchors the kidney and adrenal gland to the posterior abdominal wall and nearby structures

- Memory tip: go from the kidney outward: capsule, fat, fascia. The capsule sits on the kidney itself, the fat cushions it, and the fascia is farthest out because it ties the whole package to the body wall.
- The renal fascia is the outermost layer, not the capsule. The capsule is the smooth fibrous covering right on the kidney.
- The hilum is on the medial border, facing the spine. On a drawing, both hilums point inward, not out to the sides.

Competency 134. Internal kidney regions

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

When a kidney is cut in half from top to bottom (a frontal section), you see two kinds of tissue and a set of collecting spaces. The tissue that makes urine is called the parenchyma. The collecting spaces carry the finished urine to the ureter.

The working tissue

Renal cortex

The outer layer, just under the renal capsule. It looks grainy because it holds the renal corpuscles and the coiled tubules of the nephrons.

Renal medulla
The deeper region. It is made of cone-shaped renal pyramids, which look striped because tubules and collecting ducts run side by side through them.

Renal pyramid
A striped cone in the medulla. Each pyramid has a wide base facing the cortex and a narrow tip pointing toward the renal sinus.

Renal papilla
The narrow tip of each pyramid. Urine drains out of tiny openings on the papilla into a minor calyx.

Renal column
Cortex tissue reaching inward between neighboring pyramids. One pyramid together with the cortex around it makes up one lobe of the kidney.

Parenchyma
The cortex and the medulla together, the working tissue of the kidney. All of the nephrons are in the parenchyma.

The spaces that collect urine

Each papilla drains into a cup called a minor calyx. Several minor calyces join to form a major calyx, and the major calyces (usually two or three) join to form the renal pelvis. The renal pelvis is a funnel in the renal sinus that narrows into the ureter at the hilum.

Structure	Where it is	How to recognize it
Renal cortex	Outer layer, under the capsule	Grainy band around the edge
Renal medulla	Deep to the cortex	Made up of the pyramids
Renal pyramid	In the medulla	Striped cone with its base toward the cortex
Renal papilla	Tip of each pyramid	Points into a minor calyx
Renal column	Between neighboring pyramids	Cortex tissue reaching inward
Parenchyma	Cortex and medulla together	All the tissue that holds nephrons
Minor calyx	Around each papilla	Small cup
Major calyx	Where minor calyces join	Larger branch leading to the pelvis
Renal pelvis	In the renal sinus, at the hilum	Funnel that becomes the ureter
Renal sinus	Space just inside the hilum	Holds the pelvis, calyces, vessels, and fat

- Memory tip: renal columns are cortex. If a strip of tissue sits between two pyramids and looks like the outer layer, it is a renal column, even though it lies deep in the kidney.
- Sorting the parts: the cortex, renal columns, renal pyramids, and renal papillae are parenchyma. The minor calyces, major calyces, and renal pelvis are the spaces that collect urine. The papilla is the tip of a pyramid, so it belongs with the parenchyma, not with the urine spaces.
- The pyramids point their bases out toward the cortex and their tips in toward the calyces. An upside down pyramid is the usual slip on a drawing.

Competency 135. Nephron structure

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

The nephron is the microscopic unit that makes urine. Each kidney has roughly a million of them. A nephron has two main parts. The renal corpuscle filters the blood, and the renal tubule turns that filtrate into urine.

The renal corpuscle

The renal corpuscle is the filtering head of the nephron, and it always sits in the cortex. It is made of a ball of capillaries and the cup that surrounds it.

Glomerulus
The ball of capillaries. Its capillary walls are fenestrated (full of small pores), which makes them very leaky.

Glomerular capsule (Bowman's capsule)
The double-walled cup around the glomerulus. Its outer parietal layer is simple squamous epithelium and forms the outside wall of the cup. Its inner visceral layer hugs the glomerular capillaries and is made of podocytes.

Podocytes
The cells of the visceral layer. They have foot-like extensions (foot processes) that wrap around the capillaries. The narrow gaps between the feet are filtration slits. The slits are gaps between the podocyte feet, not holes in the capillary wall.

Capsular space
The space between the parietal and visceral layers. Filtrate collects in this space and then flows into the tubule.

The renal tubule and collecting duct

The renal tubule leaves the corpuscle as one long tube with three parts, and then empties into a collecting duct that is shared with other nephrons.

Proximal convoluted tubule (PCT)
Comes first. It is coiled, stays in the cortex, and is lined with cuboidal cells that carry a brush border of microvilli. The brush border belongs on the proximal tubule, the first coiled segment, not the distal one.

Nephron loop (loop of Henle)
A U-shaped section that dips toward or into the medulla. It has a descending limb going down and an ascending limb coming back up.

Distal convoluted tubule (DCT)
The loop returns to the cortex and becomes the DCT, a second coiled section with cuboidal cells and few microvilli.

Collecting duct
The DCT empties into a collecting duct. Many nephrons drain into the same collecting duct, so it is shared among them. The collecting ducts run down through the pyramids and merge into larger papillary ducts, which open on the renal papilla.

Structure	Where it is	Key feature
Glomerulus	Renal corpuscle, cortex	Ball of fenestrated capillaries where blood is filtered
Glomerular capsule	Renal corpuscle, cortex	Double-walled cup with a parietal layer and a visceral layer
Podocytes	Visceral layer of the capsule	Foot-like extensions wrap the capillaries and leave filtration slits
Capsular space	Between the two capsule layers	Collects the filtrate
Proximal convoluted tubule	Cortex	First coiled section; cuboidal cells with a brush border
Nephron loop	Dips into the medulla	U-shaped, with a descending limb and an ascending limb
Distal convoluted tubule	Cortex	Second coiled section; few microvilli
Collecting duct	Cortex down through the pyramids	Shared by many nephrons; leads to papillary ducts

The path of the filtrate, in order

1. Glomerulus: blood is filtered out of the capillaries.
2. Capsular space: the filtrate collects inside the glomerular capsule.
3. Proximal convoluted tubule: the first coiled section of the tubule.
4. Descending limb of the nephron loop: the filtrate moves down toward the medulla.
5. Ascending limb of the nephron loop: the filtrate moves back up toward the cortex.
6. Distal convoluted tubule: the second coiled section of the tubule.
7. Collecting duct: fluid from many nephrons joins here.
8. Papillary duct: opens on the renal papilla and drains into a minor calyx.

- Memory tip: the segments are named in the order the filtrate meets them. Proximal means near the corpuscle, distal means farther away. Learn them in that order and you also know the path of the filtrate.

Competency 136. Cortical vs juxtamedullary nephrons

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

Every nephron has the same parts. Nephrons differ in where the renal corpuscle sits and how deep the nephron loop reaches. The depth of the loop decides which capillary bed runs alongside the tubule.

Cortical nephrons
About 85 percent of all nephrons. Their renal corpuscles sit in the outer part of the cortex. Their loops are short and barely dip into the outer medulla. They are wrapped by the peritubular capillaries, a mesh of capillaries that surrounds the tubules in the cortex.

Juxtamedullary nephrons
About 15 percent. Juxta means next to, and their renal corpuscles sit in the deepest part of the cortex, right next to the medulla. Their loops are long and reach deep into the medulla, close to the renal papilla. Long, straight capillaries called the vasa recta run down and back up beside these loops. Recta means straight.

Both capillary beds are supplied by efferent arterioles leaving the glomeruli. The long loops of the juxtamedullary nephrons, together with the vasa recta, are what allow the kidney to make concentrated urine.

Feature	Cortical nephrons	Juxtamedullary nephrons
Share of all nephrons	About 85 percent	About 15 percent
Where the renal corpuscle sits	Outer cortex	Deep cortex, next to the medulla
Nephron loop	Short; barely enters the medulla	Long; reaches deep into the medulla
Capillary bed along the tubule	Peritubular capillaries	Vasa recta

- Memory tip: long loop, long straight vessels. The juxtamedullary nephron reaches deep, so its capillaries (the vasa recta) have to reach deep with it.

- Both kinds of renal corpuscle sit in the cortex. The difference is the outer cortex for the cortical nephron and the deep cortex next to the medulla for the juxtamedullary nephron.
- The juxtamedullary nephrons are the rare ones, about 15 percent, but they are the ones that concentrate the urine. Do not swap the percents, and do not swap the capillary beds.

Competency 137. JGA and filtration membrane

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

The juxtaglomerular apparatus

After the nephron loop climbs back to the cortex, the tubule passes right beside its own renal corpuscle, between the afferent and efferent arterioles. The spot where the tubule and the arterioles touch is the juxtaglomerular apparatus (JGA). It checks the fluid in the tubule and the blood pressure in the arteriole, and it helps adjust filtration.

Macula densa
A patch of tall, closely packed cells in the tubule wall, where the ascending limb of the loop ends and the distal convoluted tubule begins. Macula densa means dense spot, which describes how the crowded cells look on a slide. These cells sense the amount of sodium chloride in the fluid passing by.

Juxtaglomerular cells (JG cells, also called granular cells)
Modified smooth muscle cells, mostly in the wall of the afferent arteriole. Their granules hold the enzyme renin, which they release when blood pressure drops.

Structure	Where it is	What it does
Juxtaglomerular apparatus	Where the tubule touches the afferent and efferent arterioles of its own corpuscle	Monitors the tubule fluid and the arteriole and helps adjust filtration
Macula densa	Tubule wall, where the ascending limb meets the distal convoluted tubule	Senses the sodium chloride level of the fluid
Juxtaglomerular (granular) cells	Wall of the afferent arteriole	Release renin when blood pressure is low

- The juxtaglomerular cells sit in the afferent arteriole, the vessel going in, and the macula densa sits in the distal tubule. These two get swapped a lot.

The filtration membrane

Blood in the glomerulus is filtered into the capsular space across the filtration membrane. Fluid crosses three layers in a set order, and each layer holds back a different size of particle.

Layer (from the blood side to the capsular space)	What it is	What it holds back
Fenestrated endothelium	The pored wall of the glomerular capillary; lets plasma through	Blood cells and platelets
Basement membrane	The layer between the capillary wall and the podocytes	Large plasma proteins
Filtration slits of the podocytes	The gaps between podocyte feet, bridged by thin slit membranes	Most of the smaller (medium sized) proteins that get past the basement membrane

- What passes all three layers is the filtrate: water and small dissolved substances such as ions, glucose, and wastes like urea. Healthy filtrate has almost no protein and no blood cells.
- Protein or blood in the urine is a sign that the filtration membrane is damaged.
- Memory tip: each layer catches something smaller than the one before it: cells first, then big proteins, then the smaller proteins. Run the layers from the blood side out.

Competency 138. Kidney blood supply pathway

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

The kidneys receive about a quarter of the blood the heart pumps at rest. Blood follows a fixed chain of arteries inward, passes through two capillary beds, and leaves through a chain of veins that runs back along the same route.

The names of the vessels tell you where they run. A pyramid with its surrounding cortex is a lobe, so interlobar vessels run between lobes, up through the renal columns. Arcuate means arched, and these vessels arch over the bases of the pyramids at the border between the cortex and medulla. Cortical radiate vessels radiate outward into the cortex. Some books call them interlobular vessels.

Renal artery to renal vein, in order

1. Renal artery: branches off the abdominal aorta and enters the kidney at the hilum.
2. Segmental arteries: branches of the renal artery inside the renal sinus, before the blood reaches the tissue.
3. Interlobar arteries: run up through the renal columns, between the pyramids.
4. Arcuate arteries: arch over the bases of the pyramids, along the border between cortex and medulla.

5. Cortical radiate arteries: also called interlobular arteries; radiate outward into the cortex.
6. Afferent arteriole: carries blood into one glomerulus.
7. Glomerulus: the first capillary bed, where blood is filtered.
8. Efferent arteriole: carries blood out of the glomerulus.
9. Peritubular capillaries and vasa recta: the second capillary bed. Peritubular capillaries wrap the tubules in the cortex, and the vasa recta run beside the long loops in the medulla.
10. Cortical radiate veins: drain the cortex and run back toward the medulla.
11. Arcuate veins: arch over the bases of the pyramids.
12. Interlobar veins: run down through the renal columns toward the renal sinus.
13. Renal vein: leaves the kidney at the hilum and empties into the inferior vena cava.

- This path is unusual in one way. In most of the body, a capillary bed drains into a venule. The glomerulus has an arteriole on both sides, the afferent arteriole going in and the efferent arteriole coming out. Blood then passes through a second capillary bed before it reaches any vein.
- Memory tip: blood meets two capillary beds in the kidney, one after the other: the glomerulus, where filtration happens, and then the peritubular capillaries and vasa recta around the tubules. The efferent arteriole is the vessel between them, not the afferent arteriole.
- On the way out, the veins repeat the artery names in reverse: cortical radiate, arcuate, interlobar, renal.
- Only the interlobar arteries run between the pyramids. The segmental arteries branch earlier, in the renal sinus.

Competency 139. Ureters, bladder, urethra and urine path

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

Once fluid leaves the collecting duct it is urine, and nothing along the rest of the path changes it much. The rest of the system collects the urine, carries it, stores it, and releases it.

The path of urine, in order

1. Collecting duct: gathers fluid from many nephrons and carries it through the pyramid.
2. Papillary duct: opens on the renal papilla.
3. Minor calyx: cup that catches urine from one papilla.
4. Major calyx: forms where several minor calyces join.
5. Renal pelvis: funnel that collects urine from the major calyces and narrows into the ureter.
6. Ureter: carries urine down to the bladder.
7. Urinary bladder: stores the urine.
8. Urethra: carries urine out of the body.
9. External urethral orifice: the opening to the outside.

The ureters

- Each ureter is a muscular tube about 30 cm long. It leaves the renal pelvis at the hilum, runs down the posterior abdominal wall behind the peritoneum, and enters the posterior wall of the bladder.
- The ureter passes through the bladder wall at a slant. When the bladder fills, pressure squeezes this slanted part closed, which keeps urine from flowing back up toward the kidney.
- The ureter wall has three layers. The inner mucosa is lined with transitional epithelium (urothelium), which can stretch. The middle layer is smooth muscle, which pushes urine along by peristalsis. The outer layer is a fibrous adventitia.

The urinary bladder

The urinary bladder is a hollow, muscular sac in the pelvic cavity, just behind the pubic symphysis. In males it sits in front of the rectum and above the prostate. In females it sits in front of the vagina and below the uterus. Only its upper surface is covered by peritoneum.

Layer (from the inside out)	What it is	Key feature
Mucosa	Transitional epithelium (urothelium) on a layer of connective tissue (lamina propria)	Stretches as the bladder fills; folds into rugae when the bladder is empty
Detrusor muscle	The muscular layer, made of layers of smooth muscle	Contracts to squeeze urine out
Adventitia	Outer fibrous connective tissue	Over the upper surface, the peritoneum (a serosa) covers the bladder

Trigone
A smooth triangle on the floor of the bladder. Its three corners are the two ureteral openings at the back (the upper corners) and the internal urethral orifice at the front (the lowest point), where the urethra begins. The trigone has no rugae, so it stays smooth even when the bladder is empty.

Internal urethral sphincter
A thickening of smooth muscle at the neck of the bladder, where the urethra begins. It is involuntary, so you cannot control it.

External urethral sphincter
Skeletal muscle that surrounds the urethra as it passes through the pelvic floor. It is voluntary and gives conscious control over urination.

- The internal sphincter is smooth muscle you cannot control; the external one is skeletal muscle you can. These two get swapped, so check them.

The urethra

The urethra carries urine from the bladder to the outside. It is very different in females and males, and the male urethra is the longer one.

Feature	Female urethra	Male urethra
Length	About 4 cm	About 20 cm
What it carries	Urine only	Urine and semen
Parts	One short tube	Prostatic urethra through the prostate, membranous urethra through the deep muscles of the perineum, and spongy urethra through the corpus spongiosum of the penis, in that order from the bladder down
External urethral orifice	Between the clitoris and the vaginal opening	At the tip of the glans penis

Competency 140. Clinical anatomy of urinary disorders

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

Urinary disorders make the most sense when you place each one on the structures you already know. Some damage the filtering parts of the nephron. Others block, infect, or weaken the tubes that carry and hold urine.

Any blockage along the path makes urine back up into the parts above it. When urine backs up into the renal pelvis and calyces, they swell. This swelling is called hydronephrosis.

Disorder	Structure or pathway affected	What changes
Nephroptosis	Adipose capsule and renal fascia; the ureter below	The kidney drops out of its normal position, often when its cushion of fat is thin or lost. The ureter can kink and block urine flow
Kidney stones (renal calculi)	Form in the calyces and renal pelvis; lodge in the ureter	Minerals harden into stones. A stone tends to stick at the narrow points of the ureter: where it leaves the renal pelvis, where it crosses the iliac vessels at the pelvic brim, and where it enters the bladder
Urinary tract infection (UTI)	Urethra, bladder, and, if it climbs, the ureter and kidney	Bacteria, most often from the intestine (such as E. coli), climb up the urine path and inflame its lining. Infection of the bladder is cystitis. Infection that reaches the renal pelvis and kidney tissue is pyelonephritis
Glomerulonephritis	Glomeruli	The glomeruli become inflamed, which damages the filtration membrane. Protein and blood can leak into the urine
Renal failure	Nephrons of both kidneys	The kidneys lose their ability to filter the blood, either suddenly (acute) or slowly over time (chronic)
Polycystic kidney disease	Renal parenchyma (the tubules)	An inherited disorder in which many fluid-filled cysts form and enlarge the kidneys, crowding out working tissue
Urinary incontinence	Detrusor muscle, urethral sphincters, and pelvic floor	Loss of voluntary control over urination. In stress incontinence a weak pelvic floor and external sphincter leak with coughing or lifting. In urge incontinence the detrusor contracts too readily. In overflow incontinence the bladder cannot empty fully, so it overfills and leaks. In functional incontinence the urinary organs work, but the person cannot reach a toilet in time

What this means for patients

- Position and pathway predict the problem. The kidney depends on its adipose capsule to stay in place. When that fat is lost, the kidney loses its support and can drop, and the ureter can kink and block the flow of urine.
- Urine travels one narrow path. A kidney stone often causes little pain while it sits in the renal pelvis. Pain usually starts when the stone moves into the ureter, where it scrapes and blocks a tube only a few millimeters wide. Where the stone sits along that path shapes where the pain is felt.
- The two urethras differ, and so does the risk. The female urethra is short and opens close to the anus, which gives bacteria a shorter climb to the bladder. This is one reason urinary tract infections are more common in women.
- A urinary tract infection can sit in the urethra, the bladder, or the kidney. It is not tied to only one spot.

The male reproductive system

The written notes for the male reproductive lecture: the scrotum and testis, the sperm duct pathway, the sperm cell, the accessory glands and penis, and the disorders of each structure. Module 5 competencies 141 to 145, with eight worked questions.

Before the competencies: the four groups of parts

The male reproductive system makes sperm and delivers them. Its parts sort into four groups: the gonads, a duct system, the accessory glands, and the supporting structures. Learn the groups first, because every structure in this chapter belongs to one of them, and the order of the duct system is the order sperm travel.

Group	What it includes
Testes	The male gonads. They produce sperm and secrete hormones (the interstitial cells make testosterone)
Duct system	The epididymis, ductus deferens, ejaculatory ducts, and urethra, which store, mature, and carry sperm
Accessory glands	The seminal vesicles, prostate, and bulbourethral glands, which add fluid to form semen
Supporting structures	The scrotum and the penis

- Competency 141 covers the testes and the scrotum. Competency 142 covers the duct system. Competency 143 covers the sperm cell itself. Competency 144 covers the accessory glands and the penis. Competency 145 puts the disorders on the structures.
- Semen is sperm plus the secretions of the three accessory glands. The sperm are only a small part of it.

Competency 141. Scrotum and testis structure

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.

This section covers the gonads, the testes, and the pouch that holds them. The testes sit outside the body because sperm production needs a temperature a few degrees below the body core, and two muscles in the scrotum adjust that temperature by moving the testes closer to or farther from the body.

The scrotum and its two muscles

The scrotum is a pouch of skin and subcutaneous tissue that hangs behind the penis, outside the pelvic cavity. A ridge down its middle, the raphe, marks where the scrotal septum divides the inside into two compartments, one for each testis.

Muscle	Tissue type	Where it is	What it does when it contracts
Dartos muscle	Smooth muscle	A thin layer in the subcutaneous tissue of the scrotum, under the skin	Wrinkles and tightens the scrotal skin, which cuts heat loss
Cremaster muscle	Skeletal muscle	Comes off the internal oblique muscle of the abdominal wall, travels down in the spermatic cord, and loops around each testis	Pulls the testis up toward the warmer body wall

- The two muscles are different tissues. Dartos is smooth muscle in the scrotal wall. Cremaster is skeletal muscle in the spermatic cord.
- In warm conditions both muscles relax and the testes hang lower, away from the body.
- The two structures that set the temperature of the testes are the two muscles, the dartos and the cremaster, not the tunics.

The coverings of the testis: the two tunics

Tunica vaginalis
The outer tunic, a serous membrane with a parietal layer and a visceral layer. It comes from a pouch of peritoneum that the testis brought with it when it moved down into the scrotum before birth, and it covers the front and sides of the testis (it partly covers the testis, not the whole thing). It lets the testis move smoothly inside the scrotum.

Tunica albuginea
The inner tunic, a tough white capsule of dense connective tissue that sits directly on the testis. It sends thin walls (septa) inward that divide the testis into a few hundred wedge-shaped lobules.

Inside the testis

Each lobule holds tightly coiled seminiferous tubules. These tubules are the only place sperm are made. On a slide, a seminiferous tubule is a round or oval profile with a thick wall of cells around a small central lumen. The developing sperm cells are stacked in that wall, the least mature at the outer edge and the most mature next to the lumen.

Sustentacular cells (Sertoli cells)
Tall cells inside the wall of each seminiferous tubule, reaching from the outer edge of the tubule to its lumen. They surround, support, and nourish the developing sperm cells. Tight junctions between neighboring sustentacular cells form the blood-testis barrier, which keeps substances in the blood away from the developing sperm.

Interstitial cells (Leydig cells)
Cells in the connective tissue between the tubules, outside the tubule walls. They produce testosterone.

Straight tubules
Short tubules that carry sperm from the seminiferous tubules toward the back of the testis.

Rete testis
A network of small channels in the connective tissue at the posterior side of the testis. It collects the sperm from the tubules.

Efferent ductules
A set of tiny ducts that leave the rete testis, carry the sperm out of the testis, and deliver them into the head of the epididymis.

- Memory tip: the names tell you where to look on the slide. Sustentacular cells sustain the sperm, so they sit inside the tubule wall with the sperm. Interstitial cells sit in the interstitial spaces, between the tubules.
- Interstitial cells sit between the tubules, not in the tubule wall. The sustentacular cells are the ones in the wall.
- Inside the testis the sperm go seminiferous tubules, straight tubules, rete testis, efferent ductules, epididymis. The next competency picks up from there.

The scrotum and the testis, from the outside in

Structure	Where it is	What it does
Scrotum	Skin pouch behind the penis, outside the pelvic cavity	Holds the testes and keeps them cooler than the body core; the scrotal septum gives each testis its own compartment
Dartos muscle	Smooth muscle in the subcutaneous tissue of the scrotum	Wrinkles the scrotal skin to reduce heat loss
Cremaster muscle	Skeletal muscle in the spermatic cord, looping around the testis	Raises the testis toward the body
Tunica vaginalis	Serous membrane over the front and sides of the testis	Lets the testis move smoothly inside the scrotum
Tunica albuginea	Dense white capsule directly on the testis	Encloses the testis and divides it into lobules
Seminiferous tubules	Coiled tubules inside each lobule	Produce sperm
Sustentacular cells	Inside the wall of each seminiferous tubule	Support and nourish developing sperm; form the blood-testis barrier
Interstitial cells	In the connective tissue between the tubules	Produce testosterone
Rete testis	Network of channels at the posterior side of the testis	Collects sperm from the tubules
Efferent ductules	Small ducts between the rete testis and the epididymis	Carry sperm out of the testis into the epididymis

Competency 142. Sperm duct system and pathway

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

Only the seminiferous tubules make sperm. Every duct after them stores, matures, or carries the sperm. The route runs in one direction: up out of the scrotum, into the pelvis behind the bladder, and then down through the prostate and the penis to the outside.

The path of sperm, from the testis to the outside

1. Seminiferous tubules: sperm are made in the walls of the tubules inside the testis.
2. Rete testis: straight tubules carry the sperm into this network at the back of the testis.
3. Efferent ductules: small ducts carry the sperm out of the testis.
4. Epididymis: sperm enter the head, are stored, and mature as they move to the tail.
5. Ductus deferens: sperm travel up the spermatic cord, through the inguinal canal, over the ureter, and behind the bladder to the ampulla.
6. Ejaculatory duct: the ductus deferens joins the duct of the seminal vesicle and passes through the prostate.
7. Prostatic urethra: sperm enter the urethra inside the prostate.
8. Membranous urethra: sperm pass through the pelvic floor.
9. Spongy urethra: sperm pass through the corpus spongiosum of the penis.

10. External urethral orifice: semen leaves the body at the tip of the glans.

- Memory tip: SEVEN UP gives the main stops in order: Seminiferous tubules, Epididymis, Vas deferens, Ejaculatory duct, (N is for nothing), Urethra, Penis.
- Only the seminiferous tubules make sperm. The epididymis stores and matures them.

Epididymis

The epididymis is a comma-shaped structure on the posterior side of each testis. Its head sits at the top, where the efferent ductules enter. Its body runs down the back of the testis, and its tail sits at the bottom. Inside is a single duct about 6 meters long, coiled tightly enough to fit in a few centimeters. Sperm are stored here, and they gain the ability to swim as they move through it.

Ductus deferens

The ductus deferens (vas deferens) begins at the tail of the epididymis. It is a long duct with a thick wall of smooth muscle, so it feels like a firm cord through the scrotal skin. It climbs in the spermatic cord, passes through the inguinal canal into the pelvic cavity, crosses over the ureter, and runs down the posterior surface of the bladder. Its last part widens into the ampulla of the ductus deferens.

Ejaculatory duct

Behind the bladder, each ampulla joins the duct of a seminal vesicle. The two together form the short ejaculatory duct. There is one on each side. Each ejaculatory duct passes through the prostate and empties into the prostatic urethra. It does not end in the bladder.

The three parts of the male urethra

The male urethra carries both urine and semen. It has three parts, named for what they pass through.

Part	Where it runs	Notes
Prostatic urethra	Through the prostate, just below the bladder	First part; the ejaculatory ducts and the prostate's ducts empty into it
Membranous urethra	Through the muscles of the pelvic floor (the urogenital diaphragm), in the perineum	Short middle part
Spongy urethra (penile urethra)	Through the corpus spongiosum of the penis	Longest part; opens at the external urethral orifice on the glans

Spermatic cord

The spermatic cord is the bundle that suspends each testis and carries its duct, vessels, and nerves between the scrotum and the abdomen. It passes through the inguinal canal, a slanted passage through the lower anterior abdominal wall. The cord holds these structures:

- The ductus deferens.
- The testicular artery, a branch of the abdominal aorta.
- The pampiniform plexus, a network of veins wrapped around the testicular artery. It cools the blood flowing in to the testis, and its veins join to form the testicular vein.
- Lymphatic vessels and nerves.
- The cremaster muscle and layers of fascia that wrap the whole cord.

If you cut the cord across, you find the ductus deferens, blood vessels, nerves, and the cremaster muscle inside it.

The ducts, in order

Structure	Where it is	What it does
Epididymis	Posterior side of the testis	Stores sperm; sperm gain the ability to swim
Ductus deferens	From the epididymis, up the spermatic cord, through the inguinal canal, behind the bladder	Carries sperm into the pelvis
Ejaculatory duct	Inside the prostate	Joins the ductus deferens and seminal vesicle duct; empties into the prostatic urethra
Prostatic urethra	Inside the prostate	First part of the urethra
Membranous urethra	Through the pelvic floor muscles	Short middle part of the urethra
Spongy urethra	Inside the corpus spongiosum of the penis	Longest part; opens at the external urethral orifice
Spermatic cord	From the scrotum through the inguinal canal	Carries the ductus deferens, vessels, and nerves of the testis

Competency 143. Sperm cell structure

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

A mature sperm cell (spermatozoon) is built for one job: reaching an oocyte and delivering its chromosomes. It has two main parts, a head and a tail. The tail is a long flagellum. From front to back, the tail has four parts: the neck, the middle piece, the principal piece, and the end piece. These notes count the neck as part of the tail, so the tail starts right behind the head.

The head and its cap

Head
A flattened oval at the front of the cell. Most of it is the nucleus, which holds 23 chromosomes, half the number in a body cell. There is almost no cytoplasm.

Acrosome
A cap that covers the front of the nucleus. It is a sac of enzymes that lets the sperm digest a path through the layers around the oocyte.

The four parts of the tail

Neck
The short, narrow part just behind the head. It holds the centrioles, and the core of microtubules that runs the length of the tail starts here.

Middle piece
The first long stretch of the tail, packed with mitochondria wound in a spiral around the core of the tail. They make the ATP that powers the tail. This is the only part packed with mitochondria.

Principal piece
The longest part of the tail. It does most of the whipping that moves the cell forward.

End piece
The short, thin tip at the very end of the tail, tapering to a point.

The parts of a sperm cell, front to back

Part	Where it is	What it holds or does
Head	Front of the cell	Holds the nucleus with 23 chromosomes
Acrosome	Cap over the front of the head	Holds enzymes that help the sperm get through the layers around the oocyte
Neck	Just behind the head	Holds the centrioles; the tail core starts here
Middle piece	First long stretch of the tail	Packed with mitochondria that power movement
Principal piece	Most of the tail	Longest part; drives the swimming
End piece	Tip of the tail	Short, thin final part

- Memory tip: name the parts front to back: head with its acrosome cap, then neck, middle piece, principal piece, end piece. The middle piece is the only part packed with mitochondria.
- The mitochondria go in the middle piece, not in the head or the end piece.
- On a slide of the testis, you see sperm heads and tails crowded in the lumen of the seminiferous tubules. On a sperm smear, the head is the dark, densely stained oval, and the tail is the long thin line behind it.

Competency 144. Accessory glands and penis

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

The accessory glands

Sperm are only a small part of semen. Three sets of accessory glands add most of the fluid. The mix of sperm and these secretions is semen. Where each gland sits tells you where its fluid joins the duct system.

Seminal vesicles
A pair of glands on the posterior surface of the bladder, beside the ampullae of the ductus deferens. The duct of each seminal vesicle joins the ductus deferens to form the ejaculatory duct. Their fluid is alkaline and rich in fructose, a sugar the sperm use for energy. It makes up about 60 percent of the volume of semen, the largest share of any gland.

Prostate
A single gland about the size of a walnut. It sits just below the bladder and wraps around the prostatic urethra. Its back surface lies against the rectum. Many small ducts carry its milky fluid into the prostatic urethra, where it helps the sperm stay active.

Bulbourethral glands (Cowper glands)
A pair of pea-sized glands just below the prostate, beside the membranous urethra in the pelvic floor. Their ducts open into the spongy urethra. They release a clear alkaline mucus before ejaculation that neutralizes traces of acidic urine in the urethra and lubricates the tip of the penis.

Gland	Number	Where it is	Empties into	What it adds
Seminal vesicles	Paired	Posterior surface of the bladder	Ejaculatory duct	Alkaline, fructose-rich fluid that nourishes the sperm; about 60 percent of semen
Prostate	Single	Just below the bladder, around the prostatic urethra	Prostatic urethra	Milky fluid that helps sperm stay active
Bulbourethral glands	Paired	Below the prostate, beside the membranous urethra	Spongy urethra	Clear alkaline mucus that neutralizes urine traces and lubricates

- The glands empty in the same order the sperm travel: seminal vesicles into the ejaculatory duct, prostate into the prostatic urethra, bulbourethral glands into the spongy urethra.

The three regions of the penis

The penis carries the spongy urethra and delivers semen. It has three regions, plus the fold of skin at the tip.

Root
The attached part, fixed to the pelvis in the perineum. It is made of the bulb of the penis, which is the back end of the corpus spongiosum, and the two crura, which are the back ends of the corpora cavernosa and attach to the ischial and pubic rami.

Body (shaft)
The free, movable part.

Glans penis
The enlarged tip, formed by the expanded end of the corpus spongiosum. The external urethral orifice opens on the glans.

Prepuce (foreskin)
The loose fold of skin that covers the glans. It is the part removed in circumcision.

The three erectile columns

Three columns of erectile tissue run the length of the body. Erectile tissue is a sponge of blood spaces inside connective tissue and smooth muscle. When these spaces fill with blood, the penis becomes erect.

Column	Number	Side	What it forms
Corpora cavernosa	Two, side by side	Dorsal	Their back ends form the crura of the root
Corpus spongiosum	One	Ventral	Surrounds the spongy urethra; its back end forms the bulb and its front end forms the glans

- Memory tip: in a cross section of the penis, find the urethra first. The column wrapped around it is the corpus spongiosum. The two larger columns above it are the corpora cavernosa.
- The urethra runs through the single corpus spongiosum on the ventral side, not between the two corpora cavernosa.
- The glans is an extension of the corpus spongiosum, not of the corpora cavernosa.

The regions of the penis and its erectile columns

Structure	What it is
Root	The attached part; the bulb of the penis and the two crura
Body	The free, movable shaft
Glans penis	The enlarged tip; the expanded end of the corpus spongiosum
Prepuce	The foreskin; the loose fold of skin over the glans
Corpora cavernosa	The two dorsal columns of erectile tissue; their back ends form the crura
Corpus spongiosum	The single ventral column; surrounds the spongy urethra and forms the bulb and the glans

Competency 145. Clinical anatomy of male disorders

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

Most male reproductive disorders make sense once you know where the structure sits. The testes form inside the abdomen and move down into the scrotum. The prostate wraps around the urethra. The veins of the testis run up a long cord. Each of these positions explains a disorder.

The descent of the testes and cryptorchidism

Before birth, the testes develop high on the posterior abdominal wall, near the kidneys. That is why each testicular artery branches from the abdominal aorta. Late in fetal development, each testis moves down and passes through the inguinal canal into the scrotum, pulling its duct and vessels behind it. In cryptorchidism, one or both testes stop along this route, in the abdomen or in the inguinal canal. An undescended testis stays at core body temperature, which impairs sperm production and raises the risk of testicular cancer.

The prostate: benign prostatic hyperplasia and prostate cancer

The prostate sits just below the bladder and surrounds the prostatic urethra. In benign prostatic hyperplasia (BPH), the prostate enlarges without cancer, most often in the tissue right around the urethra. The enlarged gland squeezes the prostatic urethra, so the urine stream weakens and the bladder may not empty fully. BPH is common in older men. Prostate cancer most often starts in the outer, posterior part of the gland. Because the back of the prostate lies against the rectum, a clinician can feel it through the rectal wall.

The inguinal canal, the tunica vaginalis, and the pampiniform plexus

The inguinal canal is a weak spot in the abdominal wall, left by the descent of the testis. In an inguinal hernia, a loop of intestine pushes into the canal and can follow the spermatic cord down into the scrotum. In a hydrocele, serous fluid collects between the two layers of the tunica vaginalis. In a varicocele, the veins of the pampiniform plexus swell and twist.

Varicoceles are more common on the left, where the testicular vein drains into the left renal vein.

Male disorders by structure and position

Disorder	Structure affected	Where it is	What changes
Cryptorchidism	Testis	Abdomen or inguinal canal, instead of the scrotum	The testis fails to descend and stays at core body temperature
Benign prostatic hyperplasia	Prostate	Below the bladder, around the prostatic urethra	Non-cancerous enlargement that squeezes the urethra
Prostate cancer	Prostate	Usually the posterior part, against the rectum	Malignant tumor of the gland
Testicular cancer	Testis	In the scrotum	Malignant tumor, usually from the sperm-forming cells of the seminiferous tubules
Hydrocele	Tunica vaginalis	Around the testis	Serous fluid collects between its two layers and the scrotum swells
Varicocele	Pampiniform plexus	Spermatic cord, above the testis	The veins swell and twist, more often on the left
Inguinal hernia	Inguinal canal	Lower anterior abdominal wall	A loop of intestine pushes into the canal along the spermatic cord

What this means for patients

- Sperm production needs a temperature a few degrees below the body core. That is why the testes sit in the scrotum and why the cremaster and dartos muscles adjust their position. In cryptorchidism, an undescended testis stays at body temperature, and if it is not corrected the warmth can damage the sperm-forming cells.
- The duct system shares its final stretch with the urinary tract. Because the urethra passes straight through the prostate, an enlarged prostate squeezes the urethra and makes urination difficult. This is how a reproductive organ produces a urinary symptom. BPH squeezes the urethra, not the bladder.
- Hydrocele is fluid around the testis, between the layers of the tunica vaginalis. Varicocele is swollen veins in the spermatic cord above the testis. Do not swap them.
- The back of the prostate lies against the rectum, which is why a clinician can feel the posterior part of the gland, where prostate cancer usually starts, through the rectal wall.
- The inguinal canal is the passage the testis took on its way down, which is why an inguinal hernia can follow the spermatic cord into the scrotum.

The female reproductive system

The written notes for the female reproductive lecture: the ovary and its follicle stages, the uterine tubes, the uterus, the vagina and vulva, and the mammary glands. Module 5 competencies 146 to 150, with eight worked questions.

Before the competencies: the organs of the system

The female reproductive system makes oocytes, receives sperm, and supports a developing fetus. Six sets of organs do this work, and the five competencies in this chapter walk through them from the ovary outward.

Structure	What it is
Ovaries	The female gonads; they produce oocytes and secrete hormones
Uterine tubes	The fallopian tubes; they carry the oocyte toward the uterus
Uterus	The muscular organ where an embryo implants and a fetus develops
Vagina	The canal from the cervix to the outside; the birth canal
Vulva	The external genitalia
Mammary glands	The milk-producing glands of the breasts

- The order of the competencies follows the path an oocyte takes: ovary (146), uterine tube (147), uterus (148), vagina and vulva (149). The mammary glands (150) are last.
- A gonad is the organ that makes the gametes. In the female that is the ovary, and the gamete is the oocyte.

Competency 146. Ovary structure and follicle stages

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

The ovaries are the paired female gonads. Each is an almond-shaped organ against the lateral wall of the pelvic cavity, one on each side of the uterus, just below the open end of a uterine tube. The ovary produces oocytes and secretes hormones. To know it on a model or a slide you need three things: the ligaments that hold it, its layers from the outside in, and the follicle stages in its cortex.

The three ligaments of the ovary

The broad ligament is a large fold of peritoneum that drapes over the uterus, the uterine tubes, and the ovaries. A short fold of it, the mesovarium, attaches the ovary to the back of the broad ligament. The ovarian ligament is a cord of fibrous tissue that ties the ovary to the side of the uterus, just below where the uterine tube joins it. The suspensory ligament runs from the ovary up to the lateral pelvic wall, and it carries the ovarian artery and ovarian vein to and from the ovary. Like the testicular artery, the ovarian artery branches from the abdominal aorta.

Ligament	What it connects
Broad ligament (mesovarium)	Ovary to the back of the broad ligament, the fold of peritoneum over the uterus and tubes
Ovarian ligament	Ovary to the side of the uterus
Suspensory ligament	Ovary to the lateral pelvic wall; carries the ovarian artery and vein

- Memory tip: the suspensory ligament suspends the ovary from the pelvic wall and carries its vessels. The ovarian ligament ties the ovary to the uterus. These two get swapped a lot, so say the far end of each one out loud: suspensory to the wall, ovarian to the uterus.
- The mesovarium is not a separate ligament. It is the part of the broad ligament that holds the ovary.

The layers of the ovary, from the outside in

Germinal epithelium (ovarian surface epithelium)
A single layer of cuboidal cells covering the ovary. Despite its name, it does not make oocytes.

Tunica albuginea
A capsule of dense connective tissue under the epithelium. It has the same name as the capsule of the testis.

Ovarian cortex
The outer region of the ovary proper. It holds the follicles, so every follicle stage you name on a slide is in the cortex.

Ovarian medulla
The center: loose connective tissue with the blood vessels, lymphatic vessels, and nerves.

The follicle stages

An ovarian follicle is an oocyte plus the cells around it. The cells next to the oocyte are follicle cells, called granulosa cells once they become cuboidal. In later stages, connective tissue cells outside them form a layer called the theca. Follicles grow through a set series of stages, so on a slide you name each one by its size, its layers of cells, and whether it has a fluid space.

A primordial follicle is a primary oocyte with one layer of flat follicle cells. These are the smallest and most numerous follicles, in the outer cortex. In a primary follicle, the follicle cells become cuboidal granulosa cells and a clear layer, the zona pellucida, forms around the oocyte. A secondary follicle has several layers of granulosa cells, a theca around the outside, and small pockets of fluid between the granulosa cells. A mature follicle (vesicular or Graafian follicle) has one large fluid-filled cavity, the antrum. Its oocyte sits on a mound of granulosa cells to one side, ringed by the corona radiata. It is big enough to bulge from the surface of the ovary. Just before ovulation, its oocyte becomes a secondary oocyte.

At ovulation, the mature follicle ruptures at the ovary surface and releases the secondary oocyte with its corona radiata into the pelvic cavity. The granulosa and theca cells left behind become the corpus luteum (yellow body), a large glandular body that secretes hormones. If no pregnancy occurs, the corpus luteum breaks down into the corpus albicans (white body), a pale scar of fibrous tissue. Fluid-filled ovarian cysts often form from a follicle or a corpus luteum that does not go away.

Stage	Oocyte	Cells around it	How to spot it
Primordial follicle	Primary oocyte	One layer of flat follicle cells	Smallest; many, in the outer cortex
Primary follicle	Primary oocyte	Cuboidal granulosa cells; zona pellucida forms	Oocyte ringed by cuboidal cells
Secondary follicle	Primary oocyte	Several layers of granulosa cells and a theca	Small fluid pockets between the granulosa cells
Mature follicle	Primary oocyte; it becomes a secondary oocyte just before ovulation	Granulosa cells, corona radiata, theca	One large antrum; bulges from the ovary
Corpus luteum	None (released at ovulation)	Granulosa and theca cells left behind	Large, pale or yellow glandular body
Corpus albicans	None	Fibrous scar tissue	Pale, white scar in the cortex

- The order is fixed: primordial, primary, secondary, mature follicle, then ovulation, then corpus luteum, then corpus albicans. The corpus luteum forms after ovulation, from what the follicle left behind, so it never comes before the oocyte is released.
- Primordial and primary follicles both hold a primary oocyte. What changes is the follicle cells: flat in the primordial follicle, cuboidal in the primary follicle, with the zona pellucida appearing at the primary stage.
- The mature follicle is the only stage with one big antrum. A secondary follicle has fluid, but only in small pockets.
- Luteum means yellow and albicans means white. The yellow body is the working gland; the white body is the scar it leaves.

Competency 147. Uterine tubes and ovum pathway

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

The uterine tubes (fallopian tubes) are a pair of tubes about 10 cm long. Each runs from an upper corner of the uterus out toward an ovary, inside the upper edge of the broad ligament. This part of the broad ligament is the mesosalpinx. The tube carries the oocyte toward the uterus, and it is where fertilization usually happens.

The outer end of the tube is not attached to the ovary. At ovulation the oocyte is released into the pelvic cavity, and the tube has to catch it. The tube has four named regions, from the ovary toward the uterus.

The regions of the tube, from the ovary to the uterus

The infundibulum is the funnel-shaped outer end, open to the pelvic cavity near the ovary. Its edge is fringed with fimbriae, finger-like projections that drape over the ovary. Around ovulation the fimbriae sweep across the surface of the ovary and draw the oocyte into the funnel. The ampulla is the widest and longest part of the tube. Fertilization usually happens here. The isthmus is the narrow, thick-walled part that joins the uterus. A short intramural part then passes through the uterine wall and opens into the uterine cavity.

Region	What it is
Fimbriae	Finger-like projections on the edge of the infundibulum that sweep the oocyte into the tube
Infundibulum	The funnel-shaped outer end, open to the pelvic cavity near the ovary
Ampulla	The widest and longest part; the usual site of fertilization
Isthmus	The narrow, thick-walled part next to the uterus
Intramural part	The short part that passes through the uterine wall into the uterine cavity

- Memory tip: the tube narrows as it nears the uterus: fimbriae and infundibulum (the open funnel), then the wide ampulla, then the narrow isthmus. An isthmus of land is a narrow neck, and so is this one.
- The ampulla is the wide part and the isthmus is the narrow part next to the uterus. Students flip these two more than anything else in this competency.

What moves the oocyte

The lining of the tube has ciliated cells and secretory cells. The cilia beat toward the uterus, and a layer of smooth muscle in the wall squeezes in waves. Together they move the oocyte, or the early embryo, toward the uterus over a few days. On a drawing, the cilia arrow always points toward the uterus, never toward the ovary.

The route from the ovary to the uterus

1. Mature follicle in the ovarian cortex: the follicle ruptures at ovulation and releases the secondary oocyte into the pelvic cavity.
2. Fimbriae: the fimbriae sweep over the ovary and draw the oocyte in.
3. Infundibulum: the funnel carries the oocyte into the tube.
4. Ampulla: the oocyte moves through the widest part of the tube, where sperm usually meet and fertilize it.
5. Isthmus: the oocyte or early embryo moves through the narrow part of the tube.
6. Intramural part: it passes through the uterine wall.
7. Uterine cavity: it arrives in the cavity of the uterus. An embryo implants in the endometrium, usually in the upper part of the body of the uterus.

An open pathway to the pelvic cavity

The female tract is open to the pelvic cavity. The infundibulum opens near the ovary and does not seal onto it. That opening is how the fimbriae catch an oocyte. It is also how an infection can climb from the vagina through the cervix, uterus, and uterine tubes into the pelvic cavity. This spreading infection is pelvic inflammatory disease (PID).

In an ectopic pregnancy, an embryo implants outside the uterine cavity. Most implant in the uterine tube, usually in the ampulla. This happens when the embryo is held up inside the tube, for example where scarring from earlier PID narrows it. The thin wall of the tube cannot stretch to hold a growing embryo. In rare cases an embryo implants in the pelvic or abdominal cavity, which is possible because the infundibulum is open.

What this means for patients

- Fertilization usually happens in the ampulla, the wide part of the tube, not in the uterus. The embryo then travels the rest of the tube and implants in the endometrium.
- Because the infundibulum is open, an infection that starts in the vagina has a continuous route up through the cervix, uterus, and tubes into the pelvic cavity. That is pelvic inflammatory disease.
- Scarring from PID can narrow the tube and hold up an embryo, which is one reason ectopic pregnancies most often happen in the ampulla.

Competency 148. Uterus regions, ligaments, and wall

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

The uterus is a hollow, thick-walled, pear-shaped organ in the pelvic cavity. It sits behind the urinary bladder and in front of the rectum. In most women it tips forward over the bladder. It receives the embryo, holds the developing fetus, and contracts during childbirth.

Regions

The fundus is the rounded top, above the points where the uterine tubes enter. The body is the main central part. The space inside it is the uterine cavity. Below the body, a short narrowed region called the isthmus of the uterus leads to the cervix, the narrow lower part that projects into the vagina. The inside of the cervix is the cervical canal. It opens into the uterine cavity at the internal os and into the vagina at the external os.

Region	What it is
Fundus	The rounded top, above the entry of the uterine tubes
Body	The main central part; holds the uterine cavity
Cervix	The narrow lower part that projects into the vagina
Uterine cavity	The space inside the body of the uterus
Cervical canal	The channel through the cervix, from the internal os to the external os

- Os means mouth. The internal os is the inner mouth of the cervical canal, opening into the uterine cavity. The external os is the outer mouth, opening into the vagina. Do not flip them.
- The uterus has an isthmus and so does the uterine tube. Both are narrow necks: the isthmus of the uterus sits between the body and the cervix, and the isthmus of the tube sits next to the uterus.

Supporting ligaments

Four paired ligaments hold the uterus, with help from the muscles of the pelvic floor. The broad ligaments are a double fold of peritoneum from the sides of the uterus to the lateral pelvic walls. The part of it beside the uterus is the mesometrium; the same sheet also holds the uterine tubes (mesosalpinx) and the ovaries (mesovarium). The round ligaments are fibrous cords that leave the uterus just below each uterine tube, run forward through the inguinal canal, and

end in the labia majora. They help keep the uterus tipped forward. The cardinal ligaments (transverse cervical ligaments) run from the cervix and upper vagina to the lateral pelvic walls. They are the main support that keeps the uterus from sliding down. The uterosacral ligaments run from the cervix back to the sacrum.

Ligament	Runs from	Runs to
Broad ligaments	Sides of the uterus	Lateral pelvic walls and floor
Round ligaments	Uterus, just below each uterine tube	Through the inguinal canal to the labia majora
Cardinal ligaments	Cervix and upper vagina	Lateral pelvic walls
Uterosacral ligaments	Cervix	Sacrum

- Memory tip: the round ligament takes the same route through the inguinal canal that the spermatic cord takes in a male.
- One broad ligament, three named parts: mesometrium beside the uterus, mesosalpinx holding the tube, mesovarium holding the ovary.
- Two ligaments leave from the cervix. The uterosacral goes back to the sacrum. The cardinal goes sideways to the pelvic wall and is the main thing keeping the uterus from dropping.

The three layers of the wall, from the outside in

The perimetrium is the outer serous layer. It is part of the visceral peritoneum. The myometrium is the thick middle layer of smooth muscle, thickest in the fundus. It contracts during childbirth. The endometrium is the inner mucous lining, where an embryo implants. It has two layers. The stratum functionalis (functional layer), next to the cavity, is shed during menstruation. The stratum basalis (basal layer), next to the myometrium, stays in place and rebuilds the functional layer each cycle.

Layer	What it is
Perimetrium	The outer serous layer; part of the visceral peritoneum
Myometrium	The thick middle layer of smooth muscle that contracts during childbirth
Endometrium	The inner lining; its stratum functionalis sheds in menstruation and its stratum basalis rebuilds it

- Peri means around, myo means muscle, endo means inside. Perimetrium outside, myometrium in the middle, endometrium lining the cavity.
- The layer that contracts during childbirth is the myometrium, the muscle layer, not the endometrium.
- Inside the endometrium, the functional layer is the one next to the cavity and the one that leaves each month. The basal layer stays and regrows it.

What this means for patients

Three common disorders each belong to one part of the uterus.

Disorder	Part of the uterus	What it is
Uterine fibroids	Myometrium	Non-cancerous tumors of the smooth muscle of the myometrium
Endometriosis	Endometrium	Endometrial tissue grows outside the uterus, for example on the ovaries or the pelvic peritoneum
Cervical cancer	Cervix	A malignant tumor of the cervix

- Fibroids are muscle, so they live in the myometrium. Endometriosis is lining tissue in the wrong place, so it is the endometrium gone outside the uterus.

Competency 149. Vagina and external genitalia

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

The vagina

The vagina is a thin-walled fibromuscular tube about 10 cm long. It runs from the cervix down and forward to the outside. It lies behind the bladder and urethra and in front of the rectum. It receives the penis, carries menstrual flow out, and serves as the birth canal. Its lining is stratified squamous epithelium in folds called rugae, with a layer of smooth muscle around it. It opens to the outside at the vaginal orifice.

The cervix pokes into the top of the vagina. The recess that rings the cervix is the fornix. Its posterior part, the posterior fornix, is the deepest. The hymen is a thin fold of mucous membrane at the vaginal orifice that partly covers it. Its shape and size vary a great deal from person to person.

- Front to back in the pelvis: bladder and urethra, then the vagina, then the rectum. The vagina is the middle of the three.
- The fornix is at the top of the vagina, around the cervix. The hymen is at the bottom, at the opening. They are the two ends of the same tube.

The vulva

The external genitalia together are the vulva. The mons pubis is a rounded pad of fat over the pubic symphysis. The labia majora are two folds of skin and fat that run back from the mons pubis. They are covered with hair after puberty. The labia

minora are two thin, hairless folds inside the labia majora. At the front, they meet over the clitoris and form its prepuce.

The clitoris is a small erectile organ at the front of the vulva, where the labia minora meet. Like the penis, it is built of erectile tissue, and only its tip, the glans of the clitoris, shows on the surface. The vestibule is the space between the labia minora. It holds the two main openings: the external urethral orifice toward the front and the vaginal orifice behind it.

The greater vestibular glands (Bartholin glands) are a pair of pea-sized glands, one on each side of the vaginal orifice, slightly toward the back. Their ducts open into the vestibule, and their mucus lubricates the vestibule. The perineum is the diamond-shaped region between the pubic symphysis and the coccyx. Its front half, the urogenital triangle, holds the external genitalia. Its back half, the anal triangle, holds the anus.

Structure	Where it is	What it is
Vagina	From the cervix to the vestibule	Fibromuscular canal; the birth canal
Fornix	Top of the vagina, around the cervix	Recess that rings the cervix
Hymen	At the vaginal orifice	Thin fold of mucous membrane that partly covers the opening
Mons pubis	Over the pubic symphysis	Rounded pad of fat
Labia majora	Outer folds, back from the mons pubis	Two folds of skin and fat, hair covered after puberty
Labia minora	Inside the labia majora	Two thin, hairless folds that enclose the vestibule
Clitoris	Front of the vulva, where the labia minora meet	Small erectile organ
Vestibule	Between the labia minora	Space holding the urethral and vaginal openings
Greater vestibular glands	On each side of the vaginal orifice	Glands whose mucus lubricates the vestibule
Perineum	Between the pubic symphysis and the coccyx	Diamond-shaped region with the urogenital and anal triangles

- Memory tip: front to back along the midline: mons pubis, clitoris, external urethral orifice, vaginal orifice, then the anus behind the vestibule. A urethral opening drawn behind the vaginal opening is the usual slip.
- The vestibule is the area between the labia minora, not between the labia majora. Majora means larger; minora means smaller and sits inside.

Female structures and their male matches

Several female structures develop from the same embryonic tissue as a male structure, so they are homologous: the same starting material built into a different shape.

Female	Male
Ovaries	Testes
Labia majora	Scrotum
Clitoris	Penis
Greater vestibular glands	Bulbourethral glands

- The three vulvar homologs to know: labia majora match the scrotum, the clitoris matches the penis, and the greater vestibular glands match the bulbourethral glands. Both gland pairs are pea-sized and both secrete lubricating mucus.

Competency 150. Mammary gland structure

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

The mammary glands are modified sweat glands in the breasts. Each breast lies in the subcutaneous tissue in front of the pectoralis major muscle. Much of the breast is adipose tissue, and the amount of fat mostly sets the size of the breast.

Nested units: lobes, lobules, alveoli

The gland is built in nested units. It has 15 to 20 lobes arranged in a ring around the nipple, separated by fat. Each lobe divides into smaller lobules. Each lobule holds clusters of alveoli, small sacs lined with cells that secrete milk. Myoepithelial cells around the alveoli contract to push the milk into the ducts.

Ducts, nipple, and areola

Small ducts from the alveoli join into one lactiferous duct for each lobe. Just before the nipple, each lactiferous duct widens into a lactiferous sinus, where milk collects. The ducts open on the nipple, the raised projection near the center of the breast. The areola is the ring of pigmented skin around the nipple. Its small bumps are areolar glands, a type of sebaceous gland.

Suspensory ligaments

The suspensory ligaments (Cooper ligaments) are bands of connective tissue that run from the skin through the breast, between the lobes, to the fascia over the pectoralis major. They support the breast.

Structure	What it is
Lobes	The 15 to 20 compartments of the gland, each divided into lobules
Lobules	The subdivisions of a lobe, each holding clusters of alveoli
Alveoli	The small sacs of gland cells that secrete milk
Lactiferous ducts	One duct per lobe; carries milk toward the nipple
Lactiferous sinus	The widened end of a lactiferous duct where milk collects
Nipple	The projection where the lactiferous ducts open
Areola	The ring of pigmented skin around the nipple
Suspensory ligaments	Bands of connective tissue from the skin to the fascia over the pectoralis major that support the breast

The route of milk, from where it is made to the outside

1. Alveoli: gland cells inside each lobule secrete milk.
2. Lobule ducts: small ducts carry milk out of each lobule.
3. Lactiferous duct: one duct drains each lobe toward the nipple.
4. Lactiferous sinus: the duct widens and milk collects.
5. Nipple: milk leaves through openings on the nipple.

- Memory tip: the gland nests one way: alveoli inside lobules, lobules inside lobes. Each lobe drains by its own lactiferous duct to the nipple.
- The alveoli are the smallest sacs and the lobe is the biggest unit. Small sits inside big.
- The lactiferous ducts open at the nipple, not on the areola. The areola has its own small sebaceous glands, the areolar glands, which are not milk ducts.
- Breast alveoli and lung alveoli share a name because both are small sacs. The ones here make milk.

What this means for patients

- Most breast cancer starts in the ducts.
- A tumor that pulls on a suspensory ligament can dimple the skin over it, because the ligament runs all the way from the skin down to the fascia over the pectoralis major.

The brain

The written notes for the brain lecture: the four divisions, brain development, the cerebral surface and lobes, functional cortical areas, white matter tracts, basal nuclei, diencephalon, cerebellum, limbic system, and the arterial supply through the Circle of Willis. Module 5 competencies 151 to 160, with ten worked questions.

Competency 151. Brain gross organization

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

The brain sits in the cranial cavity. At the foramen magnum it continues downward as the spinal cord. Anatomists divide the brain into four major divisions, and you can find all four on a midsagittal section of the brain model (a cut straight down the midline, viewed from the side).

The four divisions

Division	Where it is	Main parts	Main job
Cerebrum	Upper cranial cavity; the largest division	Two cerebral hemispheres, each with cortex, white matter, and basal nuclei	Conscious thought, sensation, and voluntary movement
Diencephalon	Deep at the core, between the hemispheres and the brainstem	Thalamus, hypothalamus, epithalamus	Relays sensory signals to the cortex and controls homeostasis
Brainstem	Below the diencephalon, in front of the cerebellum; continuous with the spinal cord	Midbrain, pons, medulla oblongata	Connects the brain to the spinal cord
Cerebellum	Behind the brainstem, below the occipital lobes, in the posterior cranial fossa	Two cerebellar hemispheres joined by the vermis	Coordinates movement

- The cerebrum is the largest division. It forms the two cerebral hemispheres that fill the upper cranial cavity.
- The diencephalon sits at the core of the brain, between the hemispheres and above the brainstem. The hemispheres cover it, so you see it best on a midsagittal cut.
- The brainstem is the stalk below the diencephalon. It is made of the midbrain, pons, and medulla oblongata, and it continues as the spinal cord.
- The cerebellum sits behind the brainstem and below the occipital lobes, in the posterior cranial fossa. It is the second largest division, bigger than the brainstem and the diencephalon.

Gray matter and white matter

The brain is built from two kinds of nervous tissue. Gray matter holds neuron cell bodies, dendrites, unmyelinated axons, and synapses. It looks gray to tan. White matter is made of myelinated axons running in bundles called tracts. Myelin is fatty, so white matter looks pale.

Cerebral cortex

In the cerebrum, gray matter forms a thin outer layer called the cerebral cortex. It is only about 2 to 4 mm thick, and it follows every gyrus and sulcus.

Cerebral white matter

Beneath the cortex lies the deep core of tracts. Islands of gray matter also sit deep inside the white matter. These are the basal nuclei and the thalamus.

The same plan in the cerebellum

Gray cortex on the outside and white matter inside.

The opposite plan in the brainstem and spinal cord

Most of their white matter is on the outside, with gray matter inside.

- On a specimen, look at a coronal or horizontal slice. The cortex is the thin, darker rim that traces the outline of each fold. The pale tissue deep to it is white matter. Darker patches buried inside the white matter are the deep nuclei.
- Gray matter is the thin outer layer of the cerebrum and white matter is the core. Students reverse these on drawings often, so check yours.
- Memory tip: Cortex is Latin for bark. Like bark on a tree, the cortex is a thin outer layer, and the white matter is the wood inside.

Competency 152. Neural tube brain development

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

The whole central nervous system develops from the neural tube, which forms from ectoderm. In the early embryo a strip of ectoderm called the neural plate folds into a groove, and the edges of the groove fuse into a tube. The front (cranial) end of the tube becomes the brain. The rest of the tube becomes the spinal cord.

The tube stays hollow. Its central cavity becomes the ventricles of the brain and the central canal of the spinal cord. So as you follow each part of the tube into the adult brain, you also follow its cavity into a ventricle.

From tube to adult brain, in order

1. Neural tube: forms from ectoderm. The cranial end becomes the brain and the rest becomes the spinal cord.
2. Three primary vesicles, by about the fourth week: the cranial end of the tube swells into the prosencephalon (forebrain), the mesencephalon (midbrain), and the rhombencephalon (hindbrain).
3. Five secondary vesicles, by about the fifth week: the prosencephalon divides into the telencephalon and the diencephalon. The mesencephalon does not divide. The rhombencephalon divides into the metencephalon and the myelencephalon.
4. Adult brain: each secondary vesicle becomes a set of adult structures and keeps part of the tube cavity as a ventricle.

Secondary vesicle	Primary vesicle it came from	Adult structures	Cavity in the adult
Telencephalon	Prosencephalon	Cerebrum (cerebral hemispheres, with cortex, white matter, and basal nuclei)	Lateral ventricles
Diencephalon	Prosencephalon	Thalamus, hypothalamus, epithalamus	Third ventricle
Mesencephalon	Mesencephalon (unchanged)	Midbrain	Cerebral aqueduct
Metencephalon	Rhombencephalon	Pons and cerebellum	Upper part of the fourth ventricle
Myelencephalon	Rhombencephalon	Medulla oblongata	Lower part of the fourth ventricle, which continues into the central canal of the spinal cord

- There are five secondary vesicles because the mesencephalon carries through without splitting. If your chart has only four, you dropped it.
- The cerebellum grows from the metencephalon along with the pons, not from the myelencephalon.
- The telencephalon grows far more than the other vesicles. The cerebral hemispheres expand backward over the diencephalon and midbrain. That growth is why the diencephalon ends up hidden deep in the adult brain.
- Memory tip: Read the five secondary vesicles in order from the front of the tube to the back and you name the adult brain from top to bottom: cerebrum, diencephalon, midbrain, pons and cerebellum, medulla. The prefixes help too: tel means end, di means between, mes means middle, met means after, and myel means marrow, for the part next to the spinal cord.

Competency 153. Cerebral surface and lobes

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

The cerebrum is made of two cerebral hemispheres, right and left. A deep midline cleft called the longitudinal fissure separates them. At the bottom of that fissure, the two hemispheres are joined by the corpus callosum. A second deep cleft, the transverse fissure, separates the back of the cerebrum from the cerebellum below it.

Gyri (singular, gyrus)
The raised ridges of the folded surface.

Sulci (singular, sulcus)
The grooves between the gyri.

Fissures
The deepest grooves. Folding packs much more cortex into the skull than a smooth surface could hold.

The named grooves

A few sulci are named because they mark the lobe boundaries. The central sulcus runs from the top of the hemisphere down the side toward the lateral sulcus. The precentral gyrus lies just in front of it, bounded in front by the precentral sulcus. The postcentral gyrus lies just behind it, bounded behind by the postcentral sulcus. The lateral sulcus (Sylvian fissure) is the deep groove on the side of each hemisphere. The parieto-occipital sulcus is easiest to see on the medial surface, near the back of the hemisphere.

Groove	Where to find it	What it separates
Longitudinal fissure	Midline, between the hemispheres	The right and left cerebral hemispheres
Transverse fissure	Back of the brain, between the occipital lobes and the cerebellum	The cerebrum from the cerebellum
Central sulcus	Top and side of each hemisphere, running down toward the lateral sulcus	The frontal lobe from the parietal lobe; the precentral gyrus in front from the postcentral gyrus behind
Precentral sulcus	Just in front of the precentral gyrus	The precentral gyrus from the rest of the frontal lobe
Postcentral sulcus	Just behind the postcentral gyrus	The postcentral gyrus from the rest of the parietal lobe
Lateral sulcus (Sylvian fissure)	Deep groove on the side of each hemisphere	The temporal lobe below from the frontal and parietal lobes above
Parieto-occipital sulcus	Medial surface, near the back of the hemisphere	The parietal lobe from the occipital lobe

The five lobes

Each hemisphere has five lobes, so every lobe is paired, with one in each hemisphere. Four lobes are named for the cranial bone that covers them: the frontal, parietal, temporal, and occipital bones. The fifth, the insula, is hidden deep in the lateral sulcus, so no outside view shows it. To see it, gently pull the temporal lobe away from the frontal and parietal lobes and look into the lateral sulcus.

Lobe	Location	Boundaries
Frontal lobe	Front of each hemisphere, under the frontal bone	Central sulcus behind; lateral sulcus below
Parietal lobe	Top and upper side, under the parietal bone	Central sulcus in front; parieto-occipital sulcus behind; lateral sulcus below
Temporal lobe	Lower side, under the temporal bone	Lateral sulcus above
Occipital lobe	Back of each hemisphere, under the occipital bone	Parieto-occipital sulcus in front; transverse fissure below, separating it from the cerebellum
Insula	Deep inside the lateral sulcus	Covered by the frontal, parietal, and temporal lobes

- On a specimen, find the lateral sulcus first, because it is the deepest groove on the side. Then look for the central sulcus. It usually runs without a break from the top of the hemisphere almost down to the lateral sulcus, with a long, unbroken gyrus on each side of it. Those two gyri are the precentral and postcentral gyri.
- Seen from above, the longitudinal fissure runs down the middle and the central sulcus crosses each hemisphere. You can see the frontal, parietal, and occipital lobes from the top; the temporal lobe is on the side and the insula is hidden.
- Memory tip: Pre means before and post means after. The precentral gyrus comes before the central sulcus (in front of it) and the postcentral gyrus comes after it (behind it). These two get swapped a lot.

Competency 154. Functional cortical areas

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

The cortex is mapped into areas that each have a fixed location. There are three kinds. Motor areas control voluntary movement and sit in the frontal lobe. Sensory areas are the first cortex to receive a given kind of sensation. A primary sensory area receives that input, and an association area beside it interprets it. Association areas make up the rest of the cortex. They interpret input, connect it with memory, and plan responses.

Motor areas, frontal lobe

- The primary motor cortex is the precentral gyrus. Its neurons send commands for voluntary movement to skeletal muscles on the opposite side of the body.
- The body is mapped along the gyrus upside down. The leg and foot are on the medial surface, inside the longitudinal fissure. The hand is higher on the lateral surface, and the face is lowest, near the lateral sulcus. Body parts with fine control, such as the hand, lips, and tongue, get the largest share of cortex.
- The premotor cortex lies just in front of the precentral gyrus and plans learned, patterned movements.
- Broca's area sits in the inferior frontal gyrus, usually in the left hemisphere. It plans the movements of speech, so it counts as a motor area.

Sensory areas

- The primary somatosensory cortex (primary sensory cortex) is the postcentral gyrus of the parietal lobe. It receives touch, pressure, pain, temperature, and body position from the opposite side of the body. Its body map is laid out in the same order as the motor map next to it. The somatosensory association area lies just behind it.
- The primary visual cortex is in the occipital lobe, along the calcarine sulcus on the medial surface.
- The primary auditory cortex is on the superior temporal gyrus, along the lower edge of the lateral sulcus.
- The primary gustatory cortex (taste) is in the insula and the lower end of the postcentral gyrus that covers it.
- The primary olfactory cortex (smell) is on the medial surface of the temporal lobe, near the uncus. Smell reaches it from the olfactory bulb and olfactory tract, which lie on the inferior surface of the frontal lobe.

Association areas

- Wernicke's area sits in the posterior part of the superior temporal gyrus, where it meets the parietal lobe, usually in the left hemisphere. It is needed to understand spoken and written language. A tract called the arcuate fasciculus connects it to Broca's area.
- The prefrontal cortex covers the front of the frontal lobe. It is a large association area that handles planning, judgment, and personality.

Area	Kind	Lobe	Gyrus or landmark	Main job
Primary motor cortex	Motor	Frontal	Precentral gyrus	Sends commands for voluntary movement
Premotor cortex	Motor	Frontal	Just in front of the precentral gyrus	Plans learned, patterned movements
Frontal eye field	Motor	Frontal	Middle frontal gyrus	Controls voluntary eye movement
Broca's area	Motor	Frontal, usually left	Inferior frontal gyrus	Plans the movements of speech
Primary somatosensory cortex	Primary sensory	Parietal	Postcentral gyrus	Receives touch, pressure, pain, temperature, and body position
Somatosensory association area	Association	Parietal	Just behind the postcentral gyrus	Interprets texture, shape, and position
Primary visual cortex	Primary sensory	Occipital	Calcarine sulcus, medial surface	Receives visual input
Visual association area	Association	Occipital	Surrounding the primary visual cortex	Recognizes objects and faces
Primary auditory cortex	Primary sensory	Temporal	Superior temporal gyrus, at the lateral sulcus	Receives sound input
Auditory association area	Association	Temporal	Next to the primary auditory cortex	Interprets speech and music
Primary gustatory cortex	Primary sensory	Insula and parietal	Insula and the lower end of the postcentral gyrus	Receives taste input
Primary olfactory cortex	Primary sensory	Temporal	Medial temporal lobe, near the uncus	Receives smell input
Wernicke's area	Association	Temporal, usually left	Posterior superior temporal gyrus, next to the parietal lobe	Understanding spoken and written language
Prefrontal cortex	Association	Frontal	Front of the frontal lobe	Planning, judgment, and personality

- The thalamus relays most sensory input up to these primary areas. It is a relay station deep in the diencephalon, below the cortex, and competency 157 covers it.
- Broca's area is in the frontal lobe and Wernicke's area is back where the temporal and parietal lobes meet. Speaking is in front, understanding is behind.
- Memory tip: Front moves, back feels. The precentral gyrus, in front of the central sulcus, moves the body. The postcentral gyrus, behind it, feels the body.

Competency 155. Cerebral white matter tracts

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

The cerebral white matter is made of myelinated axons grouped into tracts. The tracts fall into three classes, sorted by what they connect.

Association fibers

Stay within one hemisphere. Short ones arc from one gyrus to the next. Long ones connect one lobe to another. The arcuate fasciculus, which links Broca's area and Wernicke's area, is an example. Because it stays in one hemisphere, it is an association tract, not a commissural one.

Commissural fibers
Cross the midline and connect matching areas of the right and left hemispheres. The largest is the corpus callosum, a thick, arched band of white matter at the bottom of the longitudinal fissure. On a midsagittal section it is the C-shaped white band above the diencephalon, and it forms the roof of the lateral ventricles. The anterior commissure and posterior commissure are smaller commissural bundles.

Projection fibers
Connect the cortex with the lower brain and spinal cord. Some carry signals down from the cortex, and some carry signals up to it. Just below the cortex they spread out in a fan called the corona radiata. Deeper down they gather into one compact band, the internal capsule. Below that, the descending fibers continue into the midbrain as the crus cerebri.

- Hanging below the corpus callosum is the septum pellucidum, a thin sheet that separates the two lateral ventricles. The fornix runs along its lower edge.
- The internal capsule passes between the thalamus and caudate nucleus on its medial side and the lentiform nucleus (putamen and globus pallidus) on its lateral side.

Fiber class	What it connects	Examples	Where to see it
Association fibers	Gyri and lobes within the same hemisphere	Arcuate fasciculus (Broca area to Wernicke area)	Inside one hemisphere; they never cross the midline
Commissural fibers	Matching areas of the right and left hemispheres	Corpus callosum (largest); anterior and posterior commissures	Corpus callosum on a midsagittal section, the arched band above the diencephalon
Projection fibers	The cortex with the diencephalon, brainstem, and spinal cord, in both directions	Corona radiata; internal capsule; crus cerebri below it	Internal capsule on a coronal or horizontal section, between the thalamus and caudate nucleus and the lentiform nucleus

- On a specimen, find the corpus callosum on the midsagittal cut first. To find the internal capsule, use a horizontal or coronal section. On a horizontal section it is a white band bent into a V that points toward the midline, with the caudate nucleus and thalamus on its inner side and the lentiform nucleus on its outer side.
- Direction is the quickest check. Association fibers stay on one side, commissural fibers cross side to side between the hemispheres, and projection fibers run up and down.
- Memory tip: Commissure means a joining, so commissural fibers join the two hemispheres. Projection fibers project down to the brainstem and spinal cord or up from them. Association fibers stay in one hemisphere.

What this means for patients

- The projection fibers of the internal capsule carry nearly all the traffic between the cortex and the body through one narrow band. A small stroke there can cut a great deal of wiring at once, which is why such a small lesion can weaken the entire opposite side of the body.

Competency 156. Basal ganglia

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

The basal nuclei, also called the basal ganglia, are paired masses of gray matter buried deep in the cerebral white matter of each hemisphere. They lie beside the lateral ventricle and the thalamus. Clusters of cell bodies in the central nervous system are usually called nuclei, so many texts use basal nuclei. Basal ganglia is the older name and is still widely used. The basal nuclei work with the motor cortex. They share three roles: they help start and regulate wanted movements, they hold back unwanted movements, and they set muscle tone.

The three core nuclei, in the cerebrum

Caudate nucleus
Long and curved, with a head, a body, and a tail. Its large head bulges into the lateral wall of the front part (anterior horn) of the lateral ventricle. The body and tail follow the curve of the ventricle backward and then down into the temporal lobe.

Putamen
The outer, lateral part of the lentiform nucleus.

Globus pallidus
The inner, medial part of the lentiform nucleus, between the putamen and the internal capsule. It looks paler.

Lentiform nucleus
The putamen and globus pallidus together, a lens-shaped mass lateral to the internal capsule.

Striatum
The caudate nucleus and putamen together, named because bands of gray matter cross the internal capsule between them and give the area a striped look.

Two nuclei grouped with them by function

Two more nuclei sit outside the cerebrum but work in the same movement circuits, so they are grouped with the basal nuclei by function. The substantia nigra is a band of dark, pigmented neurons in the midbrain, just behind the crus cerebri. Loss of these neurons is the change seen in Parkinson disease. The subthalamic nucleus is a small nucleus in the diencephalon, below the thalamus and just above the substantia nigra.

Structure	Brain region	Where it is	How to find it on a section
Caudate nucleus	Cerebrum	Curves along the lateral ventricle; head in the wall of the anterior horn	Gray mass bulging into the lateral ventricle, medial to the internal capsule
Putamen	Cerebrum	Lateral part of the lentiform nucleus	The outer, darker part of the lens-shaped mass lateral to the internal capsule
Globus pallidus	Cerebrum	Medial part of the lentiform nucleus	The inner, paler part of that mass, between the putamen and the internal capsule
Substantia nigra	Midbrain	Just behind the crus cerebri	Dark band across the midbrain on a cross section
Subthalamic nucleus	Diencephalon	Below the thalamus, above the substantia nigra	Small nucleus under the thalamus on a coronal section

- Sorting by region: the caudate nucleus, putamen, and globus pallidus are in the cerebrum; the subthalamic nucleus is in the diencephalon; the substantia nigra is in the midbrain. The last two work with the basal nuclei but do not sit in the cerebrum.
- The globus pallidus is medial, closer to the midline, and the putamen is lateral. Check your section.
- Memory tip: Pallidus means pale. The globus pallidus is the paler, inner part of the lentiform nucleus, and the putamen wraps around it on the outside.

What this means for patients

- The substantia nigra is darkly pigmented, and the loss of its neurons is the change seen in Parkinson disease. Because it works in the same movement circuits as the basal nuclei, losing it disturbs the starting, holding back, and tone-setting of movement that those circuits handle.

Competency 157. Diencephalon

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

The diencephalon lies between the cerebrum and the midbrain. The cerebral hemispheres cover it, so you see it best on a midsagittal section. It has three parts, the thalamus, the hypothalamus, and the epithalamus, and together they surround the third ventricle.

Thalamus

- A pair of egg-shaped masses of gray matter, one on each side of the third ventricle. It makes up about 80 percent of the diencephalon.
- It is made of many nuclei, and it relays most sensory input on its way to the cerebral cortex.
- The massa intermedia (interthalamic adhesion) is a small bridge of gray matter that crosses the third ventricle and joins the two thalami. On a midsagittal section it shows up as a small cut oval in the middle of the thalamus. Some people do not have one. It is part of the thalamus, not the hypothalamus.

Hypothalamus

- Lies below the thalamus. A shallow groove on the wall of the third ventricle, the hypothalamic sulcus, marks the border between them. The hypothalamus forms the floor and the lower walls of the third ventricle.
- It is the main control center for homeostasis. It controls the pituitary gland and the autonomic nervous system and helps regulate body temperature, hunger, thirst, and sleep and wake cycles.
- On the inferior surface of the brain, three landmarks belong to this area. The optic chiasm sits at its front, where the optic nerves meet and some of their fibers cross before continuing as the optic tracts. The infundibulum is the stalk behind the chiasm that connects the hypothalamus to the pituitary gland. The mammillary bodies are two small round bumps behind the infundibulum.

Epithalamus

- The small, posterior, upper part of the diencephalon. It forms part of the roof of the third ventricle.
- Its main structure is the pineal gland, a small pine-cone-shaped endocrine gland that projects backward from the roof of the third ventricle, above the midbrain. The pineal gland secretes melatonin.
- The epithalamus also holds the habenular nuclei.

Third ventricle

The third ventricle is the narrow, midline cavity enclosed by these parts. The thalami form its upper side walls, the hypothalamus forms its floor and lower walls, and the epithalamus forms the back of its roof. In front, it connects to each lateral ventricle through an interventricular foramen. Behind, it drains into the cerebral aqueduct of the midbrain.

Part	Position	Features you identify	Relation to the third ventricle
Thalamus	Upper and largest part; one on each side	Paired thalami; massa intermedia joining them	Forms the upper side walls
Hypothalamus	Below the thalamus	Optic chiasm in front, infundibulum, mammillary bodies	Forms the floor and lower side walls
Epithalamus	Back and top	Pineal gland; habenular nuclei	Forms the back of the roof
Third ventricle	Midline, between the two halves of the diencephalon	Narrow slit crossed by the massa intermedia	The cavity itself; joins the lateral ventricles in front and the cerebral aqueduct behind

- On a midsagittal view, the corpus callosum arches above the diencephalon, the optic chiasm and infundibulum hang below the hypothalamus at the front, the pineal gland projects back from the epithalamus, and the midbrain sits below.
- Memory tip: Hypo means below, so the hypothalamus sits below the thalamus. Epi means upon, so the epithalamus sits on top at the back.

Competency 158. Cerebellum

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

The cerebellum is the second largest part of the brain. It sits in the posterior cranial fossa, below the occipital lobes and behind the pons and medulla. The transverse fissure separates it from the cerebrum above. The fourth ventricle lies between the cerebellum and the brainstem. The cerebellum coordinates movement, posture, and balance, adjusting the timing and accuracy of movements without our awareness.

The outside

Cerebellar hemispheres
The two side halves, each with gray cortex over white matter.

Vermis
The narrow midline strip that joins the two hemispheres.

Folia
Thin, parallel, leaf-like folds of the surface, much finer than the gyri of the cerebrum. They are gray matter.

Lobes
A deep groove called the primary fissure divides the cerebellum into an anterior lobe and a posterior lobe, and a small flocculonodular lobe sits on its underside.

The inside

Cerebellar cortex
The outer layer of gray matter. The cerebellum follows the same plan as the cerebrum: gray outside, white inside.

Arbor vitae
The white matter under the cortex branches out into every folium. On a midsagittal cut through the vermis it looks like a branching tree, so it is called the arbor vitae, Latin for tree of life.

Deep cerebellar nuclei
Four pairs of gray matter nuclei buried in the white matter. From lateral to medial they are the dentate nucleus (the largest), the emboliform nucleus, the globose nucleus, and the fastigial nucleus. Most signals leaving the cerebellum start in these nuclei. On a shaded drawing they are gray, even though they sit inside the white arbor vitae.

The three pairs of cerebellar peduncles

Three pairs of cerebellar peduncles, thick stalks of white matter, attach the cerebellum to the brainstem. Every fiber entering or leaving the cerebellum passes through one of them. Each peduncle joins a different part of the brainstem.

Peduncle	Brainstem connection	Main traffic	Mostly in or out
Superior cerebellar peduncle	Midbrain	Mostly output, from the deep cerebellar nuclei toward the midbrain and thalamus	Out of the cerebellum
Middle cerebellar peduncle	Pons	Input from the pontine nuclei, which relay signals from the cerebral cortex; the largest peduncle	Into the cerebellum
Inferior cerebellar peduncle	Medulla oblongata	Input from the spinal cord about body position, from the vestibular nuclei about balance, and from the olive of the medulla	Into the cerebellum

- On a midsagittal specimen, find the vermis cut down the middle, the arbor vitae branching through it, and the folia along its edge. The deep nuclei and the peduncles are easiest to see on models or on sections that cut through the hemispheres.
- The superior peduncle mostly carries signals out. The middle and inferior peduncles carry signals in.
- Memory tip: The peduncles attach in order from top to bottom: superior to the midbrain, middle to the pons, inferior to the medulla.

Competency 159. Limbic system

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

The limbic system is a group of structures on the medial side of each hemisphere. Limbic comes from limbus, meaning border or ring. These structures form a ring around the diencephalon and the top of the brainstem. They work together in two jobs, emotion and memory.

The ring, structure by structure

Cingulate gyrus
The gyrus that arches just above the corpus callosum on the medial surface. Below it, the sulcus of the corpus callosum separates it from the corpus callosum. Above it, the cingulate sulcus separates it from the rest of the frontal and parietal lobes. It helps connect emotion with behavior. On a midsagittal section it is the first gyrus above the corpus callosum, so it is the part of the ring that sits above the diencephalon.

Parahippocampal gyrus
The cingulate gyrus curves around the back of the corpus callosum and continues down into the temporal lobe as the parahippocampal gyrus, on the inferior and medial surface of the temporal lobe. The front end of the parahippocampal gyrus hooks back on itself to form the uncus. Together these gyri make the C-shaped ring that gives the system its name.

Hippocampus
A curved ridge of cortex rolled into the floor of the inferior horn of the lateral ventricle (also called the temporal horn), deep to the parahippocampal gyrus in the medial temporal lobe. Its name means seahorse, which describes its shape. It is needed to form new long-term memories.

Fornix
The output of the hippocampus travels in the fornix, an arching band of white matter that runs forward under the corpus callosum, along the lower edge of the septum pellucidum, and ends at the mammillary bodies of the hypothalamus.

Amygdala
An almond-shaped nucleus deep in the front of the medial temporal lobe, just in front of the hippocampus and deep to the uncus. It is involved in fear and other emotional responses, and it links emotion to memory.

Structure	Where it is	Landmark to find it	Main role
Cingulate gyrus	Medial surface of the hemisphere, arching over the corpus callosum	First gyrus above the corpus callosum on a midsagittal section, between the sulcus of the corpus callosum and the cingulate sulcus	Connects emotion with behavior
Hippocampus	Medial temporal lobe, in the floor of the inferior horn of the lateral ventricle	Deep to the parahippocampal gyrus; its output leaves in the fornix	Forms new long-term memories
Amygdala	Front of the medial temporal lobe	Deep to the uncus, just in front of the hippocampus	Fear and emotional responses; links emotion to memory
Parahippocampal gyrus	Inferior and medial surface of the temporal lobe	Continues from the cingulate gyrus; its front end forms the uncus	Completes the limbic ring
Fornix	Arch of white matter under the corpus callosum	Lower edge of the septum pellucidum on a midsagittal section	Carries hippocampal output to the mammillary bodies

- Picture the ring on a midsagittal view: the cingulate gyrus arches over the corpus callosum above the diencephalon, and the hippocampus and amygdala curve down and forward below the diencephalon into the temporal lobe, with the amygdala at the front end.
- The hippocampus forms new memories and the amygdala handles fear. Students swap these roles often, so check yours.
- Memory tip: Amygdala means almond and hippocampus means seahorse. The names describe their shapes, and both sit in the medial temporal lobe, with the amygdala in front.

Competency 160. Cerebral arterial supply

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

The brain is supplied by two pairs of arteries. The two internal carotid arteries form the anterior circulation, which feeds most of the cerebrum. The two vertebral arteries form the posterior circulation, which feeds the brainstem, the cerebellum, and the occipital lobe. The two systems join in a ring of arteries on the inferior surface of the brain, the cerebral arterial circle, also called the Circle of Willis. The ring surrounds the optic chiasm, the infundibulum, and the mammillary bodies.

Anterior circulation

- Each internal carotid artery rises through the neck, enters the skull through the carotid canal of the temporal bone, and reaches the base of the brain beside the optic chiasm. There it divides into two branches.
- The anterior cerebral artery runs forward and medially into the longitudinal fissure and arches over the corpus callosum. It supplies the medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and somatosensory cortex.
- The anterior communicating artery is a short bridge between the right and left anterior cerebral arteries, in front of the optic chiasm.
- The middle cerebral artery is the largest branch of the internal carotid. It runs laterally into the lateral sulcus and supplies most of the lateral surface of the hemisphere. That territory includes the face and upper limb areas of the motor and somatosensory cortex, the auditory cortex, the insula, and Broca's and Wernicke's areas.

Posterior circulation

- The vertebral arteries branch from the subclavian arteries and rise through the transverse foramina of the cervical vertebrae. They enter the skull through the foramen magnum and run up the front of the medulla.
- At the junction of the medulla and pons, the two vertebral arteries unite to form the basilar artery. The basilar artery runs up the front of the pons in the midline, giving off small pontine arteries to the pons.
- At the top of the pons the basilar artery divides into the two posterior cerebral arteries. Each posterior cerebral artery curves around the midbrain to supply the occipital lobe, including the visual cortex, and the inferior surface of the temporal lobe.
- A posterior communicating artery on each side joins the internal carotid artery to the posterior cerebral artery. These two vessels link the anterior and posterior circulations.

Cerebellar and spinal branches

- Three pairs of cerebellar arteries come from the posterior circulation. The posterior inferior cerebellar artery (PICA) is the largest branch of the vertebral artery and supplies the underside of the cerebellum and the lateral medulla. The anterior inferior cerebellar artery (AICA) branches from the lower basilar artery and supplies the front part of the underside of the cerebellum. The superior cerebellar artery branches from the top of the basilar artery, just below the posterior cerebral artery, and supplies the upper surface of the cerebellum.
- The vertebral arteries also give rise to the spinal arteries. A small branch from each vertebral artery joins its partner in the midline to form the single anterior spinal artery, which runs down the front of the spinal cord. The paired posterior spinal arteries arise from the vertebral arteries or from the posterior inferior cerebellar arteries and run down the back of the cord.

Tracing the ring

The ring itself is made of the anterior communicating artery, the first part of each anterior cerebral artery, the end of each internal carotid artery, the two posterior communicating arteries, and the first part of each posterior cerebral artery. The middle cerebral arteries branch off the ring and are not part of it. Trace the route from a vertebral artery into the ring and around to the front:

1. Vertebral artery: enters the skull through the foramen magnum and runs up the front of the medulla; gives off the posterior inferior cerebellar artery.
2. Basilar artery: formed where the two vertebral arteries unite; runs up the front of the pons, giving off the anterior inferior cerebellar arteries, pontine arteries, and superior cerebellar arteries.
3. Posterior cerebral artery: one of the two end branches of the basilar artery at the top of the pons.
4. Posterior communicating artery: runs forward from the posterior cerebral artery to the internal carotid artery.
5. Internal carotid artery: reaches the base of the brain beside the optic chiasm; gives off the middle cerebral artery laterally.
6. Anterior cerebral artery: runs forward and medially toward the longitudinal fissure.
7. Anterior communicating artery: crosses the midline to the anterior cerebral artery of the other side, closing the front of the ring.

Artery	Comes from	What it supplies
Internal carotid artery	Common carotid artery in the neck	Main source of the anterior circulation; ends by dividing into the anterior and middle cerebral arteries
Anterior cerebral artery	Internal carotid artery	Medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and somatosensory cortex
Anterior communicating artery	Joins the two anterior cerebral arteries	Closes the front of the ring
Middle cerebral artery	Internal carotid artery (largest branch)	Most of the lateral surface, including the face and upper limb areas of the motor and somatosensory cortex, the auditory cortex, and the insula
Posterior communicating artery	Joins the internal carotid and posterior cerebral arteries	Links the anterior and posterior circulations
Posterior cerebral artery	Basilar artery (end branch)	Occipital lobe, including the visual cortex, and the inferior surface of the temporal lobe
Vertebral artery	Subclavian artery	Main source of the posterior circulation; the two unite to form the basilar artery
Basilar artery	Union of the two vertebral arteries	Brainstem, through the pontine arteries; gives off the anterior inferior and superior cerebellar arteries
Pontine arteries	Basilar artery	The pons
Posterior inferior cerebellar artery (PICA)	Vertebral artery	Underside of the cerebellum and the lateral medulla
Anterior inferior cerebellar artery (AICA)	Basilar artery, lower part	Front part of the underside of the cerebellum
Superior cerebellar artery	Basilar artery, top	Upper surface of the cerebellum
Anterior spinal artery	A branch from each vertebral artery, joined in the midline	Front of the spinal cord
Posterior spinal arteries	Vertebral artery or posterior inferior cerebellar artery	Back of the spinal cord

- The anterior communicating artery joins the two anterior cerebral arteries to each other. The posterior communicating artery is the one that links the front and back circulations.
- The posterior inferior cerebellar artery comes off the vertebral artery. The other two cerebellar arteries come off the basilar. Check where your three start.
- Memory tip: The cerebellar arteries are named from the bottom up: PICA from the vertebral artery, AICA from the lower basilar artery, and the superior cerebellar artery from the top of the basilar artery.

What this means for patients

- Location names the deficit. Because each lobe and cortical area has its own job, the location of an injury predicts the loss. Damage to the occipital lobe affects vision. Damage to Broca's area makes speech slow and effortful, while understanding of language is largely spared.
- The artery involved predicts the pattern too. A stroke in the middle cerebral artery affects the face and arm on the opposite side more than the leg. A stroke in the anterior cerebral artery affects the opposite leg most. A clinician can often estimate where a stroke struck from the pattern of symptoms alone.
- The projection fibers of the internal capsule carry most of the traffic between the cortex and the body through one narrow band. A small stroke there can cut a great deal of wiring at once, which is why such a small lesion can weaken an entire side of the body.
- The Circle of Willis gives the brain a backup route. If one artery feeding the ring narrows slowly, such as one internal carotid artery, blood can sometimes reach its territory through the communicating arteries. This only helps for blockages in the vessels that feed the ring. A blockage in a cerebral artery beyond the ring has no backup route through the circle, and the tissue that artery feeds is at risk. In many people one of the communicating arteries is small or missing, which also limits the backup.

The brainstem

The written notes for the brainstem lecture: the three regions in order, the midbrain, pons, and medulla structures, the cranial nerve nuclei by region, and the reticular formation. Module 5 competencies 161 to 166, with eight worked questions.

Competency 161. Brainstem regions

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

The brainstem is the stalk of the brain. It sits below the diencephalon, in front of the cerebellum, and above the spinal cord. It has three regions. From top to bottom they are the midbrain, the pons, and the medulla oblongata.

The three regions, top to bottom

Midbrain
Also called the mesencephalon. The short, narrow region just below the diencephalon. It is the most superior region of the brainstem.

Pons
The rounded bulge below the midbrain. Its name means bridge. On the model, the rounded bulge on the front of the pons is the basilar pons.

Medulla oblongata
The lowest region. It tapers down and becomes the spinal cord at the foramen magnum. The medulla, not the pons, is the region that turns into the spinal cord.

On the front of a brain model or sheep brain, look for the groove where the pons meets the medulla. This is the pontomedullary junction. It is a useful landmark, because three cranial nerves (VI, VII, and VIII) leave the brainstem along it.

Region	Where it is	What it connects
Midbrain	Most superior region; just below the diencephalon	Joins the diencephalon to the pons; the superior cerebellar peduncles attach here
Pons	Middle region; the rounded bulge in front of the cerebellum	Bridges the midbrain and the medulla, and links the cerebrum to the cerebellum through the middle cerebellar peduncles
Medulla oblongata	Most inferior region; ends at the foramen magnum	Continuous with the spinal cord below; the inferior cerebellar peduncles attach here
Pontomedullary junction	The groove between the pons and the medulla on the front surface	Marks the border of the two regions; CN VI, VII, and VIII leave here

How the brainstem connects the higher brain to the spinal cord

1. Every tract passes through it. Every ascending tract carrying sensory signals up to the brain and every descending tract carrying motor commands down to the cord runs through all three regions.
2. It holds cranial nerve nuclei. The nuclei of cranial nerves III through X and XII are in the brainstem, so ten of the twelve cranial nerves have brainstem nuclei. The nucleus of XI sits just below it, in the upper cervical spinal cord.
3. It carries the cerebellum. Three pairs of cerebellar peduncles tie the cerebellum to it: the superior peduncles to the midbrain, the middle peduncles to the pons, and the inferior peduncles to the medulla.

- Memory tip: read the brainstem from the top down: Midbrain, then Pons, then Medulla (M, P, M). The medulla is the one that turns into the spinal cord.
- One structure you can find on each region on the model: the cerebral peduncles or cerebral aqueduct on the midbrain, the basilar pons or middle cerebellar peduncle on the pons, and the pyramids or medullary olive on the medulla.

Competency 162. Midbrain structures

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

The midbrain surrounds a narrow channel, the cerebral aqueduct. The aqueduct carries cerebrospinal fluid from the third ventricle above down to the fourth ventricle below. On a cross section of the midbrain it is the small hole near the back. Use it to orient the specimen: everything behind the aqueduct is the roof, and everything in front of it is the floor and the stalks.

The roof: the tectum and its four colliculi

The roof of the midbrain is the tectum. Seen from behind, it carries four rounded bumps, together called the corpora quadrigemina. The upper pair are the superior colliculi. They are reflex centers for vision, such as turning the eyes and head toward a moving object. The lower pair are the inferior colliculi. They are reflex centers and relay stations for hearing, such as turning the head toward a sudden sound. You do not become conscious of the sight or sound in the colliculi; they run reflexes only. Awareness of what you see and hear happens in the cerebral cortex.

The front: the cerebral peduncles, the crus cerebri, and the substantia nigra

The front of the midbrain is formed by the two cerebral peduncles. These thick stalks carry the descending motor tracts from the cerebrum down to the pons, medulla, and spinal cord. The most anterior part of each peduncle is a band of white matter called the crus cerebri. Between the aqueduct and the crura lies the tegmentum, the core of the midbrain, which holds cranial nerve nuclei and part of the reticular formation.

The substantia nigra is a dark band of gray matter between the crus cerebri and the tegmentum, so on a cross section it sits just behind the crus cerebri. Its neurons contain a dark pigment, which is why it looks black in a fresh cross section. It works with the basal nuclei (basal ganglia) to help control movement.

Two cranial nerves leave the midbrain. The oculomotor nerve (CN III) leaves the front, in the gap between the two cerebral peduncles. The trochlear nerve (CN IV) leaves the back, just below the inferior colliculi.

Structure	Where it is	What it does
Tectum	The roof of the midbrain, behind the cerebral aqueduct	Carries the four colliculi
Superior colliculi	The upper pair of bumps on the tectum	Reflex centers for vision, such as tracking a moving object
Inferior colliculi	The lower pair of bumps on the tectum	Reflex centers and relays for hearing, such as turning toward a sound
Cerebral peduncles	The paired stalks on the front of the midbrain	Carry descending motor tracts from the cerebrum
Crus cerebri	The most anterior part of each cerebral peduncle	The white matter band holding the descending motor fibers
Substantia nigra	A dark band between the crus cerebri and the tegmentum	Works with the basal nuclei to help control movement
Cerebral aqueduct	The narrow channel through the midbrain, between the tectum and the tegmentum	Carries CSF from the third ventricle to the fourth ventricle

- Memory tip: superior colliculi sit higher and serve sight; inferior colliculi sit lower and serve sound. Both are reflex centers. You become aware of what you see and hear in the cerebral cortex.
- Cerebrospinal fluid runs downward through the aqueduct: in from the third ventricle above, out to the fourth ventricle below.

Competency 163. Pons structures

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

The pons is the rounded bulge between the midbrain and the medulla. The word pons means bridge, and it bridges three regions: the midbrain above it, the medulla below it, and the cerebellum behind it. Its front surface is the basilar pons, a thick band of fibers that runs crosswise and curves back on each side toward the cerebellum. The back part of the pons is its tegmentum, which carries cranial nerve nuclei and part of the reticular formation.

The pontine nuclei and the middle cerebellar peduncles

Scattered through the basilar pons are the pontine nuclei. These clusters of gray matter receive motor information from the cerebral cortex. Their axons cross to the opposite side and enter the cerebellum. This relay is how the cerebrum keeps the cerebellum informed about planned movements, so the cerebellum can coordinate them.

The axons from the pontine nuclei leave the pons as the middle cerebellar peduncles. These are the largest of the three pairs of cerebellar peduncles. They are the thick stalks that attach the side of the pons to each cerebellar hemisphere. On a model, follow the crosswise fibers of the basilar pons around the side and into the cerebellum. Signals from the pons enter the cerebellum through the middle cerebellar peduncle, not the superior or inferior one.

Motor information from the cortex to the cerebellum, in order

1. Cerebral cortex: sends the motor information.
2. Pontine nuclei: relay the motor information on to the cerebellum.
3. Middle cerebellar peduncles: carry the signals from the pontine nuclei into the cerebellum.
4. Cerebellum: uses the information to coordinate movement.

The fourth ventricle

Behind the pons lies the fourth ventricle. It is a diamond-shaped space filled with cerebrospinal fluid. Its floor is the back of the pons and the upper medulla, and the cerebellum lies over its roof. So the fourth ventricle sits between the brainstem and the cerebellum; it is not inside the pons. The cerebral aqueduct opens into it from above.

The pons also holds nuclei that help adjust the breathing rhythm set by the medulla. The trigeminal nerve (CN V) leaves the side of the pons. The abducens (CN VI), facial (CN VII), and vestibulocochlear (CN VIII) nerves leave at the pontomedullary junction.

Structure	Where it is	What it does
Basilar pons	The front surface of the pons	Crosswise fibers heading to the cerebellum, with the pontine nuclei scattered among them
Pontine nuclei	Clusters of gray matter in the basilar pons	Relay motor information from the cerebral cortex to the cerebellum
Middle cerebellar peduncles	The thick stalks on each side of the pons	Carry the axons of the pontine nuclei into the cerebellum
Fourth ventricle	Behind the pons and upper medulla, in front of the cerebellum	Holds CSF; receives it from the cerebral aqueduct

Competency 164. Medulla oblongata structures

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

The medulla oblongata is the lowest region of the brainstem. It runs from the pontomedullary junction down to the foramen magnum, where it becomes the spinal cord. Its upper half forms the lower part of the floor of the fourth ventricle. Its lower half surrounds a narrow channel that continues into the cord as the central canal.

On the surface: pyramids, decussation, and olives

Pyramids

The front of the medulla has two long ridges, one on each side of the midline. Each pyramid is made of the corticospinal tract, the descending motor pathway from the cerebral cortex to the spinal cord.

Decussation of the pyramids

Near the bottom of the medulla, just above the spinal cord, most corticospinal fibers cross the midline to the opposite side. Decussation means crossing. On a model it shows as fibers that cross over and blur the groove between the two pyramids. Because of this crossing, the left cerebral hemisphere controls the muscles on the right side of the body, and the right hemisphere controls the left side.

Olive (the medullary olive)

Just lateral to each pyramid is an oval swelling, the olive. The olives sit to the outside of the pyramids, not between them. Each is formed by the inferior olivary nucleus underneath, which relays information to the cerebellum through the inferior cerebellar peduncle.

On the front of the medulla, the hypoglossal nerve (CN XII) leaves in the groove between the pyramid and the olive. The glossopharyngeal (CN IX) and vagus (CN X) nerves leave in the groove behind the olive.

Inside: the three vital centers

The medulla also holds three vital centers in its core. They are part of the reticular formation and are not visible on the surface. The cardiac center adjusts the rate and force of the heartbeat. The respiratory center sets the basic rhythm of breathing. The vasomotor center adjusts the diameter of blood vessels and so helps set blood pressure. Only the pyramids, olives, and decussation show on the surface; the three centers are inside.

Structure	Where it is	What it is or does
Pyramids	Paired ridges on the front of the medulla (surface)	Formed by the descending corticospinal tracts
Decussation of the pyramids	The lower medulla, just above the spinal cord (surface)	Where most corticospinal fibers cross to the opposite side
Olive	Oval bulge lateral to each pyramid (surface)	Formed by the inferior olivary nucleus, a relay to the cerebellum
Cardiac center	Inside the medulla (not visible on the surface)	Adjusts the rate and force of the heartbeat
Respiratory center	Inside the medulla (not visible on the surface)	Sets the basic rhythm of breathing
Vasomotor center	Inside the medulla (not visible on the surface)	Adjusts blood vessel diameter, and so blood pressure

What this means for patients

- The medulla oblongata holds the cardiac center, respiratory center, and vasomotor center. Because these centers control the heartbeat and breathing, an injury to the medulla can be life-threatening right away, in a way that an injury to a single area of cerebral cortex is not.
- The decussation of the pyramids is the anatomical reason a stroke in the right cerebral hemisphere weakens the left side of the body. Most of the motor fibers have already crossed by the time they reach the spinal cord.
- The reticular activating system links the brainstem to consciousness. Damage to this ascending network in the upper brainstem can cause a coma, even when the cerebral cortex itself is not injured.
- The substantia nigra links the brainstem to movement: loss of its pigmented neurons is the basis of Parkinson disease.

Competency 165. Cranial nerve nuclei distribution

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

A cranial nerve nucleus is a cluster of neuron cell bodies inside the brain that gives rise to the motor fibers of a cranial nerve or receives its sensory fibers. Cranial nerves I (olfactory) and II (optic) connect to the forebrain, so they have no brainstem nucleus. The nuclei of cranial nerves III through X and XII are housed in the brainstem, and the nucleus of XI sits just below it in the upper cervical spinal cord. That makes 10 of the 12 cranial nerves with brainstem nuclei.

The nuclei line up from top to bottom in the same order as the nerve numbers. The midbrain holds III and IV. The pons holds V, VI, VII, and VIII. The medulla holds IX, X, and XII. A nerve usually leaves the brainstem near the level of its nucleus, so the attachment point on a model tells you the region.

Three nerves that need a closer look

Trigeminal (CN V)
Assigned to the pons, where its main motor and sensory nuclei sit, but its sensory nuclei stretch up into the midbrain and down through the medulla.

Vestibulocochlear (CN VIII)
Its vestibular and cochlear nuclei sit at the junction of the pons and medulla, so it is listed with both. It is the one nerve that belongs beside two regions.

Accessory (CN XI)
Grouped with the medulla in the 2-2-4-4 rule below, but its motor nucleus sits in the upper cervical spinal cord. Its rootlets rise through the foramen magnum and then leave the skull with the vagus nerve.

Cranial nerve	Region that houses the nucleus	Where the nerve leaves the brainstem
III Oculomotor	Midbrain	Front of the midbrain, between the cerebral peduncles
IV Trochlear	Midbrain	Back of the midbrain, just below the inferior colliculi
V Trigeminal	Pons (sensory nuclei extend into the midbrain and medulla)	Side of the pons
VI Abducens	Pons	Pontomedullary junction, near the midline
VII Facial	Pons	Pontomedullary junction, lateral to VI
VIII Vestibulocochlear	Pons and medulla, at their junction	Pontomedullary junction, lateral to VII
IX Glossopharyngeal	Medulla	Groove behind the olive
X Vagus	Medulla	Groove behind the olive, below IX
XI Accessory	Upper cervical spinal cord (grouped with the medulla)	Side of the upper cervical cord; rootlets rise through the foramen magnum
XII Hypoglossal	Medulla	Groove between the pyramid and the olive

- Memory tip: use the 2-2-4-4 rule. Two nerves (I and II) attach above the brainstem, two (III and IV) to the midbrain, four (V to VIII) to the pons, and four (IX to XII) to the medulla. Remember that VIII also reaches the medulla and that XI starts in the cervical cord.
- Abducens (VI) and facial (VII) belong to the pons, not the medulla. They leave at the pontomedullary junction, which is the border, but their nuclei are on the pons side of it.

Competency 166. Reticular formation

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

The reticular formation is a loose net of gray matter that runs through the core of the whole brainstem, from the medulla up through the pons and the midbrain. Its name comes from the Latin for net. It is made of small clusters of neurons scattered among bundles of white matter, so it has no sharp borders and no surface landmark. It runs through all three brainstem regions, not only the medulla.

To locate it, look at any brainstem cross section. The reticular formation fills the central core (the tegmentum), behind the large descending motor tracts and in front of the ventricle or aqueduct. Its neurons connect with the cranial nerve nuclei, the cerebellum, the thalamus, and the spinal cord.

Two components running in opposite directions

The ascending reticular activating system, or RAS, sends fibers up through the thalamus to the cerebral cortex. It keeps the cortex awake and alert, holds attention, and filters out unimportant sensory input so the cortex is not flooded. RAS activity falls during sleep: when the RAS shuts down, the result is sleep. Severe damage to it can cause a coma.

The descending component sends fibers down to the spinal cord, and through it acts on the skeletal muscles. These fibers help set the resting tone of skeletal muscles and help keep posture. The cardiac, respiratory, and vasomotor centers of the medulla are also part of the reticular formation, so this descending output links the brainstem to heart rate, breathing, and blood vessel diameter.

Component	Direction and target	What it does
Ascending reticular activating system (RAS)	Up through the thalamus to the cerebral cortex	Keeps the cortex awake and alert (consciousness and arousal), holds attention, filters out unimportant sensory input
Descending component	Down to the spinal cord, and through it the skeletal muscles	Helps set resting muscle tone and posture; includes the medullary centers for heart rate, breathing, and blood vessel diameter

- Memory tip: ascending means awake: the ascending system wakes up the cortex. Descending means down to the muscles: it sets their tone.
- Muscle tone belongs to the descending component. The RAS is the upward part that keeps the cortex awake.

The meninges and cerebrospinal fluid

The written notes for the meninges and cerebrospinal fluid lecture: the cranial and spinal meninges, the dural reflections and venous sinuses, the meningeal spaces, the end of the cord and the lumbar puncture, the ventricles, the CSF route, and the blood-brain barrier. Module 5 competencies 167 to 172, with eight worked questions.

Competency 167. Cranial meninges and dural reflections

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

Three connective tissue membranes, the meninges, wrap the brain. From outside to inside they are the dura mater, the arachnoid mater, and the pia mater. The names describe them. Dura mater means tough mother, pia mater means tender mother, and arachnoid means spider-like, for the web of fine strands it sends down to the pia.

The three layers, from the skull to the brain

Dura mater
: The outermost layer, against the skull. Around the brain the dura has two layers. The periosteal layer is attached to the inner surface of the skull. The meningeal layer lies inside it. The two layers are fused together except in a few places where they separate, and the spaces between them form the dural venous sinuses, channels that collect venous blood from the brain.

Arachnoid mater
: The thin, delicate middle membrane just inside the dura. It does not dip into the grooves of the brain. Fine web-like strands cross from it to the pia, through the CSF-filled space between them.

Pia mater
: The innermost layer, a very thin membrane rich in small blood vessels. It clings to the surface of the brain and follows it into every sulcus.

Layer	Position	Description
Dura mater	Outermost, against the skull	Tough; two layers (periosteal and meningeal) fused except where they part to form the dural venous sinuses
Arachnoid mater	Middle	Thin and delicate; sends web-like strands to the pia; does not follow the sulci
Pia mater	Innermost, on the brain surface	Very thin and rich in blood vessels; follows every gyrus and sulcus

- On a strip drawn from skull to brain the order is dura, arachnoid, pia, with the pia clinging to the brain surface.
- The dura is drawn as two layers: the periosteal layer against the skull and the meningeal layer beneath it. The arachnoid and pia are each a single layer.

The dural reflections

In three main places the meningeal layer of the dura folds inward as a sheet that divides the cranial cavity and steadies the brain. These folds are the dural reflections, also called the dural extensions. The falx cerebri drops into the longitudinal fissure between the right and left cerebral hemispheres. The tentorium cerebelli lies in the transverse fissure, like a tent over the cerebellum, separating it from the occipital lobes of the cerebrum above. The falx cerebelli is a small fold below the tentorium that separates the right and left cerebellar hemispheres.

Fold	Where it is	What it separates
Falx cerebri	In the longitudinal fissure, along the midline	Right and left cerebral hemispheres
Tentorium cerebelli	In the transverse fissure, roofing the cerebellum	Cerebrum above from cerebellum below
Falx cerebelli	Midline, below the tentorium	Right and left cerebellar hemispheres

- The tentorium cerebelli separates the cerebrum from the cerebellum. It does not split the cerebellum. Splitting the cerebellum into right and left halves is the job of the falx cerebelli.
- Falx means sickle. Both falx folds are midline sheets that separate a right half from a left half; one is for the cerebrum and the other, much smaller, is for the cerebellum.

Dural venous sinus drainage

The main sinuses run in the edges of these folds. The superior sagittal sinus runs along the top edge of the falx cerebri, where it attaches to the skull. The inferior sagittal sinus runs in the lower free edge of the falx cerebri. It drains into the straight sinus, which runs where the falx cerebri meets the tentorium cerebelli. The superior sagittal and straight sinuses meet at the

back of the skull, at the confluence of sinuses. Follow the blood out of the skull from the superior sagittal sinus.

1. Superior sagittal sinus: runs from front to back along the top edge of the falx cerebri. The arachnoid granulations return CSF into it.
2. Confluence of sinuses: the meeting point at the back of the skull, at the internal occipital protuberance. The straight sinus also drains here.
3. Transverse sinuses: paired sinuses that run laterally from the confluence along the back edge of the tentorium cerebelli.
4. Sigmoid sinuses: S-shaped sinuses that curve down from each transverse sinus to the jugular foramen.
5. Internal jugular veins: the sigmoid sinus passes through the jugular foramen and becomes the internal jugular vein, which carries the blood down the neck toward the heart.

- Blood runs front to back along the superior sagittal sinus, then out to the side in the transverse sinus, then down the sigmoid sinus and into the internal jugular vein.
- The arachnoid granulations empty CSF into the superior sagittal sinus, not into the internal jugular vein. By the time the blood reaches the jugular vein the CSF is already mixed into it.

Competency 168. Spinal meninges and meningeal spaces

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

The same three meninges continue down the vertebral canal around the spinal cord. The spinal dura mater is a single layer. It matches the meningeal layer of the cranial dura and is not attached to the vertebrae. It forms a tube, the dural sac, that ends at about the level of S2. The spinal arachnoid mater lines the dural sac. The spinal pia mater clings to the surface of the cord.

Two anchors made of pia

Denticulate ligaments

Tooth-shaped flaps of pia along each side of the cord, between the dorsal and ventral roots. They reach out through the arachnoid and attach to the dura, which keeps the cord centered in the dural sac. They anchor the cord to the dura along its length.

Filum terminale

A thin strand of pia that continues down from the bottom tip of the cord. It passes through the lumbar cistern, picks up a covering of dura at the end of the dural sac, and anchors to the coccyx. It anchors the lower end of the cord.

- Both anchors are extensions of the pia mater. The denticulate ligaments tie the sides of the cord to the dura; the filum terminale ties the end of the cord to the coccyx.

The meningeal spaces

The epidural space lies between the dura and the bone. In the vertebral canal it is a real space, filled with fat and veins, and it is where an epidural injection goes. In the skull the periosteal dura is attached to the bone, so the cranial epidural space is only a potential space. It opens only if something, such as blood, separates the dura from the skull.

The subdural space lies between the dura and the arachnoid. Normally the two membranes touch, so it is a potential space, holding only a thin film of fluid. The subarachnoid space lies between the arachnoid and the pia. It is a real space. It holds cerebrospinal fluid, the web-like strands of the arachnoid, and the larger blood vessels that run over the surface of the brain and cord.

Space	Location	Contents	Found in the cranium too?
Epidural space	Between the dura and the bone	Spinal: fat and veins (a real space). Cranial: a potential space only	Only as a potential space, because the dura is attached to the skull
Subdural space	Between the dura and the arachnoid	A potential space with a thin film of fluid	Yes
Subarachnoid space	Between the arachnoid and the pia	Cerebrospinal fluid, arachnoid strands, and the larger surface blood vessels	Yes

- On a cross section of a vertebra, the order from the outside in is vertebral bone, epidural space (fat and veins), dura mater, subdural space, arachnoid mater, subarachnoid space (CSF), pia mater on the surface of the cord.
- CSF is in the subarachnoid space, under the arachnoid, not in the epidural or subdural space.
- The epidural space is a real space only in the spine. In the cranium the dura is attached to the skull, so there is no room there unless something forces the two apart.

Disorders named by meningeal layer or space

Bleeding and infection are named by the layer or space they involve.

Disorder	Structure affected	What changes there
Epidural hematoma	Between the skull and the dura	Blood, often from an artery torn by a skull fracture, separates the dura from the bone
Subdural hematoma	Subdural space	Blood, often from torn veins crossing to the dural sinuses, collects between the dura and the arachnoid
Subarachnoid hemorrhage	Subarachnoid space	Blood from a ruptured surface artery mixes into the cerebrospinal fluid
Meningitis	The meninges, mainly the arachnoid and pia	The membranes become inflamed, usually from an infection; the CSF in the subarachnoid space shows the change

What this means for patients

- An epidural hematoma opens the cranial epidural space, a space that normally does not exist. The blood has to peel the dura off the skull, which is why it is usually arterial blood under pressure.
- A subdural hematoma sits just beneath the dura, in the potential space between the dura and the arachnoid, usually from veins torn where they cross to the dural sinuses.
- A subarachnoid hemorrhage and meningitis both change the CSF, because both involve the subarachnoid space. That is why CSF can be tested for either one.
- An epidural injection goes into the spinal epidural space, the fat-filled and vein-filled space outside the dura. It never enters the CSF.

Competency 169. Cord termination and lumbar puncture

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

The spinal cord is shorter than the vertebral column. In an adult it ends at about the level of the disc between L1 and L2. The tapered, cone-shaped end of the cord is the conus medullaris.

The nerve roots for the lumbar, sacral, and coccygeal spinal nerves leave the cord above that level. They have to travel down the vertebral canal before they reach their exit openings. This bundle of long roots below the conus medullaris is the cauda equina, which means horse's tail.

The dural sac and the subarachnoid space continue down to about S2. Below the conus medullaris the subarachnoid space is wide and holds no spinal cord, only CSF, the roots of the cauda equina, and the filum terminale. This enlarged space is the lumbar cistern.

Structure	Level	What it is
Conus medullaris	L1 to L2 disc (adult)	The tapered lower end of the spinal cord
Cauda equina	Below L1 to L2, down the vertebral canal	The bundle of lumbar, sacral, and coccygeal nerve roots
Filum terminale	From the conus medullaris to the coccyx	A strand of pia mater that anchors the lower end of the cord
Lumbar cistern	From the conus medullaris to about S2	The enlarged subarachnoid space, filled with CSF and the cauda equina

- The cord ends at the L1 to L2 disc, not at the bottom of the spine. On a drawing, the cord stops at the conus and only roots, the filum terminale, and CSF continue below it.

The lumbar puncture

A lumbar puncture, or spinal tap, draws CSF from the lumbar cistern. The needle goes in at the midline between the spinous processes of L3 and L4 or L4 and L5, well below the end of the cord. A line drawn across the tops of the iliac crests crosses the spine at about L4, which helps find the level. The roots of the cauda equina float in the CSF and slide aside from the needle. These are the layers the needle crosses, superficial to deep.

1. Skin: the epidermis and dermis of the lower back.
2. Subcutaneous tissue: the fatty layer under the skin.
3. Supraspinous ligament: runs along the tips of the spinous processes.
4. Interspinous ligament: fills the gap between neighboring spinous processes.
5. Ligamentum flavum: the tough, elastic ligament joining the laminae of neighboring vertebrae. The needle meets firm resistance here and then a give.
6. Epidural space: the fat-filled and vein-filled space between the bone and the dura.
7. Dura mater: the tough outer meningeal membrane.
8. Arachnoid mater: the thin middle membrane, pressed against the dura, so the needle passes the two together.
9. Subarachnoid space: the needle tip is now in the lumbar cistern, and CSF can be drawn.

- There are three ligaments in a row, and the ligamentum flavum is the last and deepest of them. It sits just above the epidural space.
- The firm ligamentum flavum giving way is the landmark the clinician feels. After that come the epidural space, the dura, and the arachnoid, and then CSF appears.
- The entry levels are between the L3 and L4 spines or the L4 and L5 spines. The line across the tops of the iliac crests marks about L4.

What this means for patients

- The cord ends, but the fluid does not. The spinal cord stops at the conus medullaris near the L1 to L2 disc, yet the subarachnoid space runs lower, as the lumbar cistern. This difference in length is the basis of the lumbar puncture. A needle placed between the L3 and L4 or L4 and L5 spines reaches a pool of CSF where only the floating cauda equina lies in its path, and those roots drift aside.
- Knowing the layers in order is what makes the tap safe. The needle crosses skin, subcutaneous tissue, three ligaments, the epidural space, and then the dura and arachnoid before CSF appears.
- The CSF drawn this way is tested for signs of meningitis or bleeding.
- The named spaces also tell bleeds apart: an epidural hematoma lies between the skull and the dura, a subdural hematoma lies just beneath the dura, and a subarachnoid hemorrhage mixes into the CSF itself.

Competency 170. Ventricular system

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

The ventricles are connected, fluid-filled spaces inside the brain. There are four: two lateral ventricles, the third ventricle, and the fourth ventricle. They are lined by ependymal cells, a type of glial cell, and filled with cerebrospinal fluid. Below the fourth ventricle the same system continues as the central canal of the spinal cord.

The four ventricles and the central canal

Lateral ventricles
The largest. There is one in each cerebral hemisphere. Each is C-shaped, curving from the frontal lobe back and then down into the temporal lobe. In the midline the two lateral ventricles are separated by a thin sheet called the septum pellucidum.

Third ventricle
A narrow slit in the midline of the diencephalon, between the right and left halves of the thalamus. Each lateral ventricle opens into it through a small hole, the interventricular foramen (foramen of Monro). There is one foramen on each side.

Cerebral aqueduct
Also called the aqueduct of Sylvius. The narrow channel through the midbrain. The third ventricle drains down through it into the fourth ventricle.

Fourth ventricle
Lies behind the pons and upper medulla and in front of the cerebellum, so it sits between the brainstem and the cerebellum.

Central canal
The narrow channel that runs down the center of the spinal cord for its full length. It is a direct continuation of the bottom of the fourth ventricle.

How the fourth ventricle empties

The fourth ventricle has two kinds of exits. Three apertures in its roof and sides open into the subarachnoid space around the brain: one median aperture (foramen of Magendie) in the midline and two lateral apertures (foramina of Luschka), one on each side. Separately, the bottom of the fourth ventricle narrows directly into the central canal, which runs down the center of the spinal cord. The apertures lead only to the subarachnoid space, and the central canal opens only from the fourth ventricle itself.

Space	Location	Connects to	Through
Lateral ventricles	One in each cerebral hemisphere; separated by the septum pellucidum	Third ventricle	Interventricular foramina (of Monro)
Third ventricle	Midline of the diencephalon, between the halves of the thalamus	Fourth ventricle	Cerebral aqueduct (of Sylvius)
Fourth ventricle	Behind the pons and upper medulla, in front of the cerebellum	Subarachnoid space	One median aperture and two lateral apertures
Fourth ventricle	Lower end, in the medulla	Central canal	Continues directly down into the cord
Central canal	Center of the spinal cord, full length	Fourth ventricle above	Direct continuation

- Memory tip: count the openings out of the fourth ventricle into the subarachnoid space: one median and two lateral, three in all. The central canal is a straight continuation down into the cord.
- The cerebral aqueduct joins the third ventricle to the fourth. The lateral ventricles reach the third ventricle through the interventricular foramina instead.
- The apertures open into the subarachnoid space, not into the central canal. The central canal joins the fourth ventricle directly, with no aperture.
- Region by region: the lateral ventricles sit in the cerebrum, the third ventricle in the diencephalon, the aqueduct in the midbrain, and the fourth ventricle between the brainstem and the cerebellum. Ependymal cells line every chamber.
- The three narrow connection points are the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. If one is blocked, CSF backs up behind it, the ventricles swell, and pressure in the skull rises. That condition is hydrocephalus, covered in competency 171.

Competency 171. CSF production and circulation

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

Cerebrospinal fluid (CSF) is a clear, watery fluid that fills the ventricles and central canal and surrounds the brain and spinal cord in the subarachnoid space. It floats the brain, which lowers the brain's effective weight. It cushions the brain and cord against jolts. It also keeps a stable chemical environment around the neurons.

Where CSF is made and where it leaves

CSF is made by the choroid plexus. A choroid plexus is a tuft of capillaries covered by ependymal cells that hangs into a ventricle. The ependymal cells make CSF from the blood plasma in the capillaries. There is choroid plexus in all four ventricles, and the largest amounts are in the lateral ventricles. About 150 mL of CSF surrounds the CNS at any one time, and about 500 mL is made each day, so the fluid is replaced several times a day.

CSF moves one way through the ventricles, out into the subarachnoid space, and back into the blood. It is reabsorbed at the arachnoid granulations, small knobs of arachnoid mater that push through the dura into the dural venous sinuses, mainly the superior sagittal sinus. From there the fluid leaves the skull with the venous blood.

The route, in order

1. Choroid plexus: capillaries covered by ependymal cells make CSF in each ventricle, most of it in the lateral ventricles.
2. Lateral ventricles: CSF collects in the ventricle in each cerebral hemisphere.
3. Interventricular foramina: CSF passes through these paired openings (foramina of Monro) into the third ventricle.
4. Third ventricle: in the midline of the diencephalon; its choroid plexus adds more CSF.
5. Cerebral aqueduct: CSF flows down this narrow channel through the midbrain.
6. Fourth ventricle: behind the pons and upper medulla; its choroid plexus adds more CSF. A small amount continues straight down into the central canal of the spinal cord.
7. Median and lateral apertures: CSF leaves the fourth ventricle through one median aperture and two lateral apertures.
8. Subarachnoid space: CSF bathes the outside of the brain and spinal cord, including the lumbar cistern.
9. Arachnoid granulations: CSF passes through these knobs of arachnoid mater into the venous blood.
10. Dural venous sinuses: the CSF now travels with the venous blood, mainly in the superior sagittal sinus, and drains out of the skull through the internal jugular veins.

- Only the three kinds of ventricle (lateral, third, and fourth) hold choroid plexus. The foramina, the aqueduct, and the apertures are passageways with no choroid plexus.
- The side branch into the central canal leaves from the fourth ventricle, not from the subarachnoid space.
- The ependymal cells of the choroid plexus are the cells that make the CSF. The capillaries supply the plasma they make it from.

Feature	Description
Where CSF is found	The four ventricles, the central canal, and the subarachnoid space around the brain and cord, including the lumbar cistern
Site of production	The choroid plexus in the lateral, third, and fourth ventricles
Site of reabsorption	The arachnoid granulations, into the dural venous sinuses
Buoyancy	Floats the brain so it does not press down on its own base
Protection	Cushions the brain and spinal cord against jolts and blows
Chemical stability	Carries nutrients, removes wastes, and keeps a steady environment around the neurons

A disorder of CSF circulation

The narrow points on this route are the interventricular foramina, the cerebral aqueduct, and the apertures of the fourth ventricle. If CSF is blocked at one of them, or is not reabsorbed, it builds up. This condition is hydrocephalus.

Disorder	Structure affected	What changes there
Hydrocephalus	The ventricles	CSF builds up behind a blockage or from poor reabsorption; the ventricles enlarge and pressure inside the skull often rises

- Memory tip: CSF is made inside the brain, in the ventricles, and reabsorbed outside the brain, at the arachnoid granulations in the dural sinuses. Anything that blocks a narrow point between those two places backs the fluid up.

What this means for patients

- Because about 500 mL is made each day and only about 150 mL is present at any moment, a blockage shows up quickly. The ventricles upstream of the block swell and the pressure inside the skull rises.

- Hydrocephalus can come from a block at any of the three narrow points, or from the arachnoid granulations failing to return CSF to the blood. Either way the fluid has nowhere to go.

Competency 172. Blood-brain barrier structure

You should be able to: Name the three structural components of the blood-brain barrier: the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

The blood-brain barrier (BBB) controls what can pass from the blood into the brain tissue. It keeps out many substances, including many drugs and germs, while letting in what neurons need, such as glucose and oxygen. It is a structure built around the brain's capillaries, with three components.

The three components, from the blood outward

Capillary endothelium
: The first and most important component. The endothelial cells that line brain capillaries are joined by tight junctions, which seal the gaps between neighboring cells. Brain capillaries also lack the pores and wide clefts found in many capillaries elsewhere in the body. As a result, most substances can only cross by passing through the endothelial cells themselves.

Basement membrane
: The second component, a thin layer of extracellular material that surrounds the endothelium.

Astrocyte end-feet
: The third component. Astrocytes are star-shaped glial cells. Their processes end in flattened feet that wrap around the outside of the capillary, covering the basement membrane. The end-feet support the endothelial cells and help keep the tight junctions in place.

The barrier is missing in a few small areas around the ventricles, such as parts of the hypothalamus, where the brain needs to sample the blood directly.

Component	What it is	Position from the blood	Its part in the barrier
Capillary endothelium	The cells lining the brain capillaries	First, in contact with the blood	Joined by tight junctions and lacking pores, so substances cannot slip between the cells
Basement membrane	A thin layer of extracellular material around the endothelium	Second	Surrounds and supports the capillary wall
Astrocyte end-feet	The flattened ends of astrocyte processes	Third, the outermost layer	Wrap the outside of the capillary and help maintain the tight junctions

- Memory tip: go from the blood outward: endothelium, then basement membrane, then astrocyte feet. The tight junctions in the endothelium are the actual seal.
- A substance leaving the blood for a neuron crosses the endothelial cells with their tight junctions first, the basement membrane second, and the astrocyte end-feet third. The end-feet are the outermost layer, outside the basement membrane.
- An ordinary capillary has pores in its wall. A brain capillary has none, and that is the difference you draw side by side.
- The end-feet come from astrocytes, not from oligodendrocytes or neurons.

Functional organization and nervous tissue

The written notes for the nervous tissue lecture: the divisions of the nervous system, the parts of a neuron, the chemical synapse, neuron classification, the six glial cells, and myelination and regeneration. Module 5 competencies 173 to 178, with eight worked questions.

Competency 173. Nervous system divisions

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

The nervous system is the body's fast control system. It senses changes, decides what they mean, and sends out commands. Anatomists divide it in two ways: by where the structures sit (central or peripheral), and by what the motor neurons control (skeletal muscle, the organs, or the gut). You need both divisions, and you need to know which branch sits under which.

By location: central and peripheral

Central nervous system (CNS)
: The brain and the spinal cord. The brain sits in the cranial cavity. The spinal cord runs down the vertebral canal and links the brain to the body. The CNS is where information is processed and where commands are issued.

Peripheral nervous system (PNS)
: Every neural structure outside the brain and spinal cord: the nerves, the ganglia, and the sensory receptors. There are 12 pairs of cranial nerves, which attach to the brain, and 31 pairs of spinal nerves, which attach to the spinal cord. The PNS carries information between the CNS and every other part of the body.

Structure	What it is
Central nervous system (CNS)	The brain and the spinal cord; it processes information and issues commands
Brain	The CNS organ in the cranial cavity
Spinal cord	The CNS cord in the vertebral canal that links the brain to the body
Peripheral nervous system (PNS)	Every neural structure outside the brain and spinal cord
Nerves	Bundles of axons in the PNS; 12 pairs of cranial nerves and 31 pairs of spinal nerves
Ganglia	Small clusters of neuron cell bodies outside the CNS
Sensory receptors	Structures that detect changes inside and outside the body

By direction: sensory and motor

The PNS has a sensory side and a motor side. The sensory (afferent) division carries signals from receptors toward the CNS. The motor (efferent) division carries commands away from the CNS to the effectors, the muscles and glands that respond.

The motor divisions, by the effector they control

The motor side is split by what it runs. The somatic nervous system sends commands to skeletal muscle, and you control it voluntarily. The autonomic nervous system (ANS) sends commands to smooth muscle, cardiac muscle, and glands, and it works without conscious control. The ANS has two divisions: the sympathetic division for fight or flight (exercise and emergencies), and the parasympathetic division for rest and digest (routine activities). Many organs receive both. The enteric nervous system (ENS) is a network of roughly 100 million neurons in the wall of the digestive tract. It controls the smooth muscle and glands of the gut and can run on its own, though the ANS connects to it and adjusts it. Some textbooks count it as part of the ANS.

Division	Effector it controls	Control
Somatic nervous system	Skeletal muscle	Voluntary
Autonomic nervous system (ANS)	Smooth muscle, cardiac muscle, and glands	Involuntary
Sympathetic division (ANS)	The same ANS effectors, for exercise and emergencies (fight or flight)	Involuntary
Parasympathetic division (ANS)	The same ANS effectors, for routine activities (rest and digest)	Involuntary
Enteric nervous system (ENS)	Smooth muscle and glands in the wall of the digestive tract	Involuntary; can work on its own

The outline, top to bottom

1. Nervous system.
2. Central nervous system (CNS): brain and spinal cord.
3. Peripheral nervous system (PNS): nerves, ganglia, sensory receptors.
4. Sensory (afferent) division of the PNS: toward the CNS.
5. Motor (efferent) division of the PNS: away from the CNS.
6. Somatic nervous system: skeletal muscle.
7. Autonomic nervous system: smooth muscle, cardiac muscle, glands, with sympathetic and parasympathetic divisions.
8. Enteric nervous system: the wall of the digestive tract.

- Sympathetic and parasympathetic belong under autonomic, not under somatic.
- The heart is muscle, but it is cardiac muscle, so it is autonomic, not somatic. Only skeletal muscle is somatic.

- A loop of gut wall has its own neurons, and those belong to the enteric division.

The three functions

Every task the nervous system performs runs through the same three steps.

1. Sensory function: receptors detect a change and send the input toward the CNS.
2. Integrative function: the CNS analyzes the input, stores some of it, and decides on a response.
3. Motor function: the CNS sends commands out to the effectors, which carry out the response.

- Memory tip: Afferent arrives at the CNS. Efferent exits the CNS.
- The motor divisions are named for what they run: somatic for skeletal muscle, autonomic for the organs, enteric for the gut.

Competency 174. Neuron structure

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

A neuron is the nerve cell that carries signals. Mature neurons are excitable, they can live for a whole lifetime, and they generally do not divide. Every neuron has a cell body and two kinds of processes: dendrites and an axon. The multipolar neuron on the lab slide and the neuron model has many dendrites and one axon, so that is the shape to learn these parts on.

The cell body

The cell body (also called the soma or perikaryon) holds the nucleus, which has a large, dark nucleolus, and most of the cytoplasm and organelles. It is the metabolic center of the neuron and makes the proteins the whole cell needs. Scattered through its cytoplasm are Nissl bodies, clusters of rough endoplasmic reticulum and free ribosomes where those proteins are made. Nissl bodies stain dark on a slide and appear in the cell body and the dendrites, never in the axon hillock or the axon. Neurofibrils, bundles of cytoskeleton filaments, run through the cell and give it its shape. Because the axon depends on the cell body for its proteins, the part of an axon cut away from its cell body dies.

Dendrites, hillock, axon, terminals

Dendrites
The short, heavily branched processes that receive incoming signals. They carry those signals toward the cell body.

Axon hillock
The cone-shaped region where the cell body narrows into the axon. It has no Nissl bodies, so it stains paler than the rest of the cell body. That pale cone is how you find the axon on a slide and tell it apart from a dendrite. Nerve impulses begin at the trigger zone, where the axon hillock meets the first part of the axon.

Axon
The single long process that carries the impulse away from the cell body. Its cytoplasm is the axoplasm and its plasma membrane is the axolemma. An axon can give off side branches called axon collaterals.

Axon terminals
At its far end the axon divides into many fine branches, the axon terminals. The tip of each terminal swells into a synaptic end bulb that contacts the next cell.

The direction a signal travels

1. Dendrites receive the incoming signal and carry it toward the cell body.
2. The cell body collects the signals arriving from the dendrites.
3. The axon hillock is where the impulse starts, at the trigger zone where the hillock joins the axon.
4. The axon carries the impulse away from the cell body.
5. The axon terminals branch at the end of the axon and end in synaptic end bulbs that pass the signal to the next cell.

Structure	What it is
Cell body	The soma or perikaryon; it holds the nucleus and cytoplasm and is the metabolic center of the neuron
Nucleus and nucleolus	The cell's nucleus, with a large, dark-staining nucleolus, inside the cell body
Nissl bodies	Clusters of rough ER and ribosomes in the cell body and dendrites; the site of protein synthesis
Neurofibrils	Bundles of cytoskeleton filaments that give the neuron its shape and support
Microtubules	Cytoskeletal tubes that transport materials along the neuron
Dendrites	The branched processes that receive incoming signals and carry them toward the cell body
Axon hillock	The cone-shaped region where the cell body joins the axon; it has no Nissl bodies
Axon	The single long process that carries the impulse away from the cell body
Axoplasm and axolemma	The cytoplasm of the axon and the plasma membrane of the axon
Axon terminals	The fine branches at the far end of the axon that end in synaptic end bulbs
Lipofuscin	A yellow-brown pigment that builds up in neurons with age; it does not appear to harm the cell

- Memory tip: Dendrites deliver to the cell body. The axon carries the signal away.
- On a slide, the pale cone with no Nissl bodies marks the start of the axon.
- The order is dendrites, cell body, axon hillock, axon, axon terminals. The hillock comes between the cell body and the axon, never after the axon.

Competency 175. Synapse structures

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

A synapse is the junction where a neuron passes its signal to another cell. The cell that sends the signal is the presynaptic neuron. The cell that receives it is the postsynaptic cell, which can be another neuron, a muscle fiber, or a gland cell. The

synapse between a motor neuron and a skeletal muscle fiber is the neuromuscular junction.

The parts of a chemical synapse

At a chemical synapse the two cells do not touch. A narrow gap, the synaptic cleft, separates them. It is only about 20 to 50 nanometers wide. The signal crosses this gap as a chemical, a neurotransmitter. The other kind of synapse, an electrical synapse, joins two cells directly through gap junctions and is much less common.

On the sending side, each axon terminal ends in a swelling called a synaptic end bulb. Your lab list calls it the synaptic bulb (terminal end). Inside the end bulb are many synaptic vesicles, tiny membrane sacs that store neurotransmitter. The membrane of the end bulb that faces the cleft is the presynaptic membrane. Across the cleft, the postsynaptic membrane holds the receptors that the neurotransmitter binds to. Only the presynaptic side has vesicles, so a chemical synapse passes its signal in one direction only.

How the signal crosses, in order

1. Synaptic end bulb: the nerve impulse arrives at the end of the axon terminal.
2. Synaptic vesicles fuse with the presynaptic membrane and release neurotransmitter.
3. Synaptic cleft: the neurotransmitter diffuses across the gap.
4. Postsynaptic membrane: the neurotransmitter binds receptors, and the signal passes to the next cell.

Structure	What it is
Synapse	The site where a neuron communicates with another neuron, a muscle cell, or a gland
Chemical synapse	A synapse with no direct contact; the signal crosses as neurotransmitter molecules
Synaptic end bulbs	The swellings at the tips of the axon terminals that hold the synaptic vesicles
Synaptic vesicles	The tiny sacs inside the end bulbs that store neurotransmitter
Synaptic cleft	The narrow gap between the two cells that the neurotransmitter diffuses across
Postsynaptic membrane	The membrane of the receiving cell, which holds the neurotransmitter receptors
Neurotransmitters	The chemical messengers that carry the signal across the cleft to the next cell

Three synapse types, named for the site the axon contacts

Each name starts with "axo" for the presynaptic axon and ends with the part of the next neuron it touches.

Synapse type	Where the axon lands	Notes
Axodendritic synapse	On a dendrite	The most common type
Axosomatic synapse	On a cell body (soma)	
Axoaxonic synapse	On another axon	

- Memory tip: Read the synapse names left to right. The first half is always the axon, and the second half is where it lands: dendrite, soma, or axon.
- There is a gap between the two cells. If your end bulb touches the next cell in a drawing, open up the cleft.
- Vesicles belong in the sending end bulb, not in the receiving cell.

Competency 176. Neuron classification

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

Neurons are classified in two ways. The structural classification counts how many processes leave the cell body. The functional classification asks which direction the neuron carries its signal, toward the CNS or away from it. One neuron gets a name from each system, so a touch neuron is both unipolar and sensory.

Structural classification, by number of processes

Multipolar neuron
Many dendrites and one axon. It is the most common type in the body. Motor neurons and interneurons are multipolar, and so is the neuron on your lab slide. Two multipolar neurons you will see on slides are the pyramidal cells of the cerebral cortex and the Purkinje cells of the cerebellum.

Bipolar neuron
One dendrite and one axon, leaving from opposite ends of the cell body. Bipolar neurons are rare. They are found in special sense organs: the retina of the eye, the olfactory epithelium of the nose, and the inner ear.

Unipolar neuron
One short process that leaves the cell body and then splits like the letter T into two branches. Some textbooks call it pseudounipolar. The peripheral branch starts at a receptor, and the central branch enters the CNS. Most sensory neurons are unipolar, and their cell bodies sit in the dorsal root ganglia and the cranial nerve ganglia.

Type	Processes	Example
Multipolar neuron	Many dendrites and one axon; the most common type	A motor neuron in the anterior horn of the spinal cord; pyramidal cells; Purkinje cells
Bipolar neuron	One dendrite and one axon	Neurons in the retina, the olfactory epithelium, and the inner ear
Unipolar neuron	One process that splits into a peripheral branch and a central branch	A touch or pain sensory neuron with its cell body in a dorsal root ganglion
Pyramidal cells	A large multipolar neuron with a pyramid-shaped cell body	The cerebral cortex
Purkinje cells	A large multipolar neuron with a fan of dendrites	The cerebellar cortex

Functional classification, by direction

Sensory (afferent) neurons
Carry signals from receptors toward the CNS. A neuron that carries touch from the skin of a fingertip to the spinal cord is a sensory neuron. Most are unipolar.

Motor (efferent) neurons
Carry commands away from the CNS to an effector. A motor neuron whose cell body sits in the anterior horn of the spinal cord and whose axon runs to the biceps brachii is an example. Motor neurons are multipolar.

Interneurons
Also called association neurons. They lie entirely within the CNS and connect sensory neurons to motor neurons and to each other. The neurons in the spinal cord gray matter that link the sensory and motor neurons of a withdrawal reflex are interneurons. Most neurons in the body are interneurons, and they are multipolar.

Type	Direction	Example
Sensory (afferent) neurons	Toward the CNS; most are unipolar	A touch neuron from the fingertip to the spinal cord
Motor (efferent) neurons	Away from the CNS; multipolar	An anterior horn motor neuron to the biceps brachii
Interneurons	Within the CNS, between other neurons; multipolar	Spinal cord neurons linking the sensory and motor neurons of a withdrawal reflex

- Memory tip: SAME. Sensory is Afferent, Motor is Efferent.
- For shape, count the processes leaving the cell body: many, two, or one. Bipolar has two and unipolar has one.
- In a skin-to-cord-to-muscle pathway, the sensory neuron is the unipolar one. The interneuron inside the cord and the motor neuron out to the muscle are both multipolar.

Competency 177. Neuroglia identification

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

Neuroglia, or glial cells, are the support cells of nervous tissue. They are smaller than neurons and very numerous, though textbooks differ on how many glia there are for each neuron. Unlike mature neurons, they can still divide. In the CNS they make up about half the volume of the tissue. On a slide, the many small dark nuclei around the large neuron cell bodies belong to glia. There are six types. Four are in the CNS and two are in the PNS.

The four glia of the CNS

Astrocytes
Star-shaped cells and the most numerous glia in the CNS. Their processes wrap around neurons and capillaries. They hold neurons in place, help form the blood-brain barrier, and keep the fluid around neurons at the right chemical balance. In the embryo they guide growing neurons, and after an injury they fill the damaged area with scar tissue.

Oligodendrocytes
Cells with a few long processes. Each process wraps around a segment of an axon to form myelin, so one oligodendrocyte myelinates parts of several different axons. They are the myelin-forming cells of the CNS.

Microglia
Small cells that act as phagocytes. They engulf microbes and clear away debris from dead and damaged cells. They are the CNS's own immune defense.

Ependymal cells
Cuboidal to columnar cells, many with cilia, that line the ventricles of the brain and the central canal of the spinal cord. In the choroid plexuses they help produce cerebrospinal fluid, and their cilia help it circulate.

Glial cell	What it looks like	Support role
Astrocytes	Star-shaped; the most numerous CNS glia	Support neurons, help form the blood-brain barrier, keep the chemical environment around neurons stable, guide neuron growth, form scar tissue after injury
Oligodendrocytes	A few long processes that wrap axons	Form the myelin sheaths of the CNS, each cell on several axons
Microglia	Small cells	Phagocytes that remove microbes and debris
Ependymal cells	Cuboidal to columnar, many with cilia	Line the ventricles and central canal; help produce and circulate cerebrospinal fluid

The two glia of the PNS

Schwann cells
Wrap around PNS axons. Each Schwann cell forms the myelin around one segment of a single axon. Schwann cells also enclose unmyelinated PNS axons, and they guide a damaged axon as it regrows.

Satellite cells
Flat cells that surround the cell bodies of neurons in ganglia. They support those cell bodies and regulate the exchange of materials between the cell bodies and the fluid around them.

Glial cell	What it looks like	Support role
Schwann cells	Wrapped around a segment of one axon	Form the myelin sheath around PNS axons; aid axon regeneration
Satellite cells	Flat cells around neuron cell bodies	Support the cell bodies of PNS neurons in ganglia and regulate their surroundings

- Memory tip: The two PNS glia both start with S, Schwann and Satellite. Every other glial cell is in the CNS.
- Two cells make myelin: oligodendrocytes in the CNS and Schwann cells in the PNS. One cell helps form the blood-brain barrier: the astrocyte.
- If you sort the six cells into a CNS side and a PNS side, the CNS side should have four and the PNS side two.

Competency 178. Myelination and regeneration

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

The myelin sheath and its parts

The myelin sheath is a covering made of many layers of a glial cell's plasma membrane, wrapped tightly around an axon. Because it is mostly lipid with some protein, it looks white. It insulates the axon and makes the nerve impulse travel much faster. An axon with this covering is a myelinated axon.

Myelin is laid down in segments along the axon. The short gaps between neighboring segments are the nodes of Ranvier, where the axolemma is left bare. The myelin-covered stretch between two nodes is an internode. In a myelinated axon the impulse jumps from node to node. This fast form of conduction is called saltatory conduction.

The neurolemma is the outermost layer of a Schwann cell. It holds the Schwann cell's nucleus and most of its cytoplasm, and it sits outside the myelin layers. Only Schwann cells have one, so the neurolemma is found only around PNS axons.

Structure	What it is
Myelin sheath	A multilayered covering of lipid and protein wrapped around an axon; it insulates the axon and speeds up the impulse
Nodes of Ranvier	The gaps in the myelin sheath, spaced along the axon
Internode	The myelin-covered segment between two nodes
Saltatory conduction	The fast conduction in which the impulse jumps from node to node
Neurolemma	The outer layer of a Schwann cell, holding its nucleus and cytoplasm; PNS only

PNS and CNS myelination compared

In the PNS, the myelin is made by Schwann cells. Each Schwann cell wraps one segment of one axon, so a long axon needs a row of Schwann cells along its length, with a node of Ranvier between neighboring cells. In the CNS, the myelin is made by oligodendrocytes. One oligodendrocyte sends processes to several axons and myelinates a segment of each. Oligodendrocytes have no neurolemma.

That difference decides whether a cut axon can regrow. When a PNS axon is cut, the part beyond the cut breaks down. If the cell body is intact, the Schwann cells line up and their neurolemma forms a regeneration tube that guides the regrowing axon back toward its target. In the CNS there is no neurolemma to guide regrowth. Oligodendrocytes also release substances that block regrowth, and astrocytes fill the area with scar tissue. Regeneration in the CNS is very limited.

Feature	PNS myelination	CNS myelination
Cell that makes the myelin	Schwann cell	Oligodendrocyte
Axons per glial cell	One segment of one axon	Segments of several axons
Neurolemma	Present; the outer layer of the Schwann cell	Absent
Axon regeneration	Possible if the cell body survives; the neurolemma forms a regeneration tube that guides the axon	Very limited; no neurolemma, oligodendrocytes block regrowth, and astrocytes form scar tissue

- Memory tip: Schwann cells serve one axon segment each and leave a neurolemma. Oligodendrocytes reach several axons and leave none. That is why a cut nerve in the arm can regrow and a cut tract in the spinal cord usually cannot.
- If your drawing shows one Schwann cell reaching several axons, you drew an oligodendrocyte.
- Myelination increases from birth to maturity, so an infant's nerve conduction is slower than an adult's.

What this means for patients

- Mature neurons generally do not divide, so tumors of nervous tissue rarely come from neurons. Most come from the glia, which can still divide. A glioma is a tumor of glial cells, and it is the most common primary malignant brain tumor.
- Myelin matters as much as the axon it wraps. In multiple sclerosis, the immune system destroys myelin around CNS axons, and signals slow down or fail.
- The presence of the neurolemma on Schwann cells, and its absence on oligodendrocytes, is why a cut nerve in the arm can regrow while damage in the brain or spinal cord usually cannot be repaired.
- The cell body keeps the whole neuron alive. The axon depends on the cell body for its proteins, so the part of an axon cut off from its cell body dies. An injury close to the cell body is more likely to kill the whole neuron than an injury far out along the axon.

Gross anatomy and neuronal integration

The written notes for the gross anatomy and neuronal integration lecture: nuclei, ganglia, tracts, and nerves, gray and white matter, the reflex arc and the types of reflexes, and neuronal circuits and pathways. Module 5 competencies 179 to 183, with eight worked questions.

Competency 179. CNS and PNS tissue terms

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

Four terms name groups of neurons. Two of them name clusters of neuron cell bodies, and two name bundles of axons. Which term you use depends on whether the group sits inside the CNS (the brain and spinal cord) or out in the PNS (everything else). So every one of these words answers two questions at once: cell bodies or axons, and inside or outside.

The four terms

Nucleus
A cluster of neuron cell bodies inside the CNS. The cranial nerve nuclei in the brainstem and the basal nuclei deep in the cerebrum are examples. Nuclei are made of gray matter. This meaning is different from the nucleus inside a single cell, which holds the DNA.

Ganglion (plural, ganglia)
A cluster of neuron cell bodies in the PNS. The dorsal root ganglion holds the cell bodies of sensory neurons, and the sympathetic trunk ganglia hold the cell bodies of autonomic neurons. Satellite cells surround the cell bodies in a ganglion. The basal nuclei of the brain are also called the basal ganglia, an older name that breaks this rule, since they sit in the CNS.

Tract
A bundle of axons inside the CNS. The corticospinal tract and the spinothalamic tract are examples. Tracts make up the white matter of the brain and spinal cord.

Nerve
A bundle of axons in the PNS, held together by connective tissue and supplied by blood vessels. The sciatic nerve and the vagus nerve are examples. A nerve holds axons but no cell bodies. The cell bodies of those axons sit in the CNS or in ganglia.

The same structure under two names

Structure	In the CNS	In the PNS
A cluster of cell bodies	Nucleus	Ganglion
A bundle of axons	Tract	Nerve

Term	Cell bodies or axons	CNS or PNS	Example
Nucleus	Cluster of cell bodies	CNS	A cranial nerve nucleus in the brainstem
Ganglion	Cluster of cell bodies	PNS	A dorsal root ganglion
Tract	Bundle of axons	CNS	The corticospinal tract
Nerve	Bundle of axons	PNS	The sciatic nerve

The wrappings of a nerve

On a nerve cross section, work from one axon outward. Each axon with its Schwann cell is wrapped in endoneurium. Groups of axons are bundled into a fascicle by perineurium. The whole nerve, with all its fascicles and blood vessels, is wrapped in epineurium.

Structure	What it wraps
Endoneurium	A single axon and its Schwann cell
Perineurium	One fascicle, a bundle of axons
Fascicle	A bundle of axons inside the nerve
Epineurium	The whole nerve

- Memory tip: decide first whether you are looking at cell bodies or axons. Then decide whether you are inside the CNS or outside it. Cell bodies: nucleus inside, ganglion outside. Axons: tract inside, nerve outside.
- Nucleus and ganglion are both clusters of cell bodies. The only difference between them is CNS or PNS.
- A bundle of axons inside the spinal cord is a tract, not a nerve. Nerves exist only outside the CNS. On a drawing of the cord with a nerve leaving it, the line between CNS and PNS falls at the surface of the cord where the nerve leaves, and the ganglion sits as a swelling along that nerve, outside the CNS.

Competency 180. Gray and white matter

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

Nervous tissue in the CNS sorts into two kinds by its look. Gray matter is made of neuron cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia. It is where signals are received and processed. White matter is made mostly of myelinated axons, running in bundles (the tracts), with the glia that myelinate them. It carries signals from one region to another. White matter looks white because myelin is rich in lipid.

Type	What it is made of	Where it is
Gray matter	Neuron cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia	The cerebral and cerebellar cortex and the deep nuclei of the brain; the H-shaped inner core of the spinal cord
White matter	Myelinated axons in bundles (tracts), with their glia	Beneath the cortex of the cerebrum and cerebellum (including the arbor vitae); the outer columns of the spinal cord

In the spinal cord

In the spinal cord, gray matter is on the inside and white matter is on the outside. On a cross section, the gray matter forms a shape like a butterfly or the letter H. Its parts are the posterior (dorsal) horns, the anterior (ventral) horns, and, in the thoracic and upper lumbar cord (T1 to L2), the lateral horns. The central canal runs through the middle of the gray matter. The white matter around it is divided into posterior, lateral, and anterior columns, and these columns hold the tracts.

In the brain

In the cerebrum, the arrangement is reversed. A thin outer layer of gray matter, the cerebral cortex, covers the surface, and its large neurons include the pyramidal cells. Beneath the cortex is a large mass of white matter, including the corpus callosum. Deep inside the brain, surrounded by white matter, sit islands of gray matter called deep nuclei, such as the basal nuclei of the cerebrum and the thalamus of the diencephalon.

The cerebellum has the same plan. Its outer layer, the cerebellar cortex, is gray matter and contains the Purkinje cells. Its inner white matter branches like a tree and is called the arbor vitae. In the brainstem, nuclei of gray matter are scattered among the white matter tracts.

Region	Gray matter	White matter
Spinal cord	Inside: posterior, anterior, and lateral horns around the central canal	Outside: posterior, lateral, and anterior columns
Cerebrum	Outside: the cerebral cortex; plus deep nuclei inside	Inside, beneath the cortex
Cerebellum	Outside: the cerebellar cortex	Inside: the arbor vitae
Brainstem	Nuclei scattered inside	Tracts around and between the nuclei

- Memory tip: gray is where the cell bodies are, white is where the myelin is. The two swap places between the spinal cord and the brain, so name the organ before you name the layer.
- On the models, the cord section shows a gray butterfly in the center with white matter around it, and the cerebrum slice shows a gray cortex on the outside with white matter beneath. Your shading of the cord should be the reverse of your shading of the brain.
- If you ever write cell bodies under white matter, swap the rows. White matter is white because of myelin, and myelin is on axons.

Competency 181. Reflex arc components

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

A reflex is a fast, automatic, predictable response to a stimulus. The pathway a reflex follows is the reflex arc. A reflex is fast because the arc is short and fixed. When the integration center is in the spinal cord, it is a spinal reflex. When it is in the brainstem, it is a cranial reflex. Either way, no conscious decision is needed, though the brain may become aware of the response afterward.

Every reflex arc has the same five components, in the same order. The signal comes in on the sensory side, is processed in the CNS, and goes out on the motor side.

The five components, in order

1. Receptor: detects the stimulus. It is the ending of a sensory neuron or a receptor cell next to one.
2. Sensory neuron: carries the signal to the CNS. In a spinal reflex, its cell body sits in a dorsal root ganglion, and its axon enters the spinal cord through the dorsal root.
3. Integration center: one or more synapses in the gray matter of the spinal cord or brainstem, where the signal passes to the next neuron. This is where the signal is processed, and it counts as one of the five components.
4. Motor neuron: carries the command away from the CNS. In a somatic spinal reflex, its cell body sits in the anterior horn, and its axon leaves through the ventral root.

5. Effector: the muscle or gland that carries out the response.

Trace a signal: the patellar reflex

The patellar (knee-jerk) reflex is the classic example. A tap on the patellar ligament, just below the kneecap, stretches the quadriceps femoris.

1. Receptor: muscle spindles in the quadriceps femoris detect the stretch.
2. Sensory neuron: carries the signal through the femoral nerve and dorsal roots into the spinal cord at L2 to L4.
3. Integration center: the sensory neuron synapses directly on a motor neuron in the anterior horn.
4. Motor neuron: carries the command out through the ventral root and the femoral nerve.
5. Effector: the quadriceps femoris contracts, and the leg extends at the knee.

Kinds of receptors

Receptors are named for the stimulus they detect.

Receptor	Stimulus it detects
Mechanoreceptors	Touch, pressure, and stretch
Thermoreceptors	Temperature
Photoreceptors	Light
Chemoreceptors	Chemicals
Nociceptors	Pain

- Memory tip: the arc runs in, through, and out: Receptor, Sensory neuron, Integration center, Motor neuron, Effector. The sensory neuron enters by the dorsal root, and the motor neuron leaves by the ventral root.
- The sensory neuron comes before the integration center and the motor neuron comes after it. If your chain has only four boxes, you left out the synapse in the cord.
- A clinician who taps below the kneecap is testing the whole arc at once. A missing reflex points to a break somewhere along that pathway: the receptor, the sensory neuron, the segment of cord, the motor neuron, or the muscle.

Competency 182. Types of reflexes

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

Reflexes are sorted in three ways: by the effector, by the number of synapses in the integration center, and by whether the reflex is inborn or learned. Each pair answers a different question, so one reflex fits into all three. The patellar reflex, for example, is somatic, monosynaptic, and innate.

Somatic versus autonomic: which effector?

A somatic reflex has skeletal muscle as its effector. The patellar reflex is a somatic reflex. An autonomic reflex, also called a visceral reflex, has smooth muscle, cardiac muscle, or a gland as its effector. In the pupillary light reflex, bright light makes the pupils constrict. The effector is the smooth muscle of the iris, and the integration center is in the midbrain.

Monosynaptic versus polysynaptic: how many synapses?

In a monosynaptic reflex, the sensory neuron synapses directly on the motor neuron, so there is only one synapse in the integration center. The stretch reflex, such as the patellar reflex, is monosynaptic. In a polysynaptic reflex, one or more interneurons sit between the sensory and motor neurons, so there are two or more synapses. The withdrawal reflex, pulling your hand away from a hot stove, is polysynaptic. A monosynaptic reflex is the faster of the two, because each synapse adds a short delay.

Innate versus learned: born with it or practiced?

An innate reflex is present from birth and does not have to be learned. Blinking when something touches the eye is an innate reflex. A learned reflex, also called an acquired reflex, is built through practice and repetition until it runs automatically. An experienced driver braking when the car ahead stops suddenly, or catching a suddenly dropped object, are learned reflexes.

Reflex type	Basis of the name	Example
Somatic reflex	The effector is skeletal muscle	The patellar, or knee-jerk, reflex
Autonomic reflex	The effector is smooth muscle, cardiac muscle, or a gland	The pupillary light reflex
Monosynaptic reflex	Just one synapse in the integration center	The stretch reflex, such as the patellar reflex
Polysynaptic reflex	Two or more synapses, with interneurons	The withdrawal reflex
Innate reflex	Present from birth, not learned	The blink reflex
Learned reflex	Built through practice and repetition	Braking when the car ahead stops; catching a dropped object

- Memory tip: each pair asks one question. Somatic or autonomic: which effector? Mono or poly: how many synapses? Innate or learned: born with it or practiced?
- The withdrawal reflex runs through interneurons, so it is polysynaptic, not monosynaptic.
- The pupillary light reflex is autonomic because its effector is smooth muscle in the iris, not skeletal muscle. Do not file it under somatic.

Competency 183. Neuronal pools and pathways

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

Neurons in the CNS work in groups. A neuronal pool is a group of interconnected neurons that process the same kind of information together. A pool can hold thousands of neurons. The pattern of connections inside and between pools is called a neural circuit, and the pattern shapes how a signal is handled.

Convergence and divergence

In a convergence circuit, several presynaptic neurons synapse on one postsynaptic neuron. The one neuron can then combine input from many sources. A motor neuron in the anterior horn of the spinal cord is an example. It receives input from several regions of the brain and from sensory neurons, and its single output goes to a skeletal muscle.

In a divergence circuit, one presynaptic neuron branches and synapses on many postsynaptic neurons. One signal spreads out to reach many cells. A single motor neuron in the brain, for example, can stimulate many motor neurons in the spinal cord, which in turn reach many muscle fibers.

Signals can also travel in series or in parallel. In serial processing, the signal passes along a single chain of neurons, as in a reflex arc. In parallel processing, the same signal travels along several pathways at the same time.

Structure	What it is
Neuronal pool	A functional group of interconnected neurons in the CNS that processes information together
Convergence	Many neurons feed their signals onto one neuron; the signal travels in toward one neuron
Divergence	One neuron spreads its signal to many neurons; the signal travels out to many neurons
Serial processing	A signal passing along a single chain of neurons, as in a reflex arc
Parallel processing	A signal sent along several pathways at the same time
Integration	The combining of all the incoming signals so that a neuron produces a single response

Ascending pathways, descending pathways, and decussation

Long-distance signals travel through the CNS in tracts that run in the white matter. A tract's name usually tells you where it starts and where it ends. The spinothalamic tract runs from the spinal cord to the thalamus.

Ascending pathways carry sensory information up the spinal cord to the brain. The spinothalamic tract, which carries pain and temperature, is an ascending pathway. Descending pathways carry motor commands down from the brain to the spinal cord. The corticospinal tract, which runs from the cerebral cortex to the spinal cord and controls voluntary movement, is a descending pathway.

Decussation is the crossing of a pathway from one side of the CNS to the other. The main sensory and motor pathways cross, so each side of the brain senses and controls the opposite side of the body. Most corticospinal axons cross in the medulla oblongata, at the pyramidal decussation. Spinothalamic axons cross in the spinal cord, close to where the sensory fibers enter.

Structure	What it is	Which way the signal travels	Example
Ascending pathways	Tracts that carry sensory information up the spinal cord to the brain	Up the cord to the brain	The spinothalamic tract
Descending pathways	Tracts that carry motor commands down from the brain to the spinal cord	Down from the brain to the body	The corticospinal tract
Decussation	The crossing of a pathway to the opposite side of the CNS	Across the midline	The pyramidal decussation in the medulla oblongata

- Memory tip: converge means come together onto one. Diverge means spread out from one. Ascending carries sensation up, descending carries commands down, and decussation is the crossover.
- On a drawing, convergence arrows point in toward one neuron and divergence arrows spread out from one. If both circuits look the same, one is wrong.
- Because the pathways cross, the left side of the brain controls the right side of the body. On a body outline, the half the left hemisphere controls is the right half.

What this means for patients

- A tap below the kneecap triggers the patellar reflex, and a clinician watches the leg kick. That one test checks the whole reflex arc: the receptor, the sensory and motor neurons, and the segment of spinal cord in between. A missing reflex points to a break somewhere along that pathway.
- Because the main sensory and motor pathways decussate, the right side of the brain senses and controls the left side of the body. This is why a stroke in one cerebral hemisphere causes weakness on the opposite side of the body.
- On an MRI scan, gray matter and white matter show up as different shades because myelin is rich in lipid. That contrast lets a radiologist see the cortex, the deep nuclei, and the white matter tracts, and spot damage such as the patches of lost myelin in multiple sclerosis.

The spinal cord, the peripheral nervous system, and the cranial nerves

The written notes for the spinal cord, peripheral nervous system, and cranial nerve lectures: the outside and inside of the cord, roots and spinal nerves, nerve wrappings, rami, sensory receptors, and the twelve cranial nerves. Module 5 competencies 184 to 190, with ten worked questions.

Competency 184. Spinal cord external anatomy

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

The spinal cord is the part of the central nervous system that sits inside the vertebral canal. It is continuous with the medulla oblongata at the foramen magnum. In an adult it is about 42 to 45 cm (16 to 18 inches) long. It carries signals between the brain and the body, and it is the center for spinal reflexes.

The cord is shorter than the vertebral column. The cord stops growing in length at about 4 to 5 years of age, while the vertebral column keeps growing. So in an adult the cord ends near the disc between L1 and L2, well above the end of the vertebral column. Below that level the vertebral canal holds only nerve roots and a thin anchoring thread.

The two enlargements

The width of the cord changes along its length. It is widest in the cervical region and thinner through the thoracic region. It swells in two places, where the nerves to the limbs leave. A limb has many muscles and a large area of skin, so these segments need more neurons.

Cervical enlargement

In the lower neck, from about C4 to T1. The nerves to the upper limbs (the brachial plexus) leave here.

Lumbar enlargement

Behind the lower thoracic vertebrae, about T9 to T12. Texts give slightly different levels, and some place it a little lower, near T11 to L1. The nerves to the lower limbs (the lumbar and sacral plexuses) leave here.

Where the cord ends

Conus medullaris

Below the lumbar enlargement the cord tapers to a point. This cone-shaped end is the conus medullaris. It sits near the disc between L1 and L2.

Filum terminale

A thin strand of pia mater that runs on from the tip of the conus medullaris. It runs down the lower vertebral canal and anchors the cord to the coccyx.

Cauda equina

The lumbar, sacral, and coccygeal nerve roots leave the cord high up, at or above the conus. They must travel down the vertebral canal to reach their exits lower in the column. Together these long roots look like a horse's tail, so they are called the cauda equina. The filum terminale runs down the middle of them. The cauda equina is nerve roots, not cord.

The two grooves

Two grooves run the full length of the cord and split it into right and left halves. The anterior median fissure is on the front. It is deep and wide. The posterior median sulcus is on the back. It is shallow and narrow. On a model or a cross-section, the wide groove tells you which side is anterior.

Structure	Where it is	What it is
Cervical enlargement	About C4 to T1	A widening where the nerves of the upper limbs leave the cord
Lumbar enlargement	Behind the lower thoracic vertebrae, about T9 to T12 (levels vary a little by text)	A widening where the nerves of the lower limbs leave the cord
Conus medullaris	Near the disc between L1 and L2	The tapered, cone-shaped end of the spinal cord
Filum terminale	From the tip of the conus medullaris down to the coccyx	A thread of pia mater that anchors the cord to the coccyx
Cauda equina	In the vertebral canal below the conus medullaris	The bundle of lumbar, sacral, and coccygeal nerve roots running down to their exits
Anterior median fissure	Front of the cord, full length	The deep, wide groove down the front
Posterior median sulcus	Back of the cord, full length	The shallow, narrow groove down the back

- Memory tip: a fissure is deep and a sulcus is shallow. The deep fissure is on the front of the cord, so the anterior side is the side with the wide groove.

What this means for patients

- The spinal cord ends at the conus medullaris near the L1 to L2 disc, but the vertebral column runs much lower. Below the conus, only the cauda equina and the filum terminale lie in the cerebrospinal fluid. That is why a lumbar puncture is done below the end of the cord, usually between L3 and L4 or between L4 and L5. A needle can draw cerebrospinal fluid there without reaching the cord itself, and the loose roots of the cauda equina drift aside.
- The posterior root ganglion holds the sensory neuron cell bodies for one spinal nerve. The varicella zoster virus (the chickenpox virus) can stay dormant in these ganglia for decades. When it reactivates as shingles, the rash follows the strip of skin served by that one spinal nerve.
- A complete spinal cord injury cuts the ascending tracts and descending tracts. These tracts carry the signals between the brain and the body below the injury. So sensation and movement are lost below the level of the injury, while function above it is spared.

Competency 185. Roots, rootlets, and root ganglion

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

Each segment of the spinal cord connects to a pair of spinal nerves, one on each side. On each side, two roots join the cord to the nerve: a posterior root and an anterior root. Each root starts as a row of small, fan-shaped bundles of axons called rootlets. The posterior rootlets attach in a line along the back of the cord. The anterior rootlets leave in a line along the front of the cord. The rootlets of one segment merge to form one root.

Posterior (dorsal) root
The sensory root. It carries sensory axons from receptors in the body into the cord. Along the posterior root is a swelling, the posterior root ganglion.

Posterior (dorsal) root ganglion
The swelling on the posterior root. It holds the cell bodies of the sensory neurons. These are unipolar neurons: one branch of the axon runs out to a receptor and the other runs into the cord. The sensory cell bodies are here, outside the cord, not in the cord or the rootlets.

Anterior (ventral) root
The motor root. It carries motor axons out of the cord. Somatic motor axons go to skeletal muscle. In the T1 to L2 and S2 to S4 segments, the anterior root also carries autonomic axons that lead toward smooth muscle, cardiac muscle, and glands. The cell bodies of these motor neurons sit inside the cord, in the gray matter. That is why the anterior root has no ganglion.

Spinal nerve
Just outside the cord the two roots join to form the spinal nerve. It holds both sensory and motor axons, so every spinal nerve is a mixed nerve.

Intervertebral foramen
The opening the spinal nerve passes through to leave the vertebral canal. It lies between two neighboring vertebrae, formed by the notch of the vertebra above and the notch of the vertebra below. The posterior root ganglion sits in or right next to this foramen.

Structure	Where it is	What it carries
Rootlets	Rows of small bundles along the back and the front of the cord	Posterior rootlets carry sensory axons in; anterior rootlets carry motor axons out
Posterior (dorsal) root	Back of the cord, between the posterior rootlets and the spinal nerve	Sensory axons only
Posterior (dorsal) root ganglion	The swelling on the posterior root, in or beside the intervertebral foramen	Cell bodies of the sensory neurons
Anterior (ventral) root	Front of the cord, between the anterior rootlets and the spinal nerve	Motor axons only
Spinal nerve	Where the two roots join, at the intervertebral foramen	Mixed: sensory and motor axons
Intervertebral foramen	The opening between two neighboring vertebrae	The exit for the spinal nerve; it holds the posterior root ganglion

A simple spinal reflex uses every one of these structures

Follow the signal in and back out. The sensory chain runs toward the cord, and the motor chain runs away from it. The spinal nerve is the one structure both chains travel in, which is why it is called mixed.

1. Receptor in the skin or a muscle detects a stimulus.
2. Spinal nerve: the sensory axon travels toward the cord.
3. Posterior root ganglion: the signal passes the sensory neuron's cell body.
4. Posterior root and rootlets: the sensory axon enters the back of the cord.
5. Gray matter of the cord: the sensory axon synapses on an interneuron or a motor neuron.
6. Anterior rootlets and anterior root: the motor axon leaves the front of the cord.
7. Spinal nerve: the motor axon travels out with the sensory axons.
8. Skeletal muscle: the effector contracts.

- Memory tip: posterior root, sensory in, and it has a ganglion. Anterior root, motor out, and it has no ganglion. If a question puts cell bodies on a root, it is the posterior root.
- Only the posterior root has a ganglion. If your drawing puts a swelling on the anterior root, take it off.

Competency 186. Spinal cord internal anatomy

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

A cross-section of the spinal cord shows gray matter in the center and white matter around it. This is the reverse of the cerebrum, where gray matter forms the outer cortex. The gray matter is shaped like a butterfly or the letter H. Gray matter holds neuron cell bodies, dendrites, unmyelinated axons, and neuroglia. White matter is mostly myelinated axons. The fatty myelin is what makes it look white.

The gray horns

Each half of the gray matter sticks out into projections called horns. Each horn holds a different kind of neuron.

Horn	Where it is	Contents
Posterior (dorsal) horn	Points toward the back of the cord; thin and narrow	Incoming sensory axons from the posterior root, and the cell bodies of interneurons
Lateral horn	A small point out to the side, between the posterior and anterior horns; only in the T1 to L2 segments	Cell bodies of autonomic (sympathetic) motor neurons
Anterior (ventral) horn	Points toward the front of the cord; wide and rounded	Cell bodies of somatic motor neurons, whose axons leave in the anterior root to skeletal muscle

- The lateral horn is found only from T1 to L2 because only those segments send out sympathetic neurons. The S2 to S4 segments hold parasympathetic neurons in a similar spot in the gray matter, but they do not form a visible horn. On a sketch, the lateral horn belongs on a thoracic section, not a cervical one.

The central canal and the commissures

Gray commissure
The two halves of the cord are joined across the middle by two bridges, called commissures. The gray commissure is the crossbar of the H. It joins the left and right gray matter.

Central canal
Runs down the middle of the gray commissure. It is a tiny channel filled with cerebrospinal fluid and lined by ependymal cells. It runs the length of the cord and is continuous with the fourth ventricle above. On a section it is the small hole at the very center.

Anterior white commissure
A band of white matter just in front of the gray commissure, toward the anterior median fissure. Here axons cross from one side of the cord to the other.

The white columns

The white matter on each side is divided into three columns (also called funiculi). The horns of the gray matter mark their borders.

Posterior column
Between the posterior median sulcus and the posterior horn.

Lateral column
On the side, between the posterior horn and the anterior horn.

Anterior column
Between the anterior horn and the anterior median fissure.

Each column is made of tracts. A tract is a bundle of axons in the CNS that share the same origin and destination. Ascending tracts carry sensory signals up to the brain. Descending tracts carry motor signals down from the brain. The posterior columns hold mainly ascending tracts. The lateral and anterior columns hold both ascending and descending tracts.

Location	Cluster of cell bodies	Bundle of axons
Central nervous system	Nucleus	Tract
Peripheral nervous system	Ganglion	Nerve

Telling the cord segments apart

Cross-sections from different levels of the cord do not look the same. The size, the shape, and the amount of gray and white matter change from top to bottom.

Segment	Distinguishing features
Cervical	Large diameter, oval shape, and a large amount of white matter; no lateral horn
Thoracic	Small diameter and a small amount of gray matter; has a small lateral horn
Lumbar	Almost circular, with large anterior horns; less white matter than the cervical segments
Sacral	Relatively small, with more gray matter than white matter
Coccygeal	The smallest; looks like the lower sacral segments

- White matter is thickest near the top of the cord. At the cervical level the ascending tracts from the whole body have gathered, and the descending tracts have not yet left. Lower down, fewer tracts are still running, so the white matter thins. Gray matter is largest in the enlargements, where the limbs need many motor neurons.
- Memory tip: to orient a slide, find the anterior median fissure first. It is the wide groove, and the wide anterior horns sit on either side of it. The thin posterior horns point toward the back of the cord, one on each side of the shallow posterior median sulcus.

Competency 187. Nerve connective tissue wrappings

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

A nerve is an organ of the peripheral nervous system. It is a bundle of many axons held together by connective tissue, with blood vessels running through it. Some of its axons are myelinated and some are not. A nerve is the PNS version of a tract in the CNS.

The smallest unit in a nerve is the single axon, also called a nerve fiber. In the PNS, myelinated axons are wrapped by Schwann cells. The small gaps between neighboring Schwann cells are the nodes of Ranvier.

Three layers at three levels, inside to outside

Endoneurium
: A thin layer of loose connective tissue around each single axon, outside its Schwann cell. The innermost wrapping.

Fascicle
: A bundle of axons. Axons are grouped into fascicles, and each fascicle has its own wrapping.

Perineurium
: The middle wrapping. It wraps each fascicle.

Epineurium
: A tough layer of dense irregular connective tissue around the whole nerve. The outermost wrapping. It also fills the spaces between the fascicles and carries the nerve's blood vessels.

Structure	What it is	Level it wraps
Axon	A single nerve fiber, with its Schwann cell if myelinated	Wrapped by the endoneurium
Endoneurium	The innermost wrapping, thin and delicate	One axon
Fascicle	A bundle of axons	Wrapped by the perineurium
Perineurium	The middle wrapping	One fascicle
Epineurium	The outermost wrapping, thick and tough	The whole nerve

- On a nerve cross-section, the thick outer ring is the epineurium. The round clusters inside it are fascicles, each outlined by a thin ring of perineurium. The tiny circles packed inside a fascicle are axons, each surrounded by endoneurium.
- Endo is inside and epi is outside. If your endoneurium is around the whole nerve, the labels are flipped. The perineurium goes around a fascicle, a group of axons, not around one axon.
- Memory tip: endo means within, peri means around, and epi means upon. Skeletal muscle uses the same pattern: endomysium around one fiber, perimysium around a fascicle, epimysium around the whole muscle.

What this means for patients

- Because a nerve is wrapped in connective tissue, it has some protection and some ability to heal. The epineurium and perineurium hold a cut nerve's shape. A surgeon stitches these layers to line up the cut ends, fascicle to fascicle, and repair the nerve.

Competency 188. Spinal nerves and rami

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

Thirty one pairs of spinal nerves leave the spinal cord. Each one is named for the region of the vertebral column where it leaves and is numbered within that region. Every spinal nerve is a mixed nerve, carrying both sensory and motor axons.

Region	Number of pairs	Names
Cervical	8	C1 to C8
Thoracic	12	T1 to T12
Lumbar	5	L1 to L5
Sacral	5	S1 to S5
Coccygeal	1	Co1
Total	31	

- There are 8 cervical nerves but only 7 cervical vertebrae. C1 leaves between the occipital bone and the atlas. C1 to C7 each leave above the vertebra with the same number. C8 leaves between C7 and T1. From T1 down, each nerve leaves below the vertebra with the same number.

The branches of a spinal nerve

A spinal nerve is very short. Just after it passes through the intervertebral foramen, it divides into branches. The two large branches are the rami (singular, ramus). Both rami are mixed, carrying sensory and motor axons.

Posterior (dorsal) ramus
Turns toward the back. It supplies the deep muscles of the back and the skin over the back of the neck and trunk. Damage here stays in the back.

Anterior (ventral) ramus
The larger branch. It runs forward and around the body. It supplies the muscles and skin of the front and sides of the trunk and all of the upper and lower limbs. Most anterior rami join into nerve plexuses. The anterior rami of T2 to T12 are the exception and run between the ribs as the intercostal nerves.

Meningeal branch
Turns back in through the intervertebral foramen. It supplies the vertebrae, the vertebral ligaments, the blood vessels of the cord, and the meninges.

Rami communicantes
Small branches that link the spinal nerve to a ganglion of the sympathetic trunk. They carry sympathetic axons to and from the trunk.

Branch	Territory it supplies	What it carries
Posterior (dorsal) ramus	Deep muscles of the back and the skin of the back of the neck and trunk	Mixed: sensory and motor axons
Anterior (ventral) ramus	Muscles and skin of the front and sides of the trunk and the limbs; forms the plexuses	Mixed: sensory and motor axons
Meningeal branch	Vertebrae, vertebral ligaments, blood vessels of the cord, and the meninges	Mostly sensory axons, plus sympathetic axons to the blood vessels
White ramus communicans	Runs from the spinal nerve into a sympathetic trunk ganglion; only T1 to L2	Myelinated preganglionic sympathetic axons
Gray ramus communicans	Runs from a sympathetic trunk ganglion back to the spinal nerve; on every spinal nerve	Unmyelinated postganglionic sympathetic axons

White and gray rami communicantes

Each spinal nerve has one or more rami communicantes, and they come in two kinds. The white ramus communicans carries sympathetic axons from the spinal nerve into the sympathetic trunk ganglion (sympathetic chain ganglion). These axons are myelinated, which makes this ramus look white. White rami are found only on the T1 to L2 spinal nerves, because only those cord segments have lateral horns.

The gray ramus communicans carries axons the other way, from the sympathetic trunk ganglion back into the spinal nerve. These axons are unmyelinated, so this ramus looks gray. Every spinal nerve has a gray ramus. Its axons ride out in the posterior and anterior rami to the sweat glands, the arrector pili muscles, and the blood vessels of the skin and body wall in that nerve's territory.

- The sympathetic trunk ganglia sit in a chain on each side of the vertebral column. Axons link them into a long cord called the sympathetic trunk. Competency 193 covers the trunk and the rest of the sympathetic pathway.
- The anterior ramus forms the plexuses, not the posterior ramus.
- Memory tip: white means myelinated and goes in to the ganglion; it is found only from T1 to L2. Gray means unmyelinated and comes back out to the spinal nerve; it is found on all 31 pairs.

What this means for patients

- The two rami serve different territories. The anterior rami supply the limbs and the front and sides of the trunk, so damage to the anterior rami or the plexuses they form shows up in a limb. A posterior ramus supplies only the deep back muscles and the skin over them, so damage there stays in the back.

Competency 189. Sensory receptor classification

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

A sensory receptor is a structure that detects a change, called a stimulus, and starts a signal in a sensory neuron. Receptors are classified in three ways: by the kind of stimulus they detect, by where they are, and by how they are built. Every receptor gets one label from each scheme. For example, a muscle spindle is a mechanoreceptor (stimulus), a proprioceptor (location), and an encapsulated receptor (structure).

By stimulus

Type	Stimulus it detects
Mechanoreceptors	Mechanical force, such as touch, pressure, vibration, and stretch
Thermoreceptors	Changes in temperature
Photoreceptors	Light; found in the retina of the eye
Chemoreceptors	Chemicals in solution, such as the molecules of smell and taste
Nociceptors	Stimuli that could damage tissue; they produce the sensation of pain

By location

Type	Where it is and what it monitors
Exteroceptors	At or near the body surface; they respond to stimuli from outside the body, such as touch, temperature, and the special senses. Tactile corpuscles are an example
Interoceptors	Also called visceroceptors; inside the body, in the organs and blood vessels. Free nerve endings in the organs are an example
Proprioceptors	In skeletal muscles, tendons, and joint capsules; they report the position and movement of the body. Muscle spindles are an example

By structure

Some receptors are the bare ends of a sensory neuron's dendrites. Others have a capsule of connective tissue around the nerve ending. A third group uses a separate receptor cell that passes the signal on to a sensory neuron.

Type	How it is built	Examples
Free (nonencapsulated) nerve endings	Bare dendrite tips with no capsule	Free nerve endings, tactile (Merkel) discs, and hair follicle receptors
Encapsulated nerve endings	A dendrite wrapped in a capsule of connective tissue	Tactile corpuscles, lamellar corpuscles, bulbous corpuscles, muscle spindles, and tendon organs
Specialized receptor cells	A separate cell that detects the stimulus and signals a sensory neuron	Photoreceptors of the retina, hair cells of the inner ear, and taste receptor cells

The named receptors

The receptors below are the ones to know by name. Most are in the skin or in the muscles and joints. The skin receptors are exteroceptors; the muscle spindle, tendon organ, and joint receptors are proprioceptors.

Receptor	Location	Stimulus	Class (stimulus, location, structure)
Free nerve endings	Epithelium and connective tissue throughout the body	Pain, temperature, and some crude touch	Nociceptor, thermoreceptor, or mechanoreceptor; exteroceptor or interoceptor; nonencapsulated
Tactile (Merkel) discs	The deepest layer of the epidermis (stratum basale)	Light touch	Mechanoreceptor; exteroceptor; nonencapsulated
Hair follicle receptors	Wrapped around the base of hair follicles	Bending of a hair (light touch)	Mechanoreceptor; exteroceptor; nonencapsulated
Tactile (Meissner) corpuscles	The dermal papillae of hairless skin, such as the fingertips, palms, soles, and lips	Light, fine (discriminative) touch	Mechanoreceptor; exteroceptor; encapsulated
Lamellar (Pacinian) corpuscles	The deep dermis and the hypodermis; also joint capsules	Deep pressure and vibration	Mechanoreceptor; exteroceptor in the skin, proprioceptor in joint capsules; encapsulated
Bulbous (Ruffini) corpuscles	The deep dermis, the hypodermis, and joint capsules	Steady deep pressure and stretching of the skin	Mechanoreceptor; exteroceptor in the skin, proprioceptor in joint capsules; encapsulated
Muscle spindles	Inside skeletal muscles, among the muscle fibers	Stretch of the muscle	Mechanoreceptor; proprioceptor; encapsulated
Tendon organs	In tendons, near where the muscle joins the tendon	Tension the muscle puts on its tendon	Mechanoreceptor; proprioceptor; encapsulated
Joint kinesthetic receptors	The capsules of synovial joints	Joint position and movement	Mechanoreceptor; proprioceptor; a mix of encapsulated and free endings

- Free nerve endings are the only receptors in the table that sense pain. Everything else in the table is a mechanoreceptor.
- The tactile corpuscle sits high, in the dermal papillae, and the lamellar corpuscle sits deep. If your drawing puts them at the same depth, move one.
- Memory tip: lamellar means layered. On a slide a lamellar corpuscle looks like a cut onion, with many layers of capsule around one dendrite, and it sits deep. A tactile corpuscle is small and sits up in a dermal papilla, close to the surface, where it senses light touch.

What this means for patients

- Receptors are not spread evenly. The fingertips and lips are crowded with tactile corpuscles and Merkel discs, which is why those areas feel fine detail so well. The skin of the back has far fewer, so it senses touch only crudely.

Competency 190. Cranial nerves

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

Twelve pairs of cranial nerves attach to the brain. They are part of the peripheral nervous system, even though they attach to the brain rather than the cord. They are numbered with Roman numerals, I to XII, from front to back in the order they attach to the brain. CN I and CN II attach to the forebrain. CN III to CN XII attach to the brainstem, except that most of CN XI comes from the upper cervical spinal cord.

The names in order

1. Olfactory (CN I)
2. Optic (CN II)
3. Oculomotor (CN III)
4. Trochlear (CN IV)
5. Trigeminal (CN V)
6. Abducens (CN VI)
7. Facial (CN VII)
8. Vestibulocochlear (CN VIII)
9. Glossopharyngeal (CN IX)
10. Vagus (CN X)
11. Accessory, also called spinal accessory (CN XI)
12. Hypoglossal (CN XII)

- Memory tip for the order of the names: On Old Olympus' Towering Tops A Finn And German Viewed Some Hops. The first letters stand for olfactory, optic, oculomotor, trochlear, trigeminal, abducens, facial, vestibulocochlear, glossopharyngeal, vagus, spinal accessory, and hypoglossal.

Sensory, motor, or mixed

Each cranial nerve is classed by the kind of axons it carries. A sensory nerve carries signals only into the brain. A motor nerve carries signals mainly out to muscles. A mixed nerve carries both.

Sensory

CN I, CN II, and CN VIII. These carry the special senses of smell, vision, hearing, and balance.

Motor

CN III, CN IV, CN VI, CN XI, and CN XII.

Mixed
CN V, CN VII, CN IX, and CN X. Only these four are both sensory and motor; count your Bs on a drawing.

- Four cranial nerves also carry parasympathetic motor axons to smooth muscle and glands: CN III, CN VII, CN IX, and CN X.
- Memory tip for the sensory, motor, or both pattern in order: Some Say Marry Money But My Brother Says Big Brains Matter Most.

The twelve cranial nerves: class, function, and skull opening

Number	Name	Class	Primary function	Skull opening
CN I	Olfactory	Sensory	Smell	Olfactory foramina in the cribriform plate of the ethmoid
CN II	Optic	Sensory	Vision	Optic canal of the sphenoid
CN III	Oculomotor	Motor	Moves four of the six eye muscles, raises the upper eyelid, constricts the pupil, and focuses the lens	Superior orbital fissure
CN IV	Trochlear	Motor	Moves the superior oblique muscle of the eye; the smallest cranial nerve	Superior orbital fissure
CN V	Trigeminal	Mixed	Sensation from the face, scalp, teeth, and the mouth and nose; moves the muscles of chewing; the largest cranial nerve, with three branches	Superior orbital fissure (V1, ophthalmic), foramen rotundum (V2, maxillary), and foramen ovale (V3, mandibular)
CN VI	Abducens	Motor	Moves the lateral rectus muscle of the eye	Superior orbital fissure
CN VII	Facial	Mixed	Moves the muscles of facial expression; taste from the front two thirds of the tongue; tears from the lacrimal gland and saliva from the submandibular and sublingual glands	Internal acoustic meatus, then out through the stylomastoid foramen
CN VIII	Vestibulocochlear	Sensory	Hearing (cochlear branch) and balance (vestibular branch)	Internal acoustic meatus
CN IX	Glossopharyngeal	Mixed	Taste and general sensation from the back third of the tongue and the pharynx; helps with swallowing; saliva from the parotid gland	Jugular foramen
CN X	Vagus	Mixed	Parasympathetic supply to the thoracic and abdominal organs; sensation from those organs; muscles of the pharynx and larynx for swallowing and voice	Jugular foramen
CN XI	Accessory	Motor	Moves the sternocleidomastoid and trapezius muscles	Jugular foramen (its spinal part first enters the skull through the foramen magnum)
CN XII	Hypoglossal	Motor	Moves the tongue	Hypoglossal canal

The openings and the structures along the way

The olfactory nerve is made of many small bundles of axons. They start in the lining of the nasal cavity and pass up through the olfactory foramina in the cribriform plate. They end in the olfactory bulb on the underside of the frontal lobe. From the bulb, the olfactory tract carries the signal back to the olfactory areas of the cerebrum.

The optic nerve leaves the back of the eye and passes through the optic canal. The two optic nerves meet at the optic chiasm, in front of the pituitary gland. Here some axons from each eye cross to the other side. Past the chiasm, the axons continue as the optic tract to the thalamus.

Three openings each carry more than one nerve. Learn them as three groups.

Shared opening	Nerves that pass through it	Where it leads
Superior orbital fissure	CN III, CN IV, V1 (ophthalmic branch of CN V), and CN VI	Into the orbit
Internal acoustic meatus	CN VII and CN VIII	Into the temporal bone. The facial nerve then runs through the temporal bone and comes out at the stylomastoid foramen
Jugular foramen	CN IX, CN X, and CN XI	Out of the skull

- Only V1 of the trigeminal passes through the superior orbital fissure. V2 uses the foramen rotundum and V3 the foramen ovale.

Where each cranial nerve attaches to the brain

Part of the brain	Cranial nerves attached
Cerebrum (olfactory bulb)	CN I
Diencephalon (through the optic chiasm)	CN II
Midbrain	CN III and CN IV (CN IV is the only cranial nerve that leaves the back of the brainstem; on an inferior view it is the thin nerve curving around the side)
Pons	CN V, large, on the side of the pons
Pontomedullary junction (where the pons meets the medulla)	CN VI, CN VII, and CN VIII, with CN VI nearest the midline
Medulla oblongata	CN IX, CN X, and CN XII
Upper cervical spinal cord, with a small part from the medulla	CN XI

The nerves to the eye muscles

Three cranial nerves move the eye and the upper eyelid. The abducens nerve supplies the lateral rectus, which abducts the eye (turns it outward). The trochlear nerve supplies the superior oblique, whose tendon runs through a pulley called the trochlea. The oculomotor nerve supplies all the rest.

Muscle	Nerve
Lateral rectus	CN VI, abducens
Superior oblique	CN IV, trochlear
Medial rectus	CN III, oculomotor
Superior rectus	CN III, oculomotor
Inferior rectus	CN III, oculomotor
Inferior oblique	CN III, oculomotor
Levator palpebrae superioris (raises the upper eyelid)	CN III, oculomotor

- Memory tip: for the eye muscles, use LR6 SO4. Lateral rectus is CN VI, superior oblique is CN IV, and all the rest are CN III.
- Memory tip: for the shared openings, learn three groups. III, IV, V1, and VI through the superior orbital fissure; VII and VIII through the internal acoustic meatus; IX, X, and XI through the jugular foramen.

What this means for patients

- Each cranial nerve has its own job, so a clinician can test the nerves one at a time. A weak or missing function points to the nerve that carries it. The nerve's attachment to the brain and its path through the skull then point to where the problem may be.
- The openings group nerves together. The jugular foramen carries CN IX, X, and XI side by side, so a tumor or fracture there can affect all three at once. The result is a recognizable cluster of swallowing, voice, and shoulder signs.
- Two common nerve problems are cranial. Bell's palsy is a sudden weakness of CN VII that makes one side of the face droop. Trigeminal neuralgia is a stabbing facial pain along the branches of CN V. The path of each nerve predicts where each one shows up.

The nerve plexuses and the autonomic nervous system

The written notes for the nerve plexuses and autonomic nervous system lectures: the five spinal nerve plexuses and the regions they serve, the somatic and autonomic motor pathways, and the two autonomic divisions with their ganglia and plexuses. Module 5 competencies 191 to 193, with eight worked questions.

Competency 191. Nerve plexuses

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

There are 31 pairs of spinal nerves: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal. The anterior rami of most of them do not run straight to their targets. Instead the anterior rami of neighboring spinal nerves join, swap axons, and branch again to form a network called a nerve plexus. Inside the plexus the axons are sorted and regrouped, and they leave in named nerves. Because of this regrouping, each nerve that leaves a plexus carries axons from several spinal nerves, and each body region gets axons from more than one spinal nerve. Damage to one spinal nerve weakens a region but does not usually wipe out all function there.

The thoracic region is the exception. The anterior rami of T2 to T12 do not form plexuses. They run straight out between the ribs as the intercostal nerves (T12 runs below the last rib, so it is called the subcostal nerve). They supply the intercostal muscles, the muscles of the abdominal wall, and the skin of the chest and abdomen. T1 is the one thoracic nerve that joins a plexus: most of it goes into the brachial plexus.

The five plexuses at a glance

Plexus	Formed by (anterior rami)	Main nerves	Region served
Cervical plexus	C1 to C4, with a small contribution from C5	Phrenic nerve (C3 to C5) and small nerves to the neck	The skin and muscles of the neck and upper shoulder; the phrenic nerve is the motor supply to the diaphragm
Brachial plexus	C5 to C8 and T1	Axillary, musculocutaneous, radial, median, and ulnar nerves	The shoulder and the entire upper limb
Lumbar plexus	L1 to L4	Femoral and obturator nerves	The lower abdominal wall, the external genitals, and the front and inner (medial) thigh
Sacral plexus	L4 to L5 and S1 to S4	Sciatic nerve (the largest nerve in the body), gluteal nerves, and pudendal nerve	The buttock, the perineum, the back of the thigh, and most of the leg and foot
Coccygeal plexus	S4, S5, and the coccygeal nerve (Co1)	Small nerves to the skin	A small area of skin over the coccyx

- Memory tip: the plexuses run head to tail in the same order as the cord regions that build them. Cervical for the neck and diaphragm, brachial for the upper limb, lumbar for the abdominal wall and front of the thigh, sacral for the buttock and the rest of the lower limb, and coccygeal for the skin over the coccyx.
- The lumbar and sacral plexuses share L4, and the sacral and coccygeal plexuses share S4. That is why the lower limb is sometimes described as being served by one lumbosacral plexus.
- The brachial plexus takes in the first thoracic nerve. When you bracket it on a drawing, make sure the bracket reaches T1.

Cervical plexus (C1 to C4, with a little of C5)

The cervical plexus lies deep in the side of the neck, under the sternocleidomastoid muscle. Its branches are of two kinds. The cutaneous branches come out around the back edge of the sternocleidomastoid and spread to the skin of the head, neck, and upper shoulder. The deep branches run to the muscles of the neck and, through the phrenic nerve, to the diaphragm.

Nerve	Spinal nerves	Where it goes
Lesser occipital nerve	C2	Skin of the scalp behind and above the ear
Great auricular nerve	C2, C3	Skin over the ear and the angle of the jaw
Transverse cervical nerve	C2, C3	Skin of the front and side of the neck
Supraclavicular nerves	C3, C4	Skin of the upper shoulder and the upper chest, above and below the clavicle
Ansa cervicalis	C1 to C3	The strap muscles in the front of the neck below the hyoid
Phrenic nerve	C3 to C5	The diaphragm. It runs down through the thorax, passing in front of the root of the lung, and is the only motor supply to the diaphragm
Segmental branches	C1 to C4	The deep muscles of the neck, with help to the sternocleidomastoid and trapezius

- The phrenic nerve is the branch that matters most. C3, 4, 5 keeps the diaphragm alive: an injury to the cord at or above that level can stop breathing.
- The phrenic nerve comes from the cervical plexus, not the brachial plexus. When you draw it, start it in the neck above the brachial plexus.

Brachial plexus (C5 to T1)

The brachial plexus starts in the lower neck, passes behind the clavicle, and runs into the axilla (armpit). It is the largest and most rearranged of the plexuses. The five anterior rami (the roots) join into three trunks (superior, middle, inferior). Each trunk splits into an anterior and a posterior division. The divisions regroup into three cords (lateral, posterior, medial), named for their position around the axillary artery. The cords give off the five main nerves of the upper limb. Roots, trunks, divisions, cords, branches: Really Tired, Drink Coffee, Be awake.

Nerve	Spinal nerves (approximate)	Where it goes
Axillary nerve	C5, C6	The shoulder: the muscles that cap the shoulder joint and the skin over the top of the shoulder
Musculocutaneous nerve	C5 to C7	The muscles on the front of the arm and the skin of the lateral (thumb side) forearm
Radial nerve	C5 to T1	The back of the whole limb: the muscles on the back of the arm and forearm and the skin of the back of the arm, forearm, and hand
Median nerve	C5 to T1	Most of the muscles on the front of the forearm, some muscles of the thumb side of the hand, and the skin of the lateral palm and the thumb side fingers
Ulnar nerve	C8, T1	Some muscles on the front of the forearm, most of the small muscles of the hand, and the skin of the medial (little finger side) hand

- Smaller branches leave the roots, trunks, and cords before the five main nerves form. They supply the muscles that move the scapula and the muscles of the chest wall and back that attach to the arm, and the skin of the medial arm and forearm. Books list the exact spinal nerves for each branch a little differently, so learn the range for the whole plexus (C5 to T1) and the main nerve names.
- Where to find them on a model: the axillary nerve wraps around the back of the top of the humerus. The musculocutaneous nerve runs down the front of the arm. The radial nerve spirals around the back of the humerus. The median nerve runs down the middle of the front of the arm and forearm and through the carpal tunnel at the wrist. The ulnar nerve passes behind the medial epicondyle of the humerus, the funny bone spot, and then down the medial forearm.

Lumbar plexus (L1 to L4)

The lumbar plexus forms inside the psoas major muscle, beside the lumbar vertebrae. Its branches run to the lower abdominal wall, the external genitals, and the front and inner thigh. The two large nerves are the femoral and obturator nerves.

Nerve	Spinal nerves	Where it goes
Iliohypogastric nerve	L1	The muscles and skin of the lower abdominal wall and the skin of the upper buttock
Ilioinguinal nerve	L1	The lower abdominal wall, the skin of the upper inner thigh, and the skin of the external genitals
Genitofemoral nerve	L1, L2	The skin of the external genitals and the skin of the upper front thigh
Lateral femoral cutaneous nerve	L2, L3	The skin of the outer (lateral) thigh
Femoral nerve	L2 to L4	The muscles on the front of the thigh and the skin of the front and inner thigh; its long cutaneous branch, the saphenous nerve, supplies the skin of the inner leg and foot
Obturator nerve	L2 to L4	The muscles on the inner thigh and the skin of the inner thigh. It leaves the pelvis through the obturator foramen

- The femoral nerve passes under the inguinal ligament into the front of the thigh. The obturator nerve goes through the obturator foramen of the hip bone to the inner thigh. The names tell you the route.

Sacral plexus (L4 to S4)

The sacral plexus lies on the back wall of the pelvis, in front of the sacrum and the piriformis muscle. It serves the buttock, the perineum, the back of the thigh, and almost all of the leg and foot. Its largest branch, the sciatic nerve, is the largest and longest nerve in the body.

Nerve	Spinal nerves	Where it goes
Superior gluteal nerve	L4 to S1	The muscles on the side of the buttock that steady the pelvis when you walk
Inferior gluteal nerve	L5 to S2	The large muscle of the buttock
Posterior femoral cutaneous nerve	S1 to S3	The skin of the lower buttock and the back of the thigh
Sciatic nerve	L4 to S3	Leaves the pelvis through the greater sciatic foramen, passes under the gluteus maximus, and runs down the back of the thigh to the muscles there. Above the knee it splits into the tibial and common fibular nerves
Tibial nerve (from the sciatic)	L4 to S3	The back of the leg and the sole of the foot, muscles and skin
Common fibular (peroneal) nerve (from the sciatic)	L4 to S2	The side and front of the leg and the top of the foot, muscles and skin. It wraps around the neck of the fibula just below the knee, where it sits close under the skin
Pudendal nerve	S2 to S4	The perineum: the muscles of the pelvic floor and the external anal and urethral sphincters, and the skin of the external genitals

- The sciatic nerve is really two nerves, the tibial and the common fibular, wrapped in one sheath. The tibial half serves the back of the leg and the sole; the fibular half serves the front and side of the leg and the top of the foot.
- The pudendal nerve is the nerve of the perineum. It carries sensation from the external genitals and controls the sphincters you consciously close.

Coccygeal plexus (S4, S5, and Co1)

The smallest plexus. It is formed by the anterior rami of S4 and S5 and the single coccygeal nerve, and it lies on the coccygeus muscle at the bottom of the pelvis. Its branches, the anococcygeal nerves, supply a small area of skin over the coccyx and around it.

Predicting the effect of plexus damage

Each plexus serves one region, so the plexus that is damaged predicts the region that loses movement and sensation. To answer a question, first name the plexus, then name the region it serves. The damage shows up on the same side of the body as the injured plexus.

Plexus damaged	Region affected
Cervical plexus	Weakness or numbness of the neck and shoulder; if the phrenic nerve is involved, the diaphragm on that side is weak and breathing is affected
Brachial plexus	Weakness and sensory loss across the shoulder and the whole upper limb on that side
Lumbar plexus	Weakness and sensory loss of the lower abdominal wall and the front and inner surfaces of the thigh
Sacral plexus	Weakness and sensory loss of the buttock, the back of the thigh, and the leg and foot
Coccygeal plexus	Numbness of the small patch of skin over the coccyx

Why the network matters

- The plexus design is a safety feature. Each limb region draws axons from several spinal nerves, so a single damaged spinal nerve root weakens an area but rarely paralyzes it completely.
- Damage to the plexus itself is different. It can damage axons from several spinal nerves at once, so it can take out a whole region. A brachial plexus injury affects the upper limb, and a lumbar or sacral plexus injury affects the lower limb.
- The phrenic nerve leaves the cervical plexus and is the only motor supply to the diaphragm. An injury to the cord at or above C3 to C5 can stop breathing, which is why a high neck injury is an emergency for the airway, not just for the limbs.

What this means for patients

- Damage to a plexus shows up as weakness and numbness in the region that plexus serves, on the same side as the injury. Name the plexus, then name the region.
- A cut or crushed phrenic nerve weakens the diaphragm on that side, so the patient's breathing suffers. If the cord is damaged at or above C3 to C5, both phrenic nerves lose their supply and the patient cannot breathe without help.
- Because a thoracic nerve does not join a plexus, damage to one intercostal nerve affects only one strip of chest or abdominal wall, not a whole region.

Competency 192. Autonomic two-neuron pathway

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

The motor side of the peripheral nervous system has two parts. The somatic nervous system controls skeletal muscle, and that control is usually voluntary. The autonomic nervous system (ANS) controls cardiac muscle, smooth muscle, and glands, and that control is involuntary. Both start in the central nervous system. The big structural difference is how many neurons it takes to get from the CNS to the effector: one for somatic, two for autonomic.

Somatic and autonomic motor systems compared

Feature	Somatic nervous system	Autonomic nervous system
Effectors	Skeletal muscle	Cardiac muscle, smooth muscle, and glands
Control	Voluntary	Involuntary
Motor neurons from CNS to effector	One	Two, with a synapse in an autonomic ganglion
Where the motor cell bodies are	Anterior horn of the cord or a motor nucleus of the brainstem	First neuron in the brainstem or the cord; second neuron in an autonomic ganglion
Myelin	The axon is myelinated	The first axon is lightly myelinated; the second axon is unmyelinated
Effect on the effector	Always excites; causes contraction	Can excite or inhibit
Awareness	Normally conscious	Normally not conscious

- The somatic system only ever excites its muscle. If a skeletal muscle needs to relax, its motor neuron just stops firing. Inhibition belongs in the autonomic column: the ANS can slow the heart as well as speed it up.
- The somatic pathway is one myelinated axon from a cell body in the anterior horn of the spinal cord (or a brainstem motor nucleus) straight to the skeletal muscle. That is the whole pathway.

The two-neuron pathway, in order

The autonomic system always uses two motor neurons in a row, and they meet in a ganglion. Trace the signal from the CNS to the effector.

1. **Preganglionic neuron.** Its cell body is in the CNS, in a brainstem nucleus or in the lateral gray matter of the spinal cord. Its axon is thin and lightly myelinated. It leaves the CNS in a cranial nerve or an anterior root and ends in an autonomic ganglion.
2. **Autonomic ganglion.** A cluster of neuron cell bodies outside the CNS. Here the preganglionic axon synapses with the second neuron. This is the only synapse on the way out.
3. **Postganglionic neuron.** Its cell body sits in the ganglion. Its unmyelinated axon runs from the ganglion out to the effector and carries the signal the rest of the way.
4. **Effector.** Cardiac muscle, smooth muscle, or a gland receives the signal.

Preganglionic neuron
The first neuron. Cell body in the brainstem or the lateral gray matter of the cord; lightly myelinated axon that leaves the CNS and ends in a ganglion.

Autonomic ganglion
A cluster of motor cell bodies outside the CNS where the preganglionic axon synapses with the postganglionic neuron.

Postganglionic neuron
The second neuron. Cell body in the ganglion; unmyelinated axon that runs to the effector.

Effector
The tissue that responds: cardiac muscle, smooth muscle, or a gland.

- The preganglionic axon is the lightly myelinated one and the postganglionic axon is unmyelinated. Do not swap them on your drawing.
- Memory tip: pre means before the ganglion and post means after it.
- Do not mix up the two kinds of ganglia. A posterior root ganglion holds sensory cell bodies and has no synapses. An autonomic ganglion holds motor cell bodies and is where the synapse happens.
- The ANS also has sensory neurons. They carry signals from receptors in the organs and blood vessels back to the CNS, and those signals let the CNS adjust the organs. The two-neuron pathway is the motor side only.

Competency 193. Autonomic ganglia and divisions

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

The ANS has two divisions, the sympathetic and the parasympathetic. Most organs get axons from both, and the two usually have opposite effects. The two divisions differ in where their preganglionic neurons start in the CNS and in where their ganglia sit. Those two facts, origin and ganglion location, set the lengths of the axons and give the divisions their other names.

The sympathetic division

The sympathetic division is the fight or flight division. It takes over during exercise, excitement, and emergencies. It is also called the thoracolumbar division, because its preganglionic cell bodies sit in the lateral horns of the T1 to L2 spinal segments. Its ganglia lie close to the spinal cord, so its preganglionic axons are short and its postganglionic axons are long.

Sympathetic trunk
A chain of ganglia on each side of the vertebral column, running from the base of the skull to the coccyx. Axons link one ganglion to the next. The trunk connects to the spinal nerves through the white and gray rami communicantes.

Sympathetic trunk ganglia
The ganglia of the chain, also called paravertebral or chain ganglia. Postganglionic neurons here send their axons back into the spinal nerves to reach the skin, blood vessels, and the organs of the thorax.

Prevertebral ganglia
Also called collateral ganglia. They lie in front of the vertebral column, close to the large branches of the abdominal aorta. They are the celiac, superior mesenteric, inferior mesenteric, aorticorenal, and renal ganglia. Preganglionic axons reach them by passing through the sympathetic trunk without synapsing, in nerves called splanchnic nerves.

White ramus communicans
The short branch that carries the lightly myelinated preganglionic axon from the spinal nerve into the sympathetic trunk. White because of the myelin. Only present at T1 to L2, where the preganglionic neurons are.

Gray ramus communicans
The short branch that carries the unmyelinated postganglionic axon from the trunk ganglion back into the spinal nerve. Gray because there is no myelin. Present at every spinal nerve, since every spinal nerve carries sympathetic axons to skin and vessels.

Follow a sympathetic signal to a sweat gland in the skin

1. Lateral horn (T1 to L2): the preganglionic cell body sends out its axon.
2. Anterior root and spinal nerve: the preganglionic axon leaves the cord.
3. White ramus communicans: the preganglionic axon enters the sympathetic trunk.
4. Sympathetic trunk ganglion: the preganglionic axon synapses with the postganglionic neuron.
5. Gray ramus communicans: the postganglionic axon returns to the spinal nerve.
6. Effector: the axon runs out in a ramus of the spinal nerve to a sweat gland, an arrector pili muscle, or a blood vessel in the skin.

For the abdominal and pelvic organs, the path differs after the white ramus. The preganglionic axon passes through the trunk ganglion without synapsing and travels in a splanchnic nerve to a prevertebral ganglion. It synapses there. The postganglionic axon then follows the arteries to the organ.

- A preganglionic sympathetic axon that enters the trunk has three choices: synapse in the trunk ganglion at its own level, travel up or down the chain and synapse in a trunk ganglion at another level, or pass straight through and synapse in a prevertebral ganglion. That is how T1 to L2 can reach every part of the body, from the pupil to the pelvis.

The parasympathetic division

The parasympathetic division is the rest and digest division. It conserves energy and runs routine work such as digestion, urination, and defecation. It is also called the craniosacral division. Its preganglionic cell bodies sit in the brainstem nuclei of four cranial nerves (CN III, VII, IX, and X) and in the lateral gray matter of the S2 to S4 spinal segments.

Terminal ganglia
The ganglia of the parasympathetic division. They lie close to the organ or inside its wall (intramural ganglia). So the preganglionic axons are long and the postganglionic axons are short.

Vagus nerve (CN X)

Carries preganglionic axons to terminal ganglia in the walls of the thoracic and abdominal organs. It carries most of the parasympathetic supply to the chest and abdomen, which is why one nerve can slow the heart, narrow the airways, and speed digestion.

Pelvic splanchnic nerves

From S2 to S4. They carry preganglionic axons to the pelvic organs and the last part of the large intestine.

The four named terminal ganglia of the head

Ganglion	Cranial nerve that brings the preganglionic axons	Target
Ciliary ganglion	CN III	The muscles that constrict the pupil and focus the lens
Pterygopalatine ganglion	CN VII	The lacrimal gland and the glands of the nasal cavity
Submandibular ganglion	CN VII	The submandibular and sublingual salivary glands
Otic ganglion	CN IX	The parotid salivary gland

- CN VII reaches two ganglia, one for tears and nasal mucus and one for the two salivary glands under the jaw. CN IX reaches the parotid. CN III reaches the eye. CN X skips the head and takes care of the chest and abdomen, synapsing in ganglia in the organ walls.

Sympathetic and parasympathetic divisions compared

Feature	Sympathetic	Parasympathetic
Other name	Thoracolumbar division	Craniosacral division
Origin in the CNS	Lateral horns of the T1 to L2 spinal segments	Brainstem nuclei of CN III, VII, IX, and X, and the S2 to S4 spinal segments
Location of ganglia	Close to the spinal cord: sympathetic trunk ganglia and prevertebral ganglia	Close to or in the wall of the organ: terminal ganglia
Named ganglia	Sympathetic trunk (chain) ganglia; prevertebral ganglia: celiac, superior mesenteric, inferior mesenteric, aorticorenal, renal	Terminal ganglia, including the ciliary, pterygopalatine, submandibular, and otic
Preganglionic axon	Short	Long
Postganglionic axon	Long	Short
Overall role	Fight or flight; prepares the body for action	Rest and digest; conserves energy

- Memory tip: the other two names map the origins. Thoracolumbar means the preganglionic neurons start from T1 to L2. Craniosacral means they start in the brainstem and in S2 to S4. The ganglia follow: sympathetic near the cord, parasympathetic at the organ.
- Sympathetic leaves the middle of the cord, T1 to L2. Parasympathetic leaves the two ends, the brainstem and S2 to S4. Thoracolumbar is sympathetic and craniosacral is parasympathetic; check that you did not swap the names.
- Ganglion location sets axon length. Ganglia near the cord mean a short first axon and a long second axon (sympathetic). Ganglia at the organ mean a long first axon and a short second axon (parasympathetic).

The autonomic plexuses

In the thorax, abdomen, and pelvis, autonomic axons form networks around the large blood vessels. These are the autonomic plexuses. They hold sympathetic and parasympathetic axons together, along with sensory axons from the organs. Some also hold sympathetic ganglia. For example, the celiac ganglia sit inside the celiac plexus.

Plexus	Where it is and what it supplies
Cardiac plexus	Near the base of the heart; supplies the heart
Pulmonary plexus	Around the bronchi at the root of each lung; supplies the bronchial tree
Celiac (solar) plexus	The largest; surrounds the celiac trunk and supplies most of the upper abdominal organs
Superior and inferior mesenteric plexuses	Around the mesenteric arteries; supply the small and large intestines
Renal plexus	Around the renal arteries; supplies the kidneys and ureters
Hypogastric plexus	In front of the lower lumbar vertebrae and in the pelvis; supplies the pelvic organs

- An autonomic plexus is a mixed network around a vessel. It is not the same thing as a ganglion, though a ganglion can sit inside one. And it is not the same thing as a spinal nerve plexus from competency 191, which is a regrouping of anterior rami for skin and skeletal muscle.

The enteric plexuses

The wall of the digestive tract has its own networks of neurons, the enteric plexuses. They run from the esophagus to the anus and form the enteric nervous system, which has about 100 million neurons. These networks can run the gut wall on their own, and the ANS adjusts their activity.

Myenteric plexus
: Lies in the muscularis externa, between the circular and longitudinal muscle layers. It mainly controls the movement of the gut.

Submucosal plexus
: Lies in the submucosa, between the muscularis mucosae and the circular muscle layer. It mainly controls the secretions of the gut.

- Myenteric means in the muscle, and it runs the muscle. Submucosal means under the mucosa, and it runs the glands of the mucosa.
- The enteric nervous system is independent enough to be called the brain of the gut. Its plexuses can keep the gut wall contracting in rhythm and its glands secreting even when the connections to the CNS are cut.

What this means for patients

- The two divisions usually pull an organ in opposite directions, so the body can adjust an organ by easing off one division as well as by pushing the other. Heart rate, for example, rises when parasympathetic input is withdrawn, even before the sympathetic division speeds the heart up.
- The vagus nerve carries most of the parasympathetic supply to the chest and abdomen. Its long reach is why one nerve can slow the heart, narrow the airways, and speed digestion, and why damage to it affects so many organs at once.
- Because the gut has its own nervous system, digestion keeps going even when the autonomic connections to it are cut. The ANS turns the enteric plexuses up or down; it does not run them.